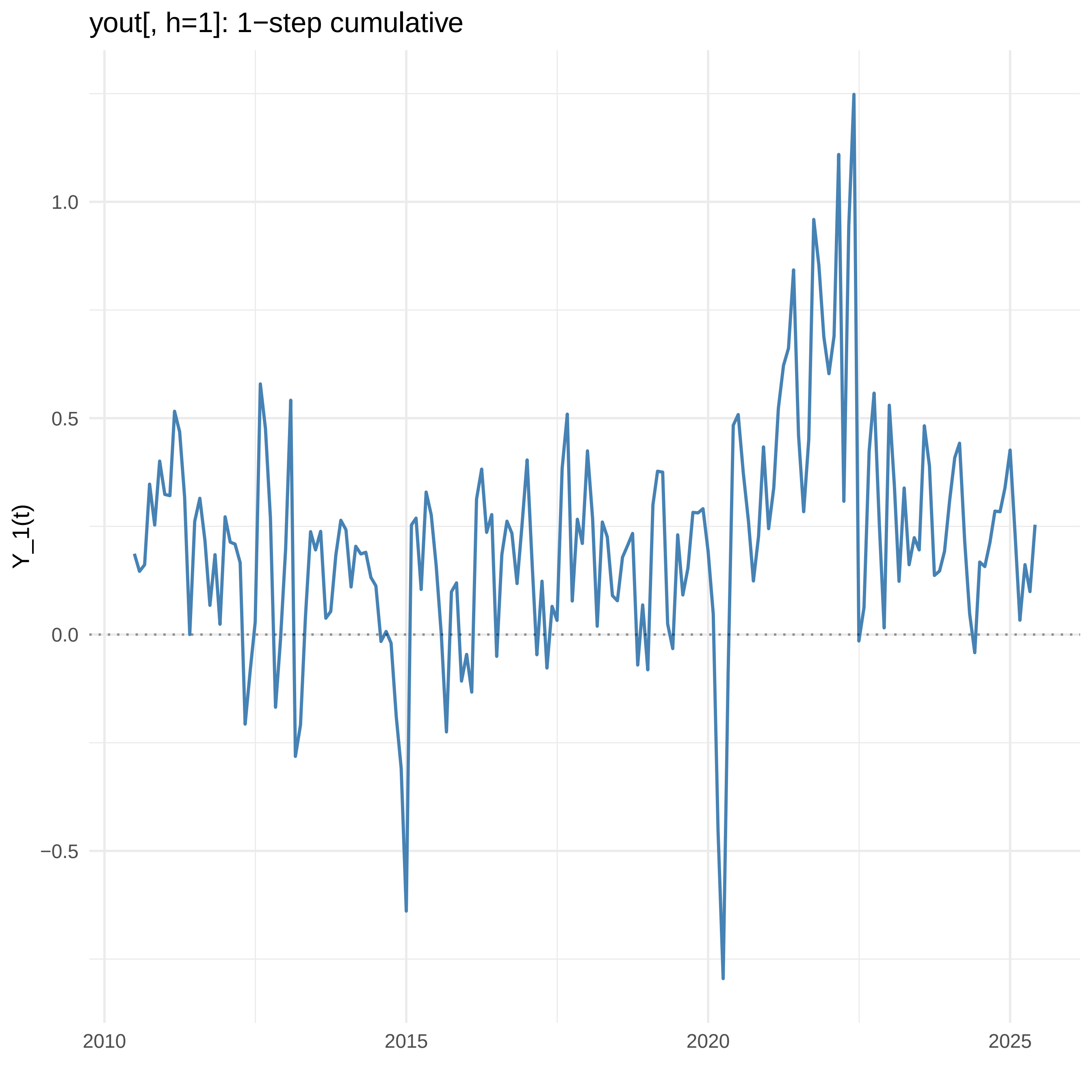
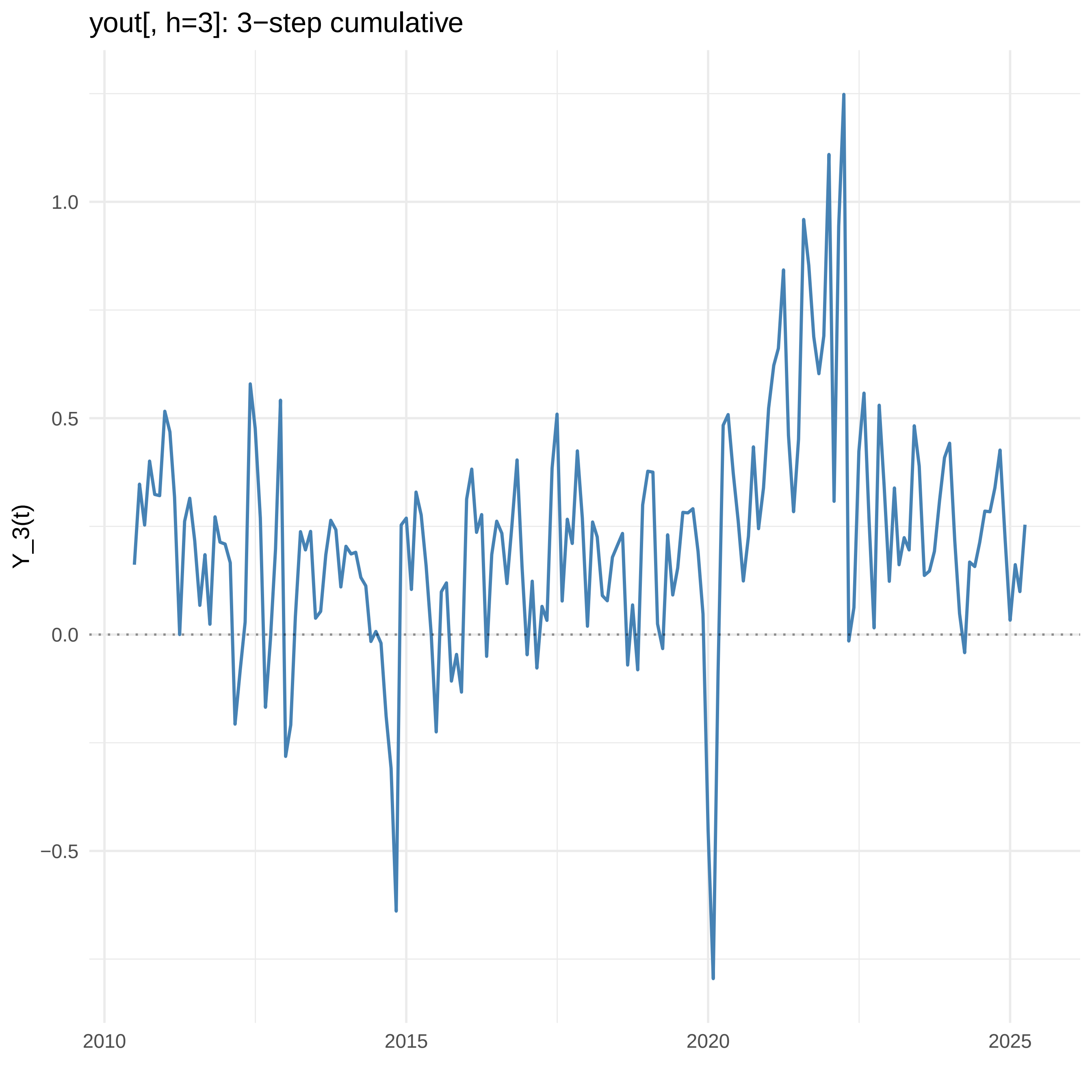
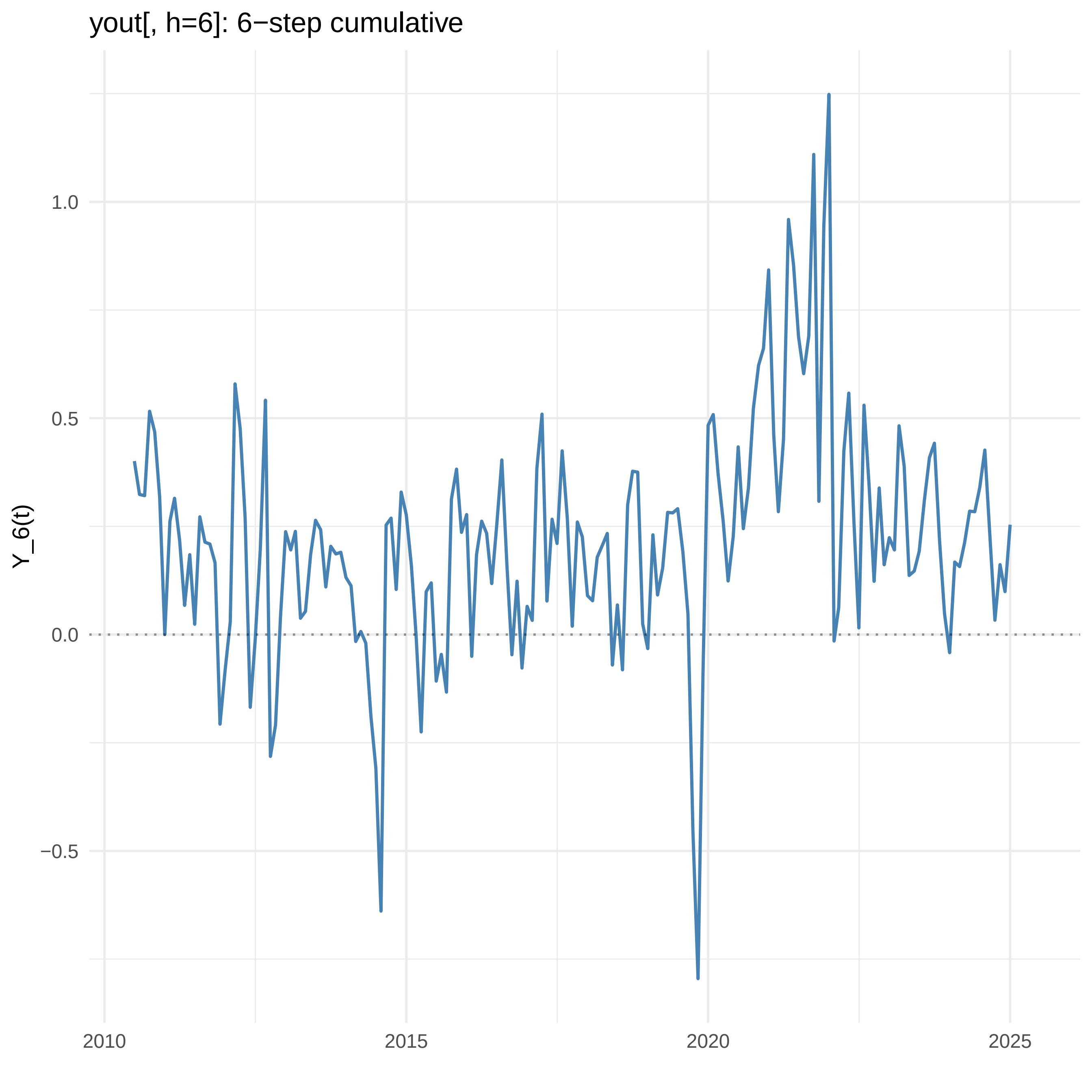
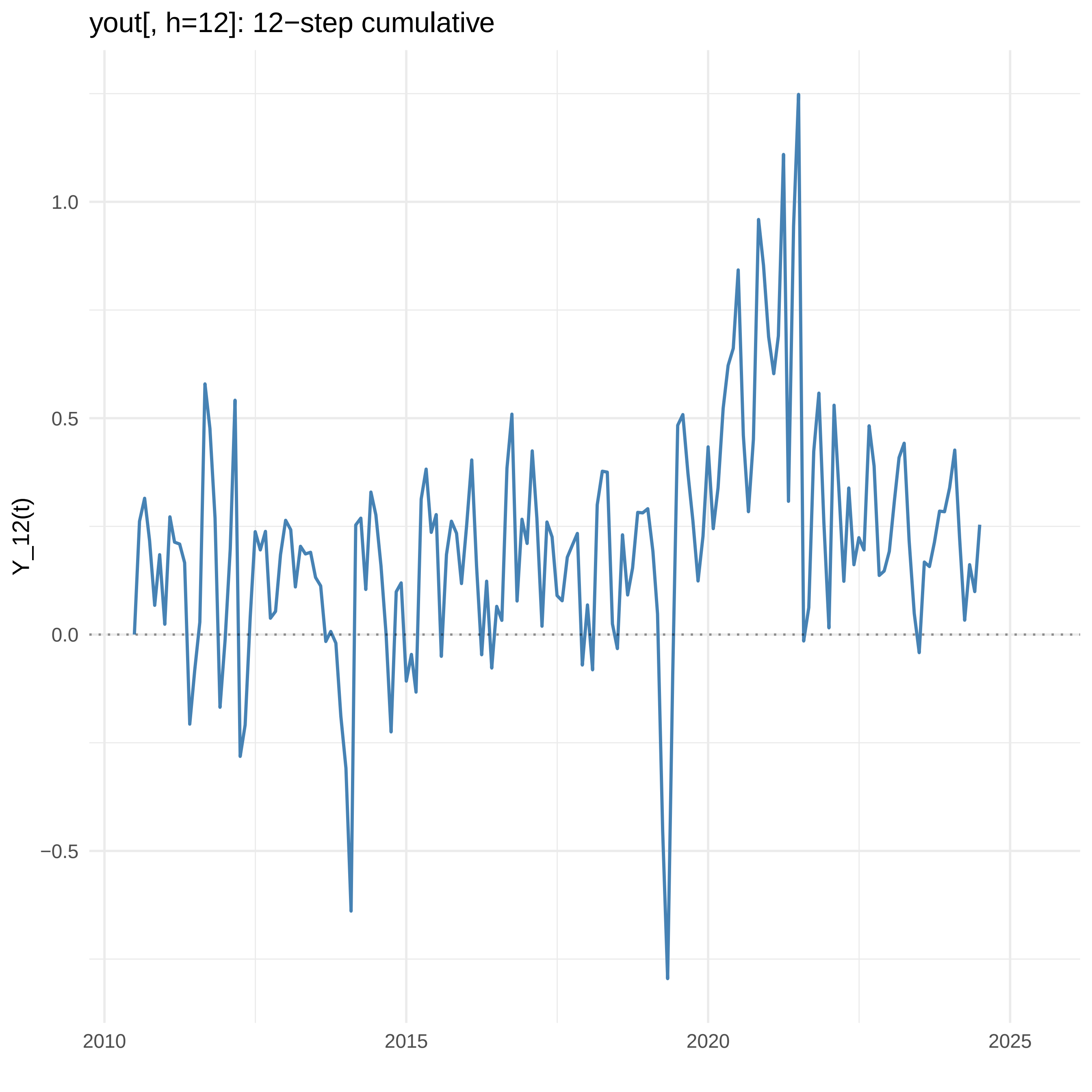

yout[, h=1]: 1−step cumulative
Y_1(t)
1.0
0.5
0.0
−0.5
2010
2015
2020
2025

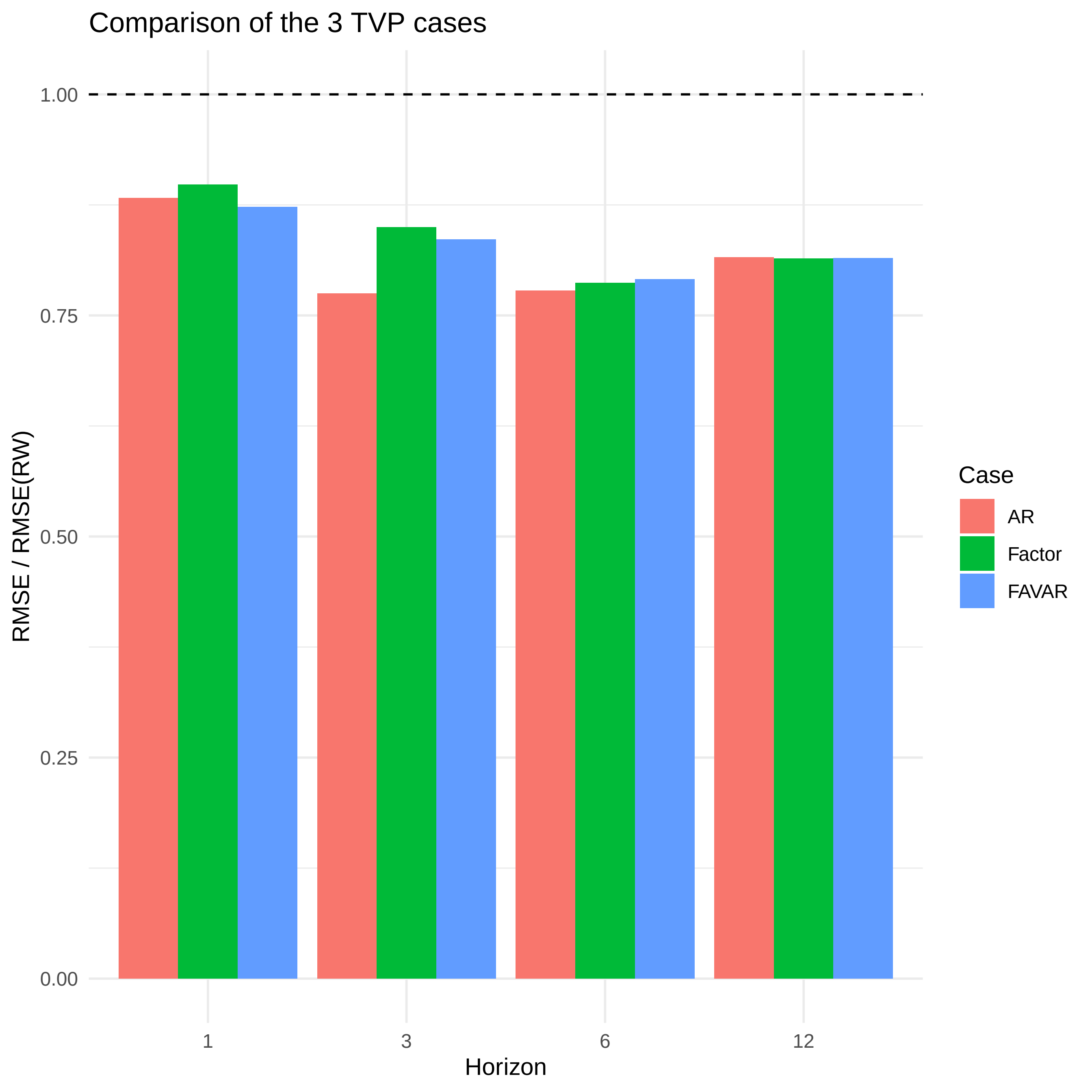

yout[, h=3]: 3−step cumulative
Y_3(t)
1.0
0.5
0.0
−0.5
2010
2015
2020
2025



yout[, h=6]: 6−step cumulative
Y_6(t)
1.0
0.5
0.0
−0.5
2010
2015
2020
2025



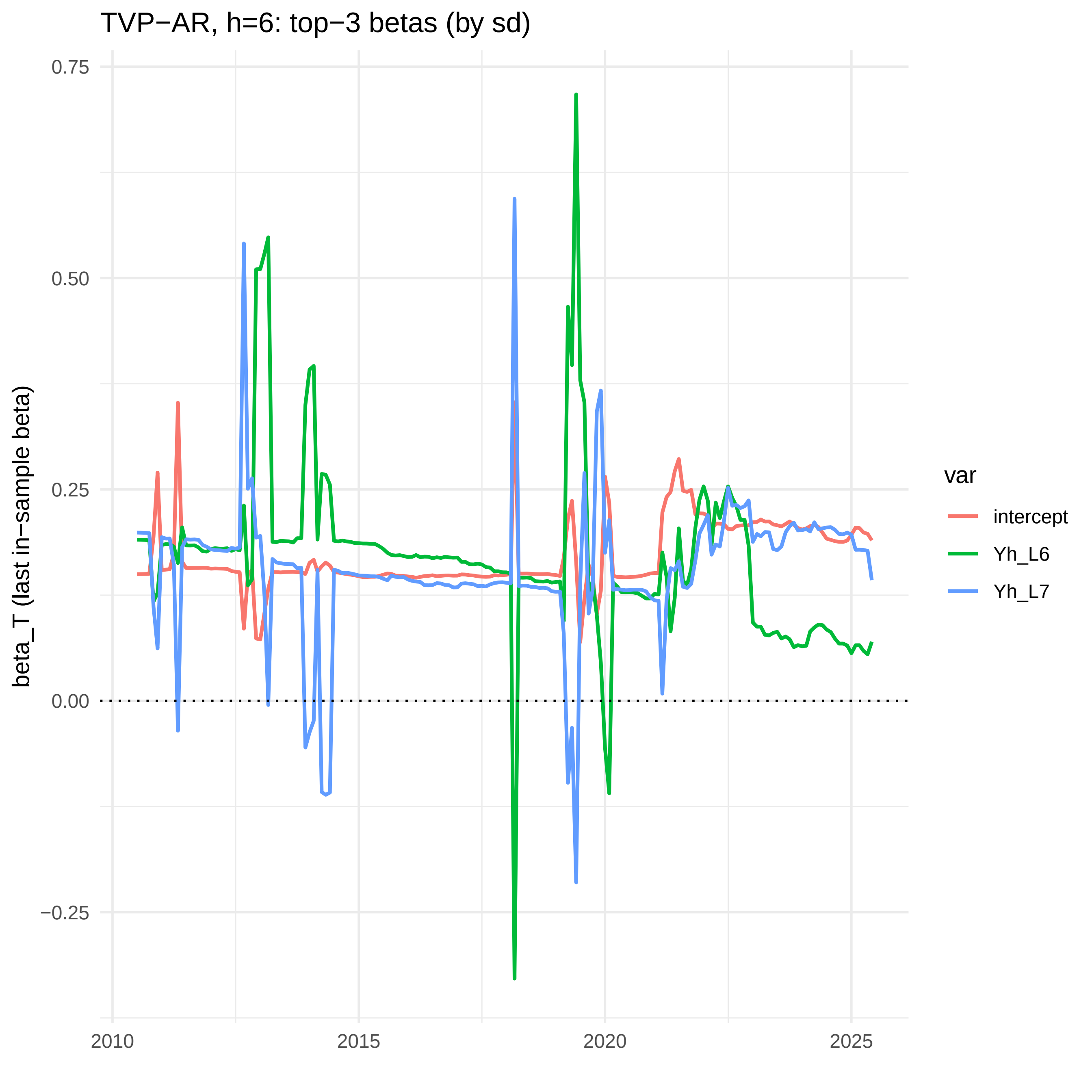
yout[, h=12]: 12−step cumulative
Y_12(t)
1.0
0.5
0.0
−0.5
2010
2015
2020
2025

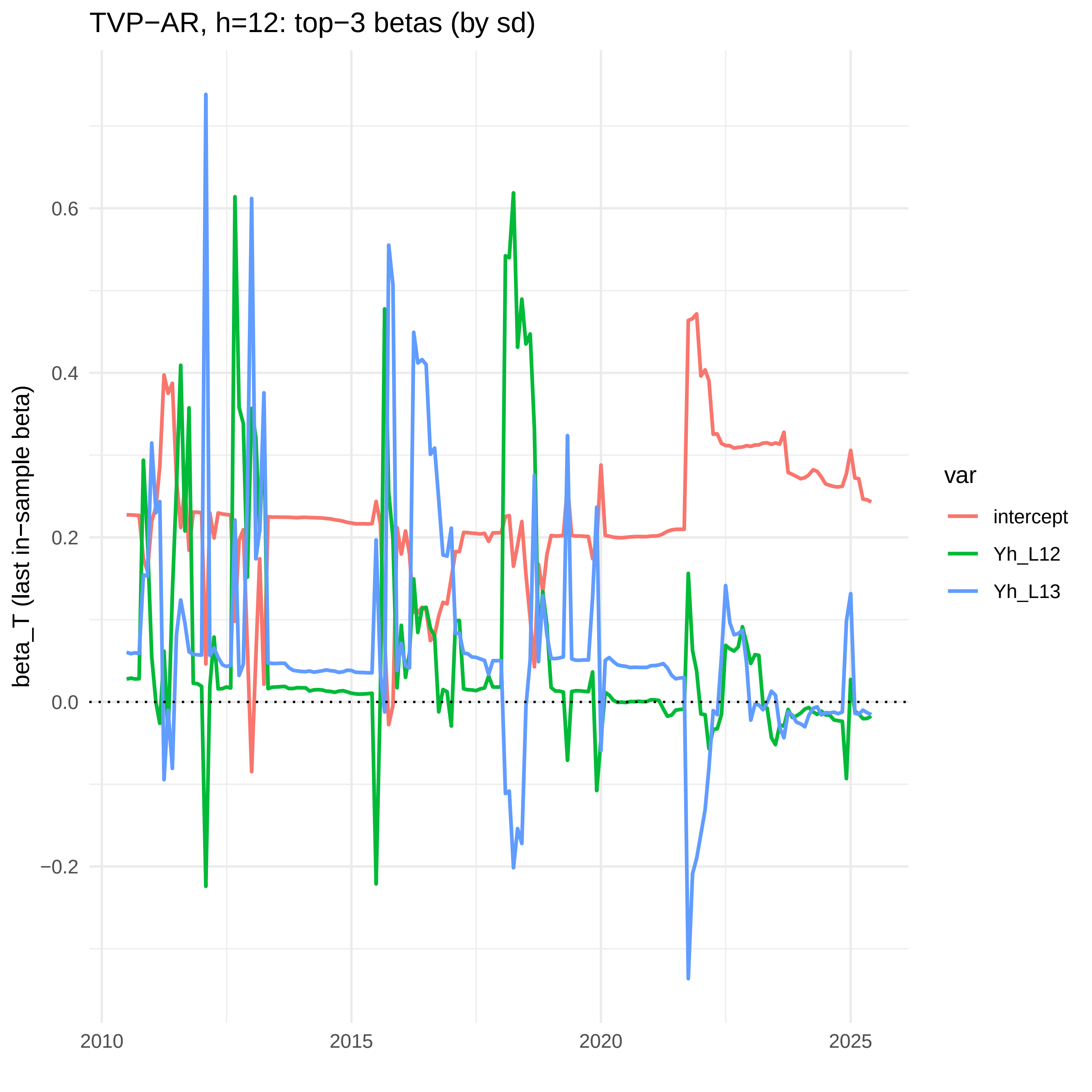


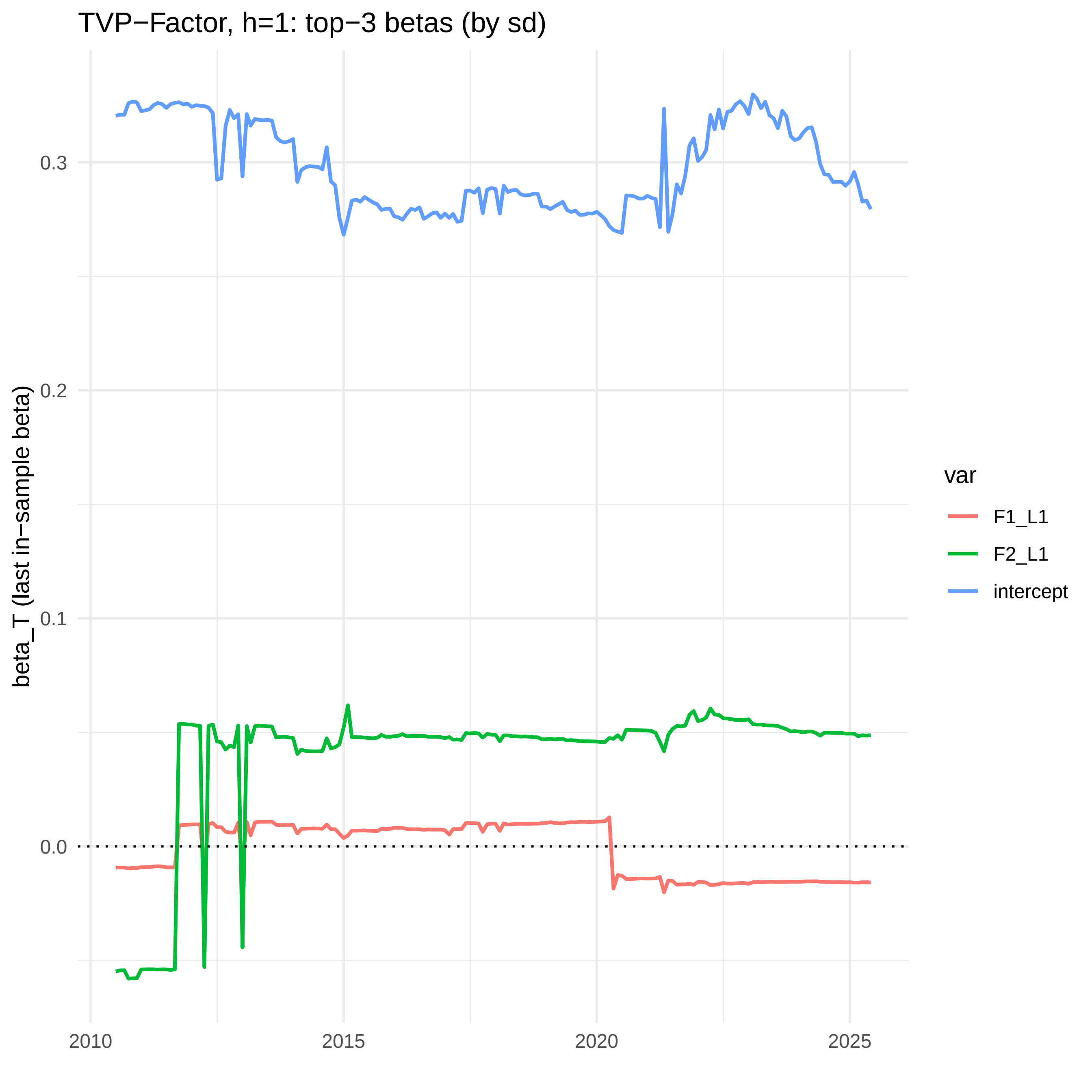
Comparison of the 3 TVP cases
RMSE / RMSE(RW)
1.00
0.75
0.50
0.25
0.00
1
3
6
12
Horizon
Case
AR
Factor
FAVAR


TVP−AR, h=1: top−3 betas (by sd)
beta_T (last in−sample beta)
0.9
0.6
0.3
0.0
−0.3
2010
2015
2020
2025
var
intercept
Yh_L1
Yh_L2

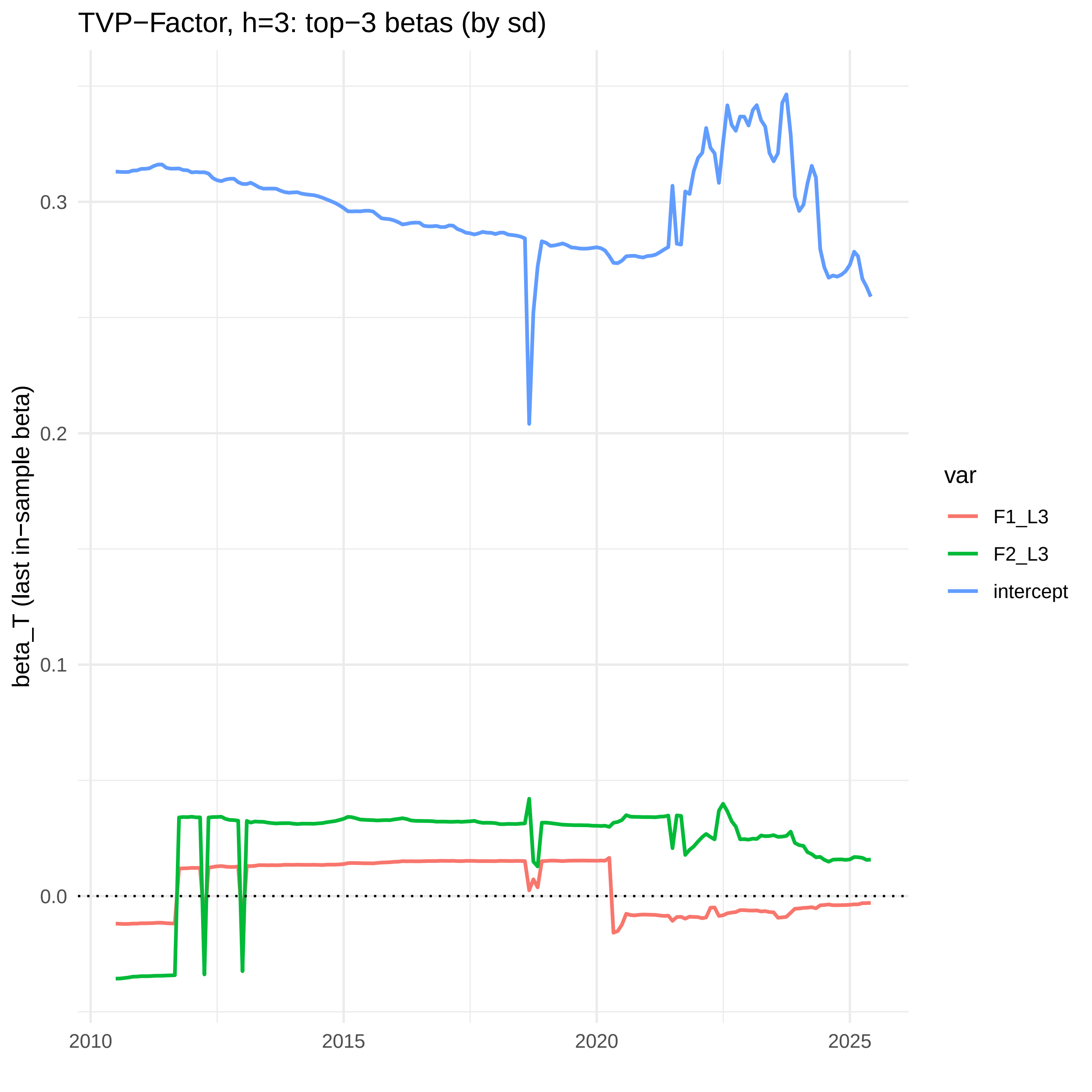
TVP−AR, h=3: top−3 betas (by sd)
beta_T (last in−sample beta)
0.4
0.2
0.0
2010
2015
2020
2025
var
intercept
Yh_L3
Yh_L4


TVP−AR, h=6: top−3 betas (by sd)
beta_T (last in−sample beta)
0.75
0.50
0.25
0.00
−0.25
2010
2015
2020
2025
var
intercept
Yh_L6
Yh_L7

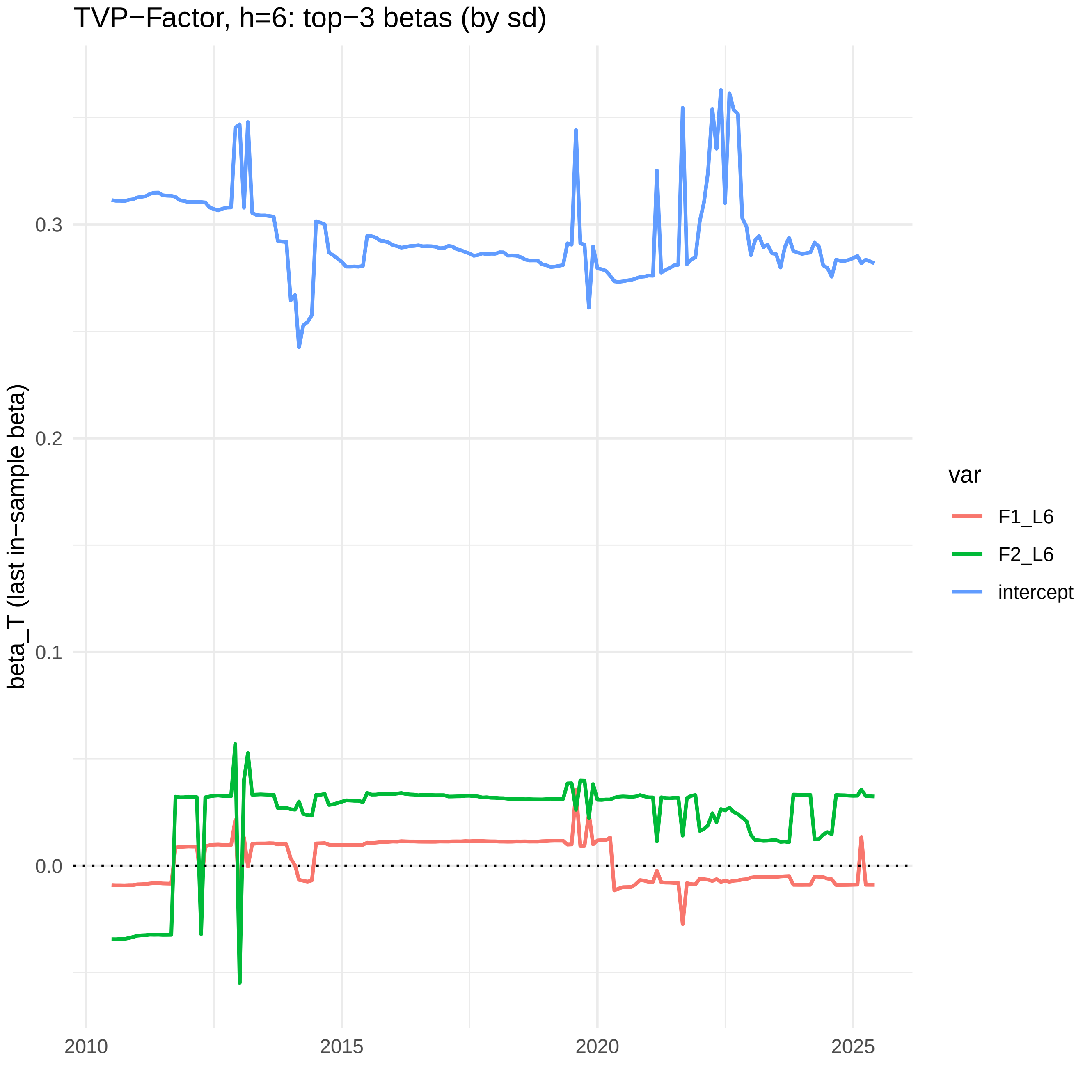

TVP−AR, h=12: top−3 betas (by sd)
beta_T (last in−sample beta)
0.6
0.4
0.2
0.0
−0.2
2010
2015
2020
2025
var
intercept
Yh_L12
Yh_L13



TVP−Factor, h=1: top−3 betas (by sd)
beta_T (last in−sample beta)
0.3
0.2
0.1
0.0
2010
2015
2020
2025
var
F1_L1
F2_L1
intercept

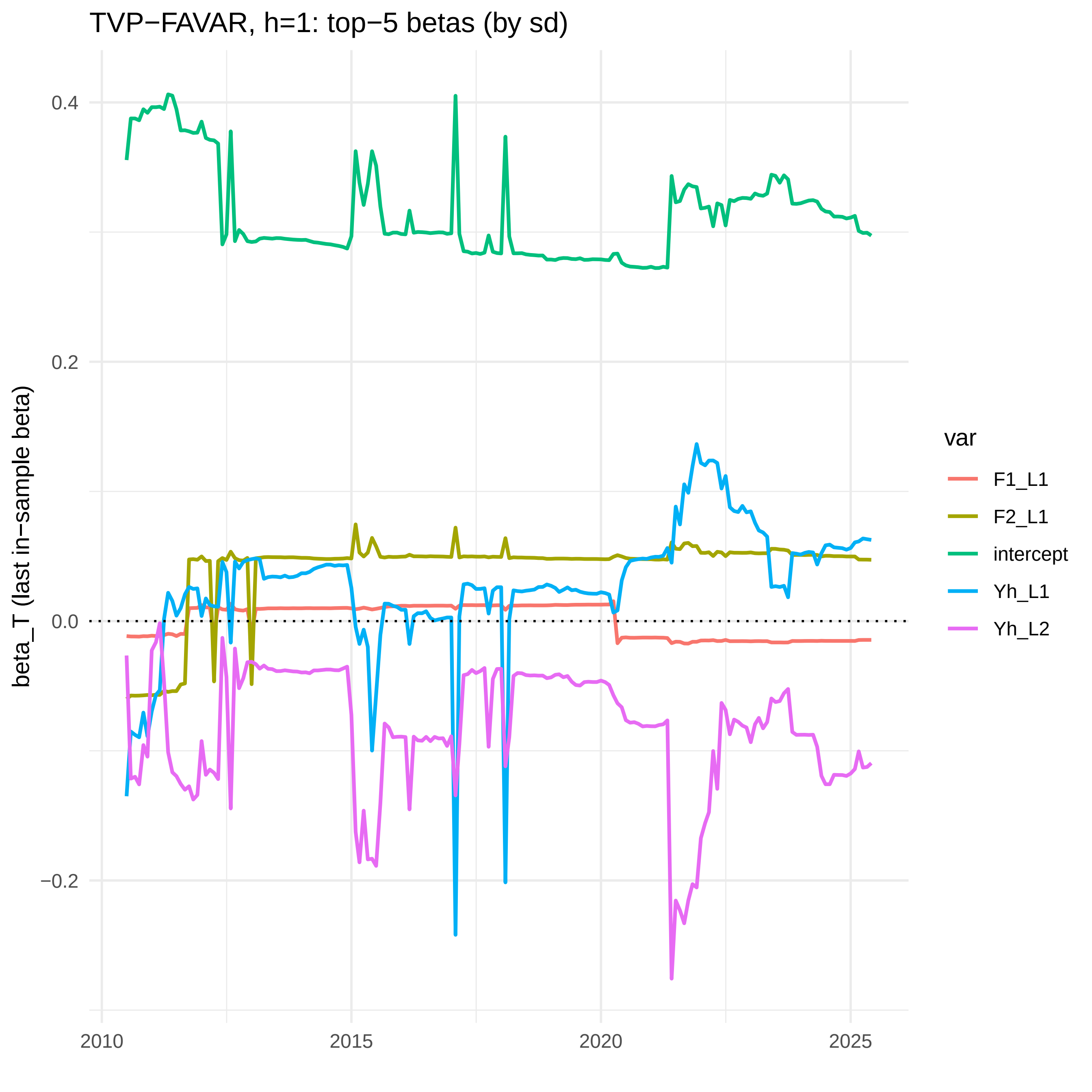


TVP−Factor, h=3: top−3 betas (by sd)
beta_T (last in−sample beta)
0.3
0.2
0.1
0.0
2010
2015
2020
2025
var
F1_L3
F2_L3
intercept

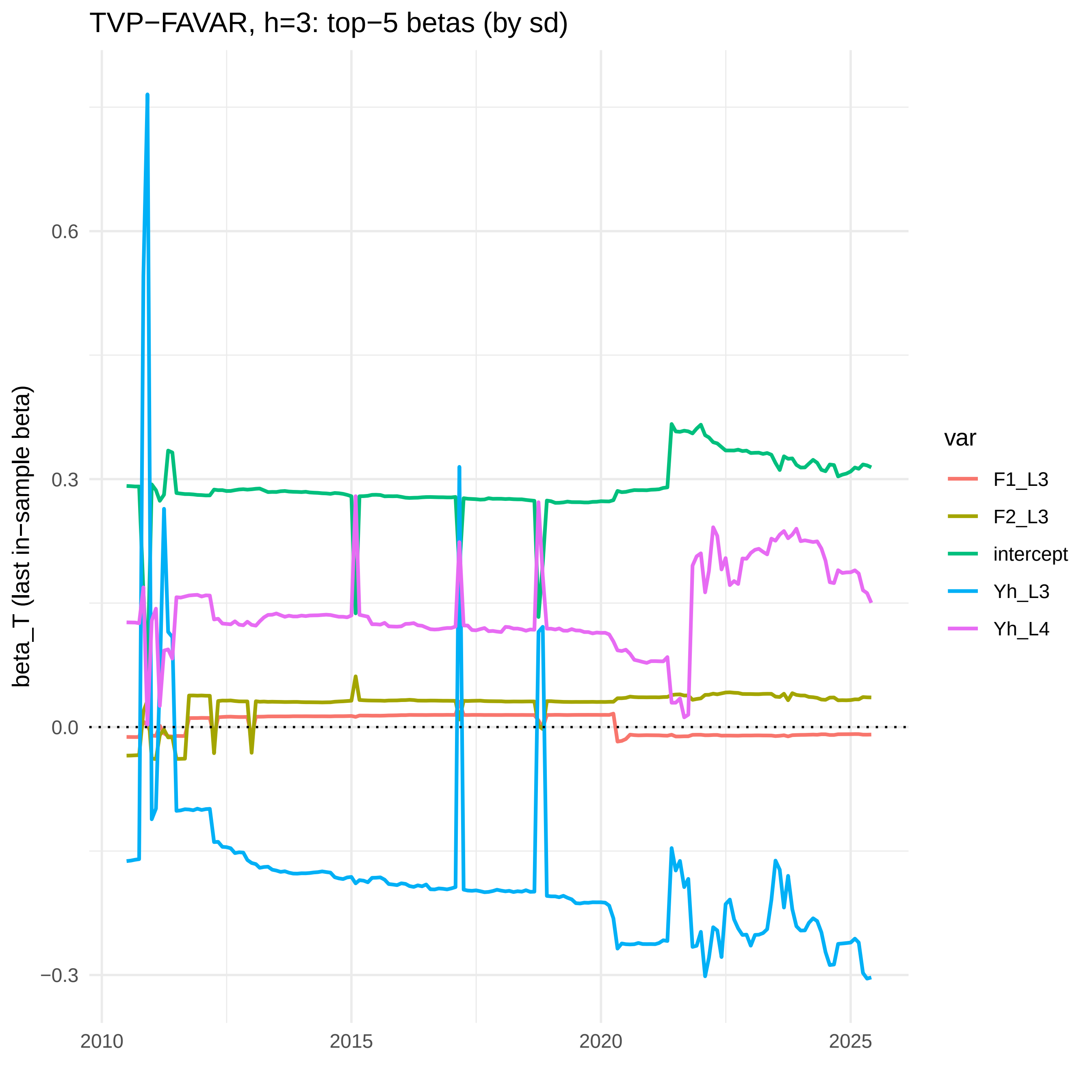


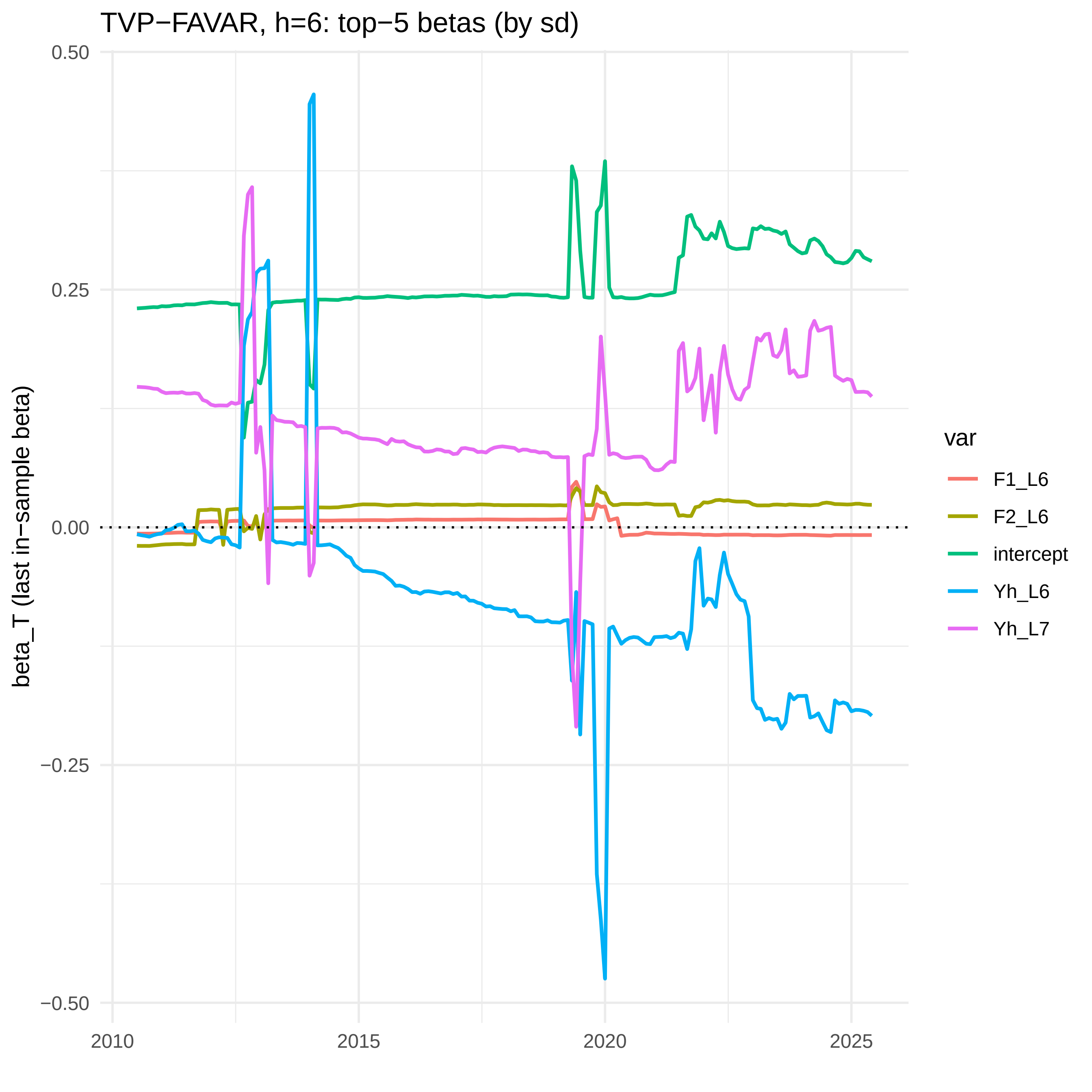
TVP−Factor, h=6: top−3 betas (by sd)
beta_T (last in−sample beta)
0.3
0.2
0.1
0.0
2010
2015
2020
2025
var
F1_L6
F2_L6
intercept

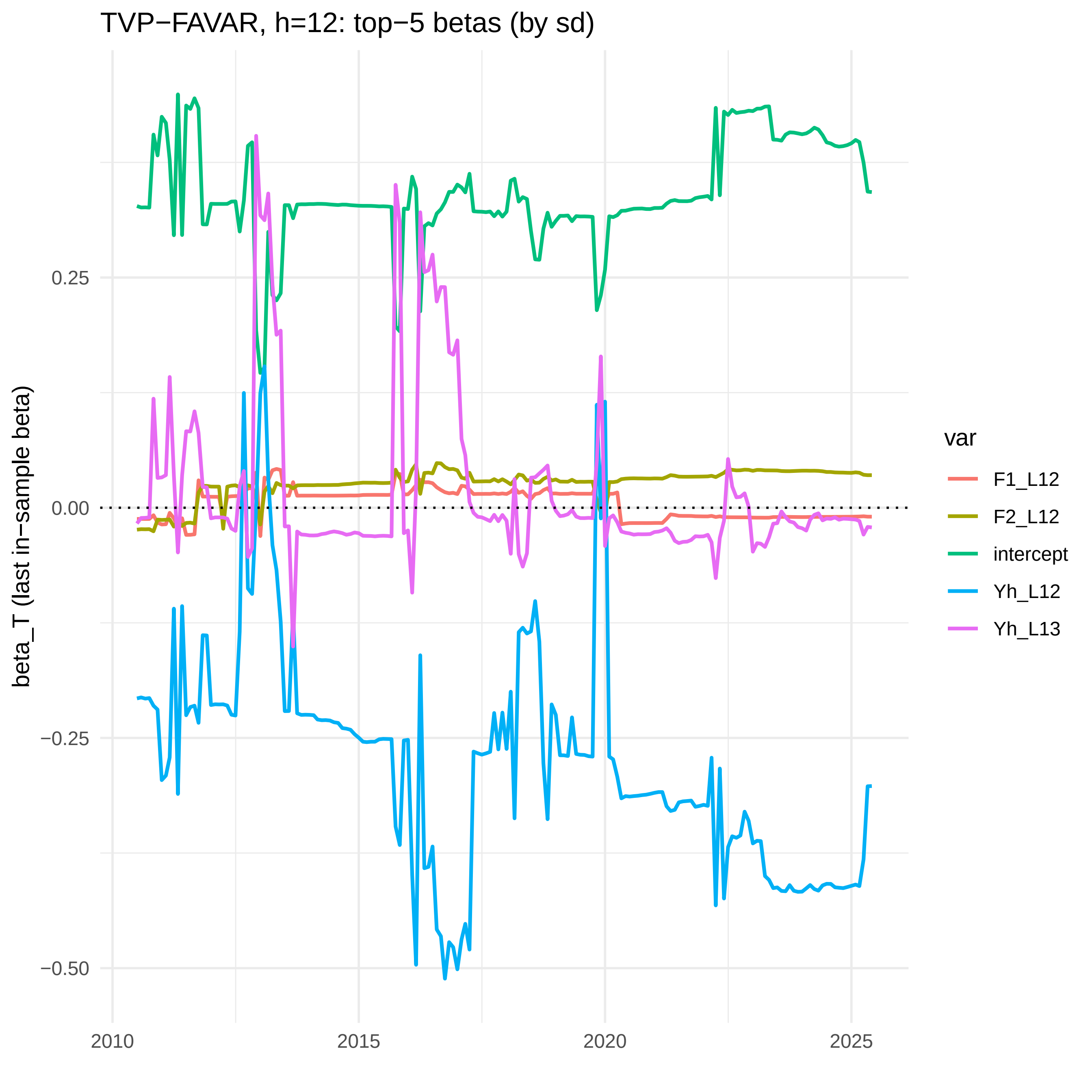

TVP−Factor, h=12: top−3 betas (by sd)
beta_T (last in−sample beta)
0.4
0.3
0.2
0.1
0.0
2010
2015
2020
2025
var
F1_L12
F2_L12
intercept



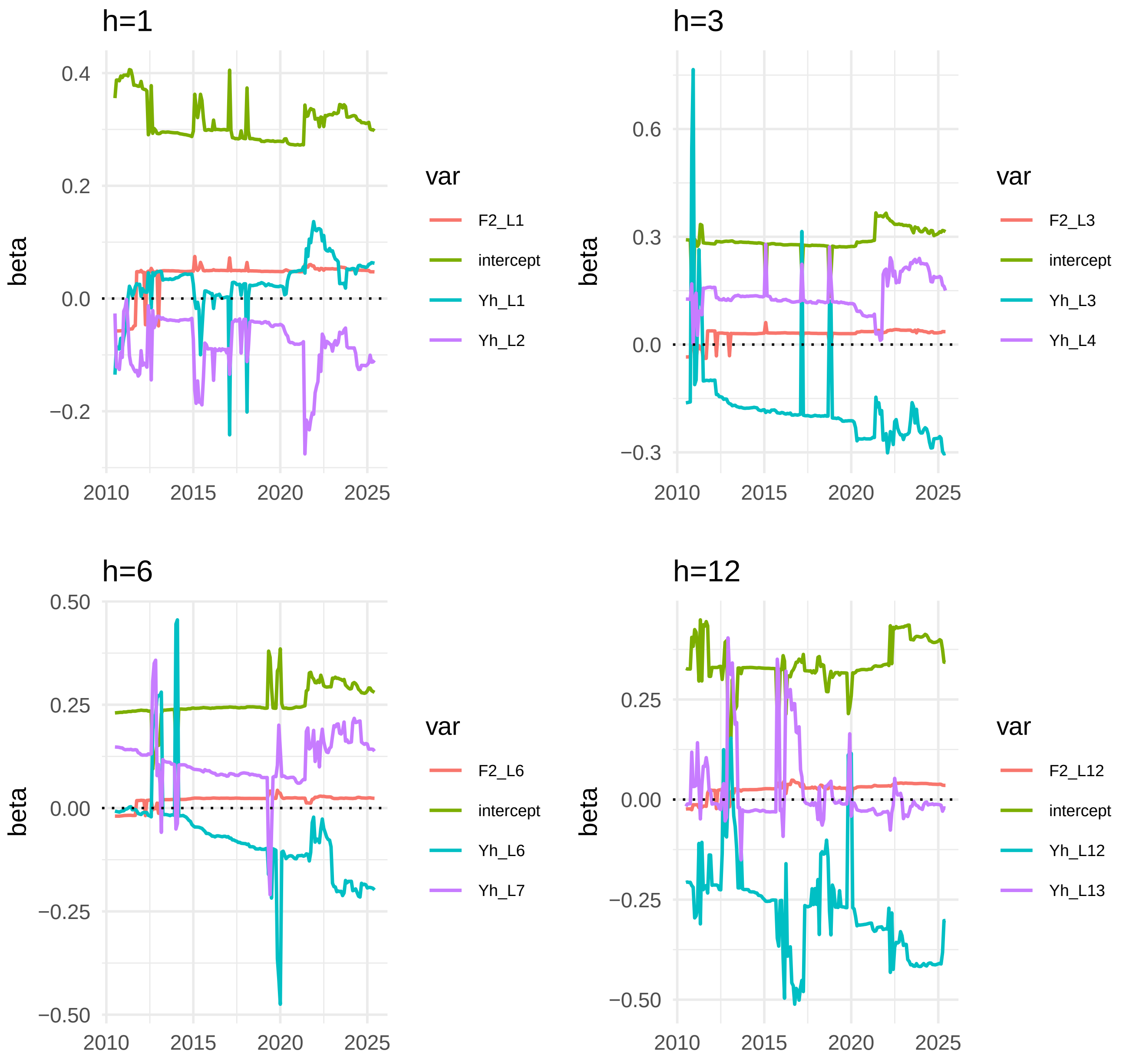
TVP−FAVAR, h=1: top−5 betas (by sd)
beta_T (last in−sample beta)
0.4
0.2
0.0
−0.2
2010
2015
2020
2025
var
F1_L1
F2_L1
intercept
Yh_L1
Yh_L2

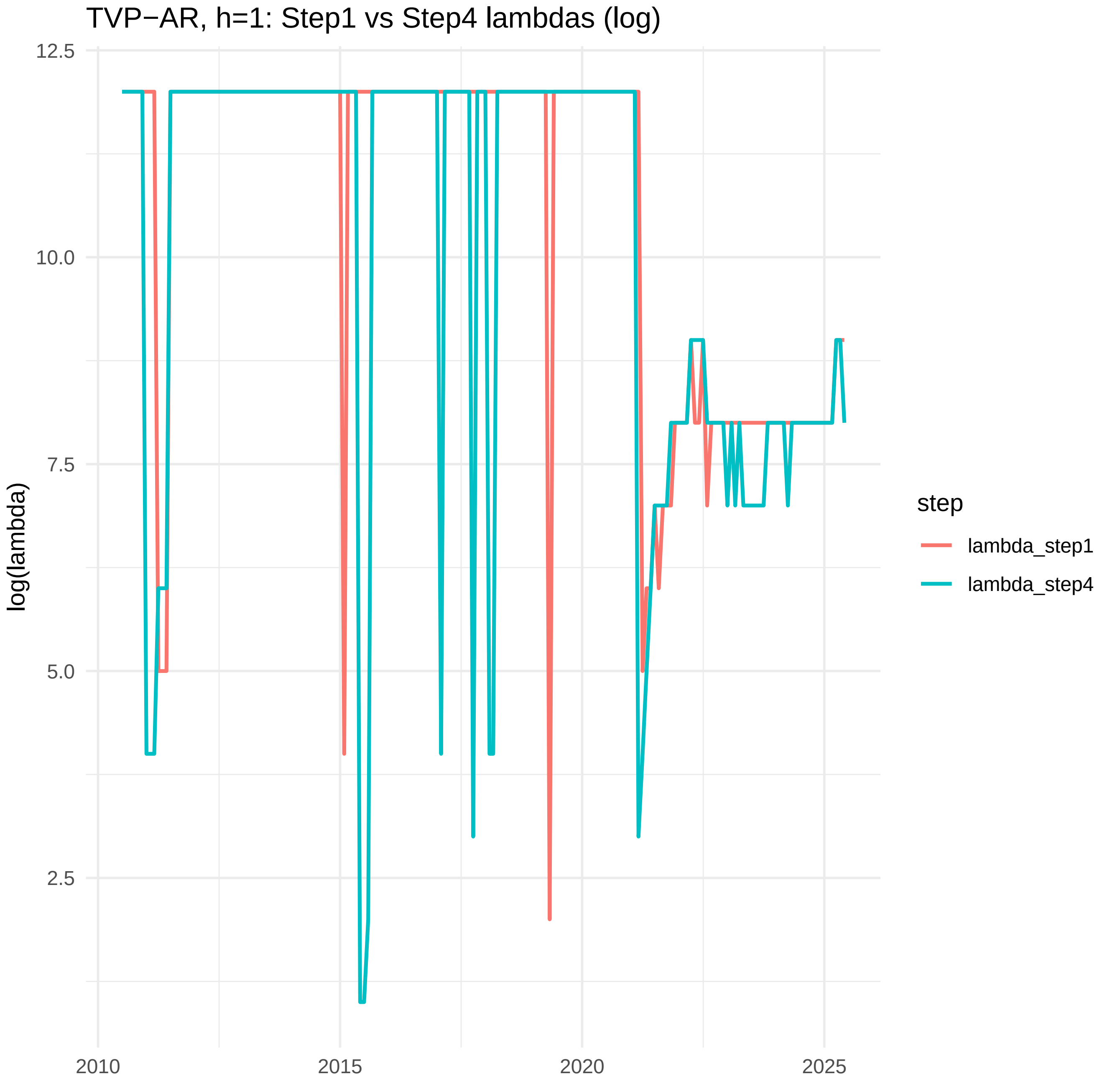

TVP−FAVAR, h=3: top−5 betas (by sd)
beta_T (last in−sample beta)
0.6
0.3
0.0
−0.3
2010
2015
2020
2025
var
F1_L3
F2_L3
intercept
Yh_L3
Yh_L4


TVP−FAVAR, h=6: top−5 betas (by sd)
0.50
0.25
0.00
−0.25
−0.50
beta_T (last in−sample beta)
2010
2015
2020
2025
var
F1_L6
F2_L6
intercept
Yh_L6
Yh_L7

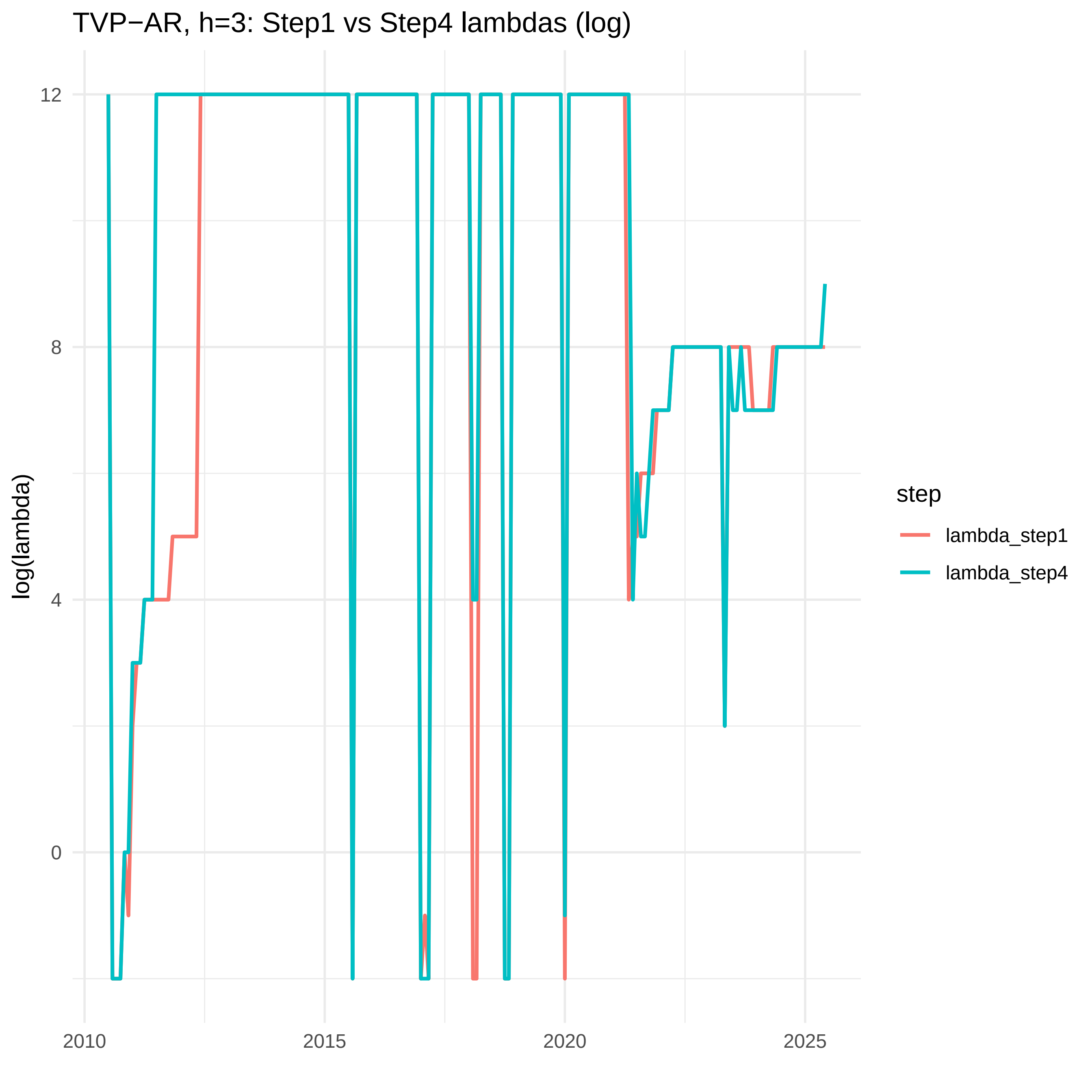

TVP−FAVAR, h=12: top−5 betas (by sd)
beta_T (last in−sample beta)
0.25
0.00
−0.25
−0.50
2010
2015
2020
2025
var
F1_L12
F2_L12
intercept
Yh_L12
Yh_L13

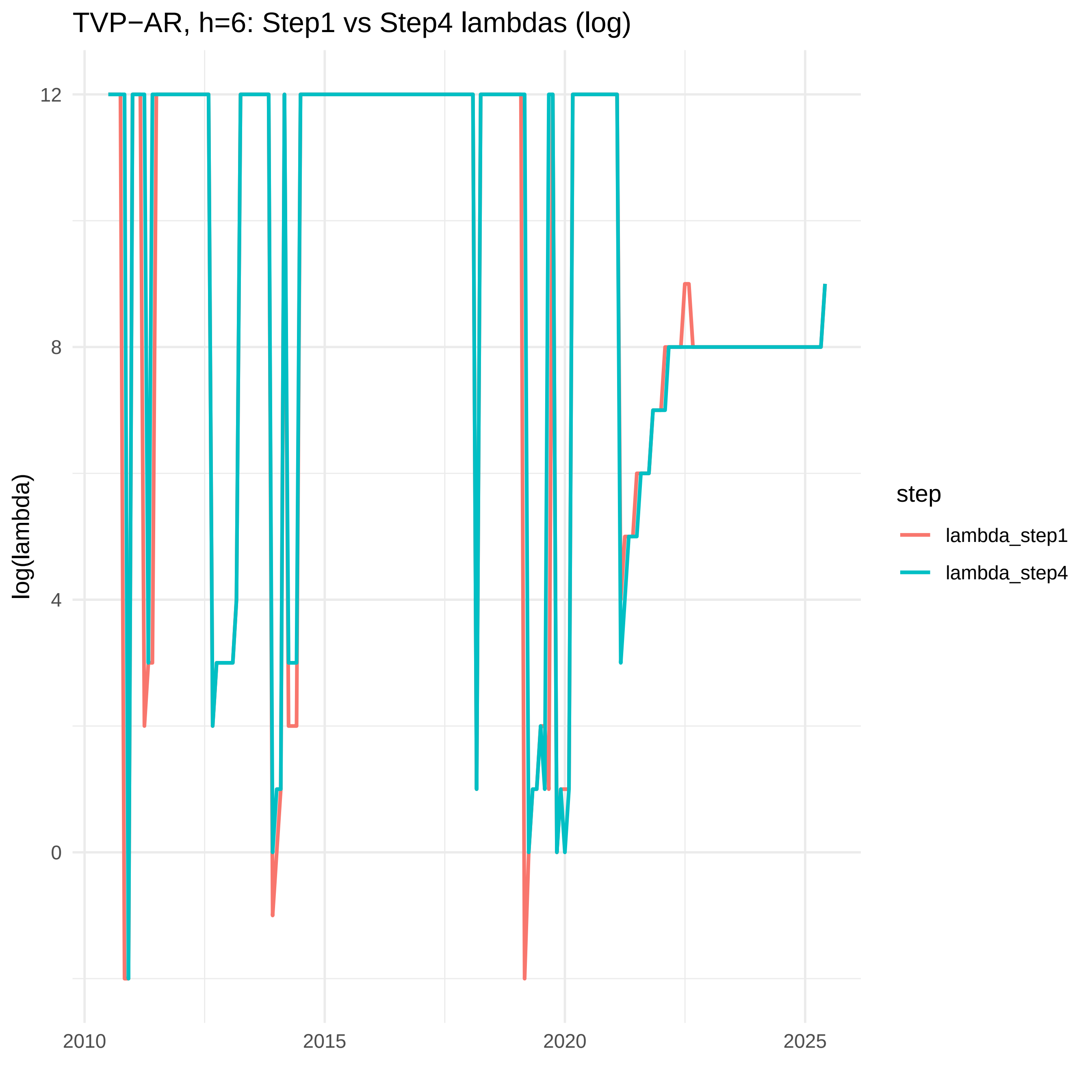
TVP−FAVAR: top−4 betas per horizon



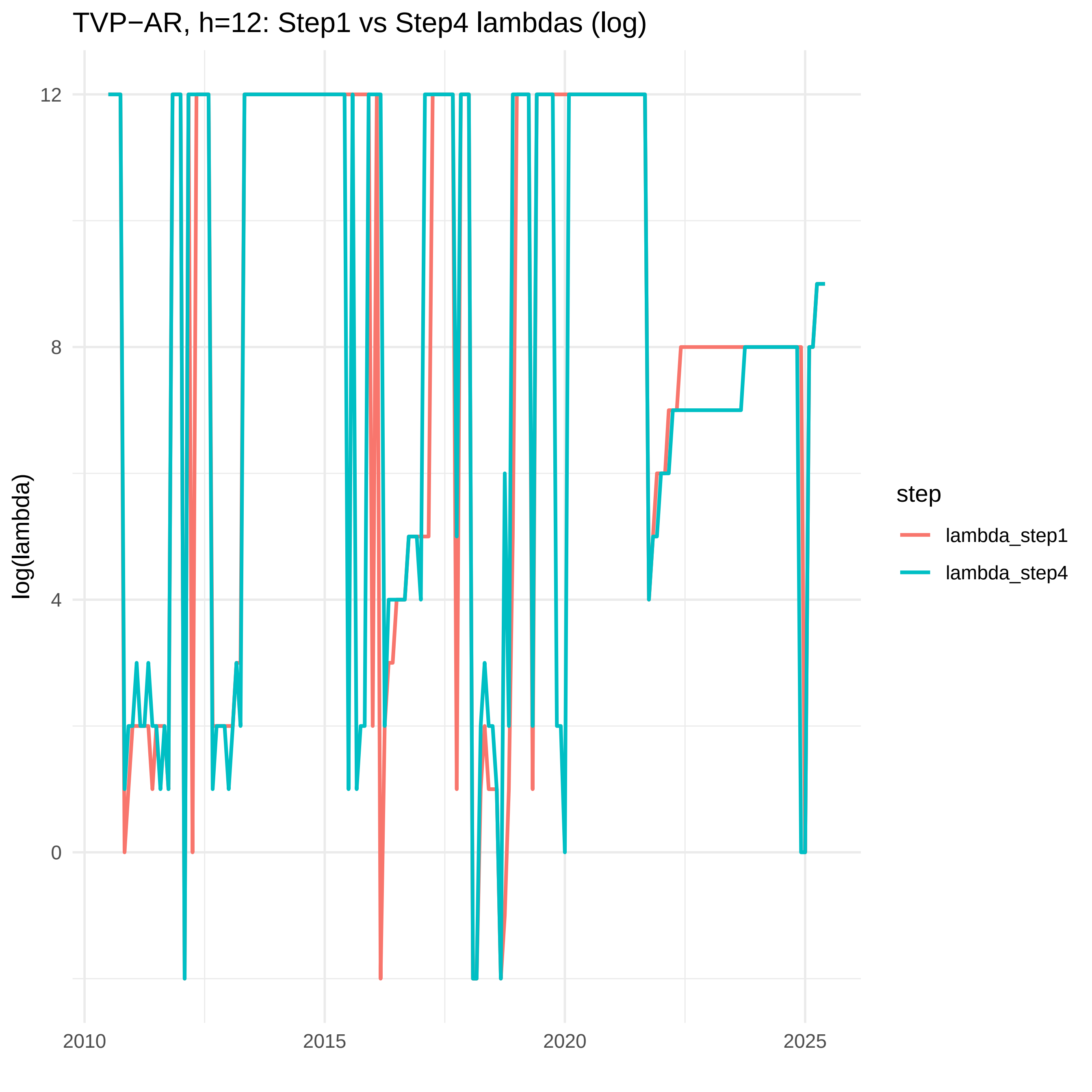


TVP−AR, h=1: Step1 vs Step4 lambdas (log)
12.5
10.0
7.5
5.0
2.5
log(lambda)
2010
2015
2020
2025
step
lambda_step1
lambda_step4

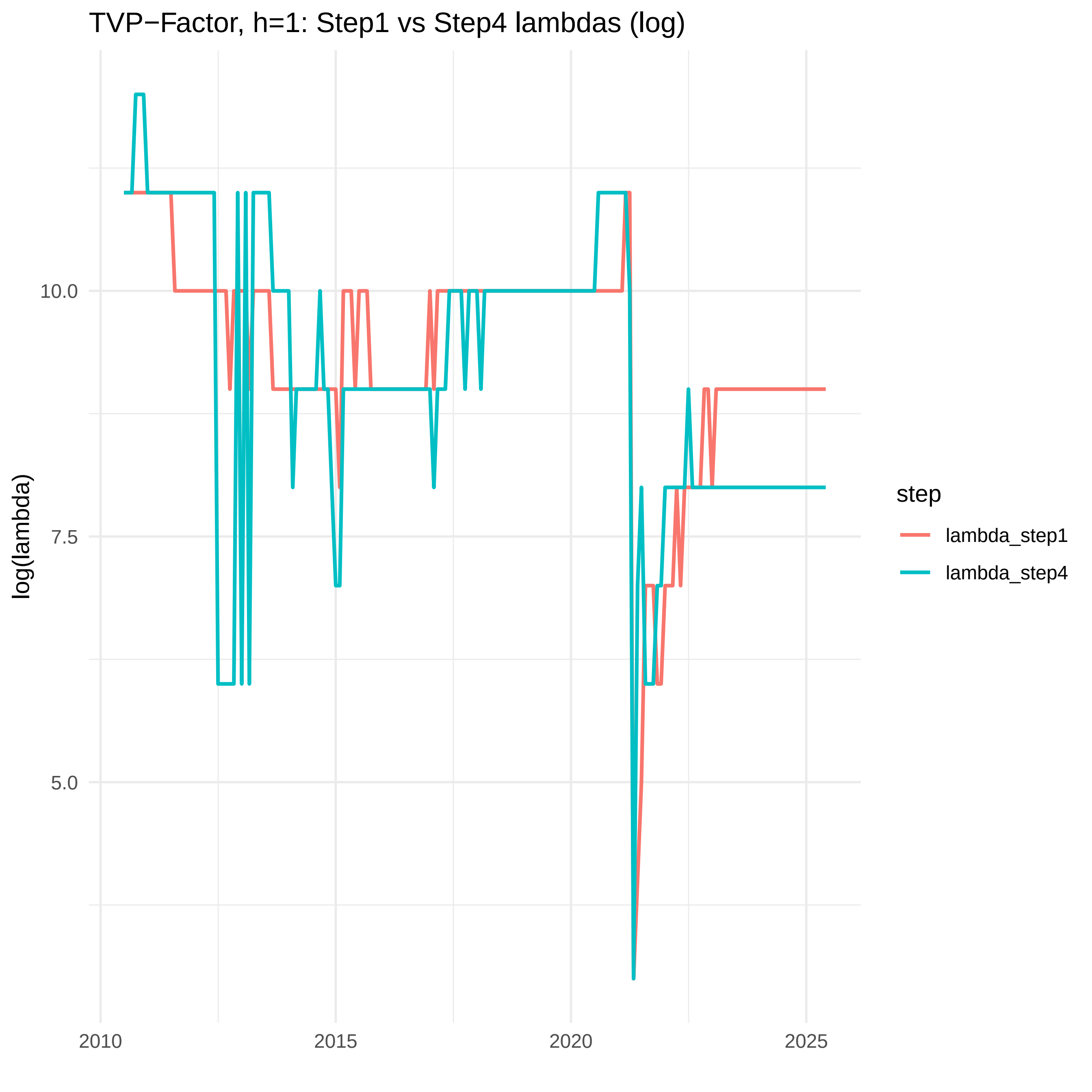

TVP−AR, h=3: Step1 vs Step4 lambdas (log)
log(lambda)
12
8
4
0
2010
2015
2020
2025
step
lambda_step1
lambda_step4

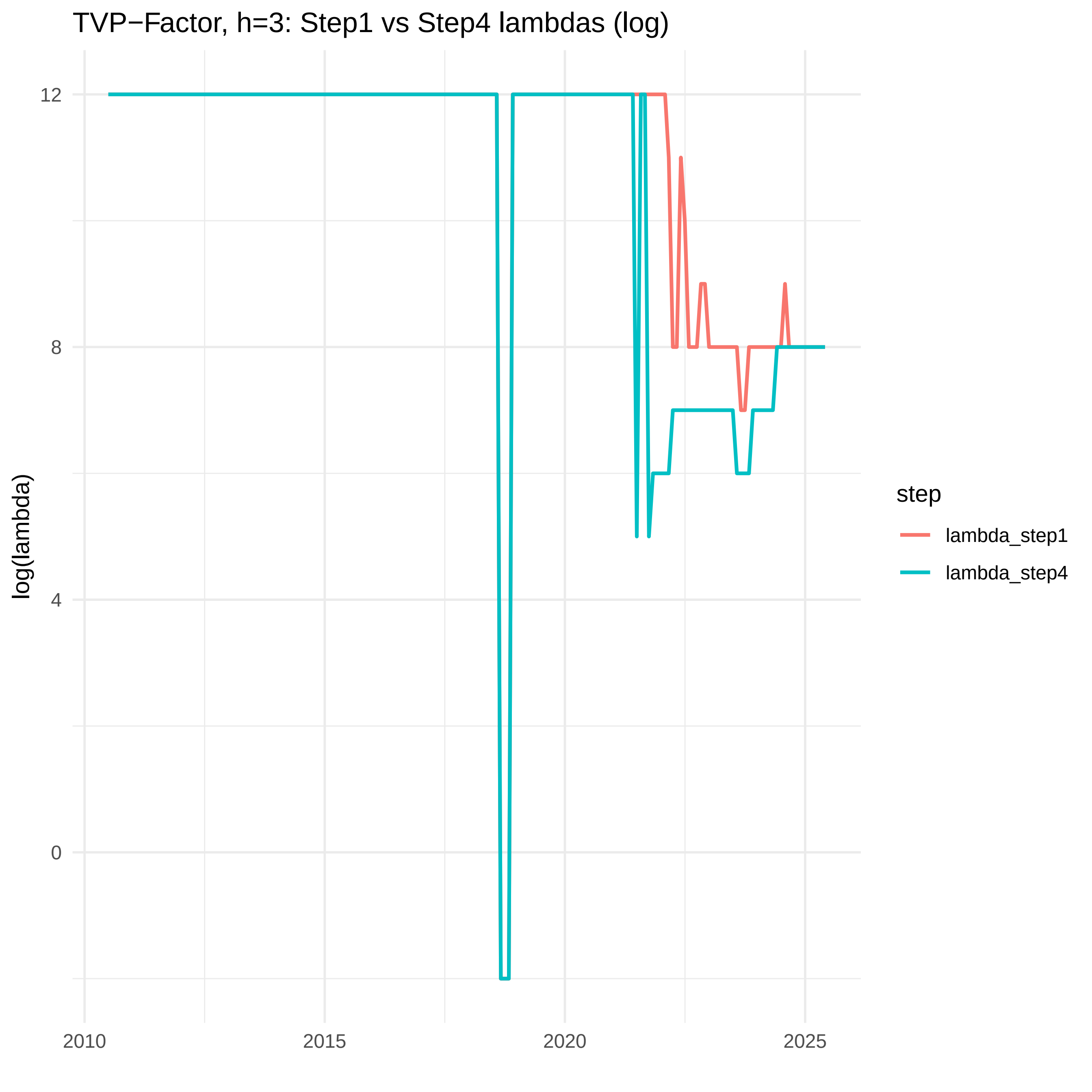


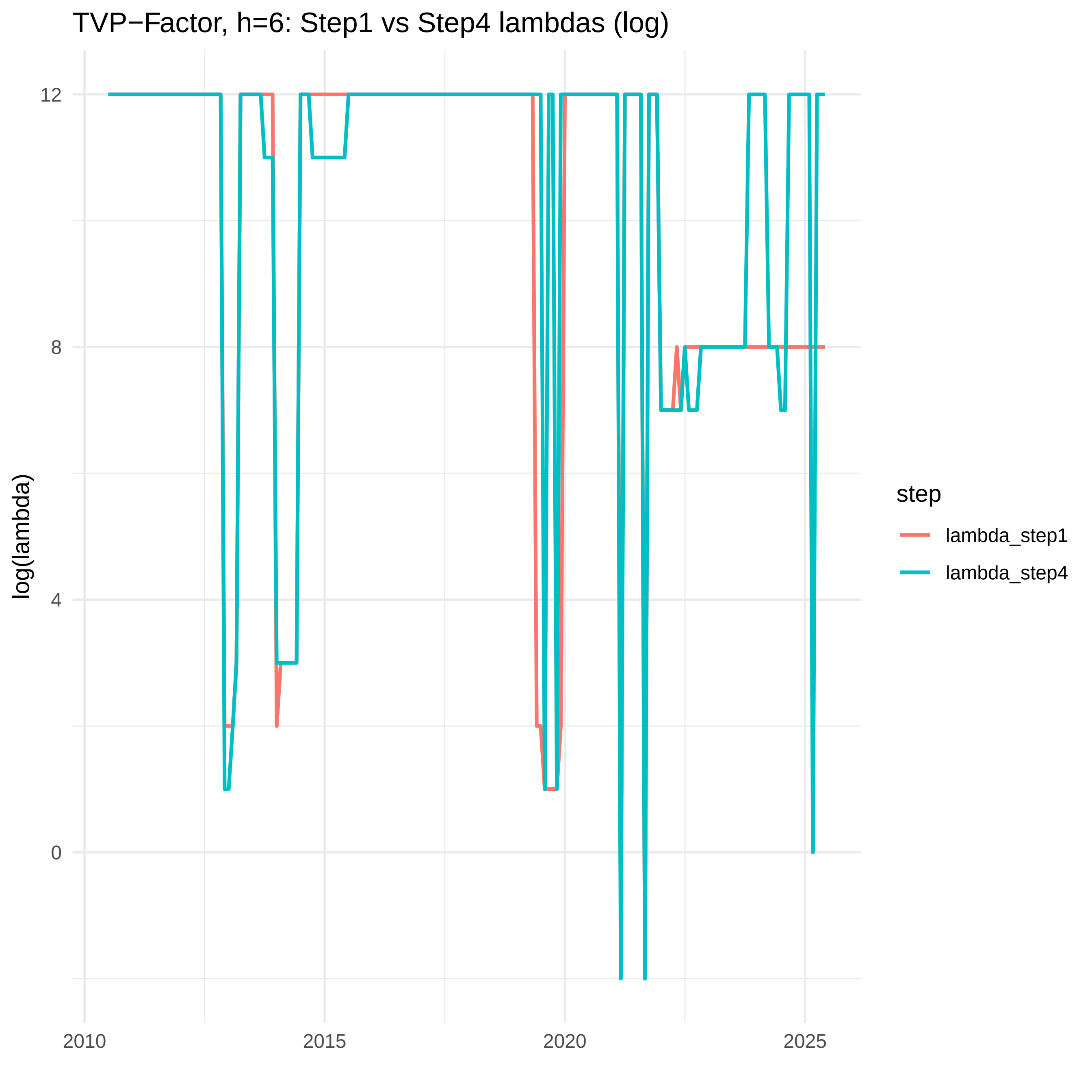
TVP−AR, h=6: Step1 vs Step4 lambdas (log)
log(lambda)
12
8
4
0
2010
2015
2020
2025
step
lambda_step1
lambda_step4



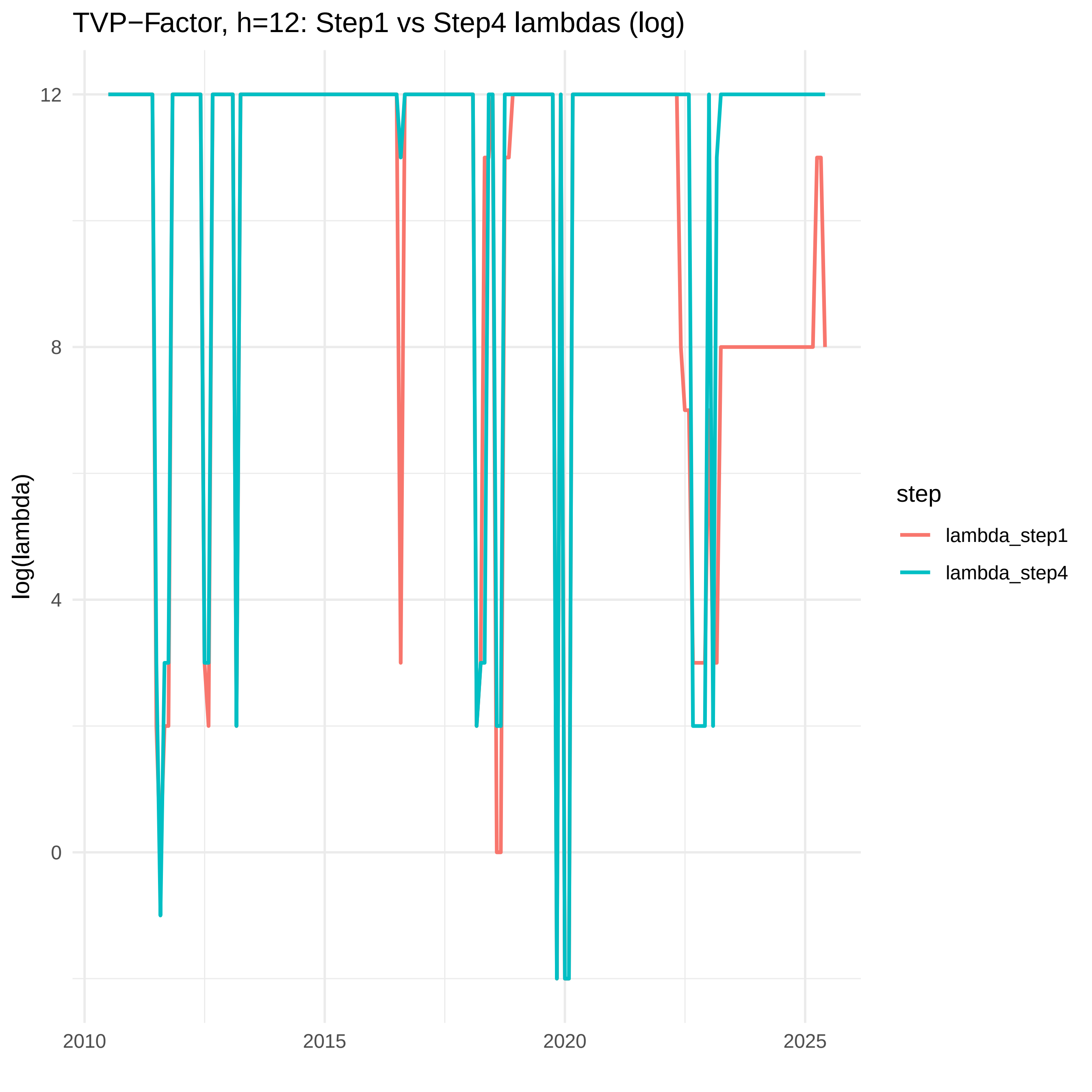
TVP−AR, h=12: Step1 vs Step4 lambdas (log)
log(lambda)
12
8
4
0
2010
2015
2020
2025
step
lambda_step1
lambda_step4


TVP−Factor, h=1: Step1 vs Step4 lambdas (log)
log(lambda)
10.0
7.5
5.0
2010
2015
2020
2025
step
lambda_step1
lambda_step4

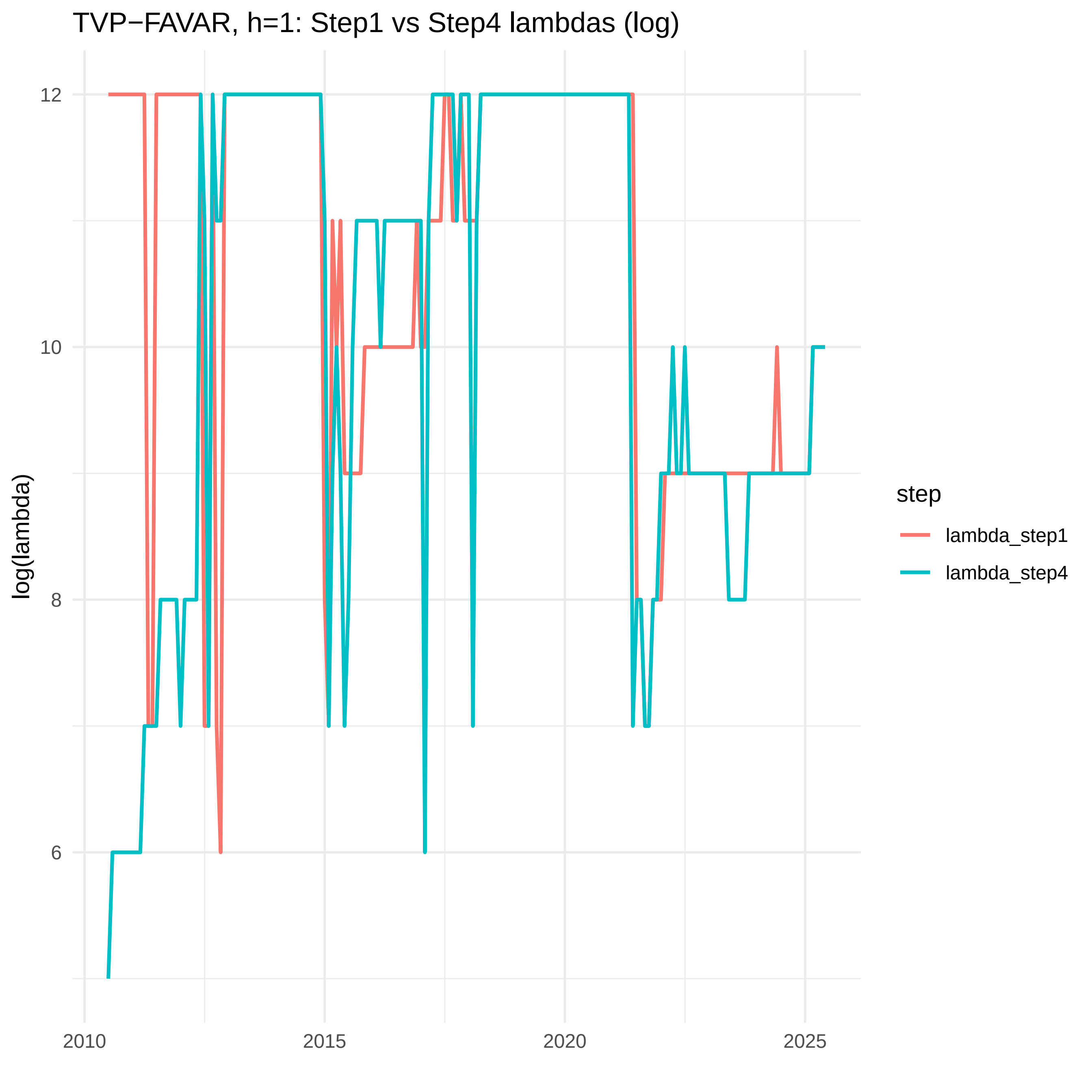

TVP−Factor, h=3: Step1 vs Step4 lambdas (log)
log(lambda)
12
8
4
0
2010
2015
2020
2025
step
lambda_step1
lambda_step4

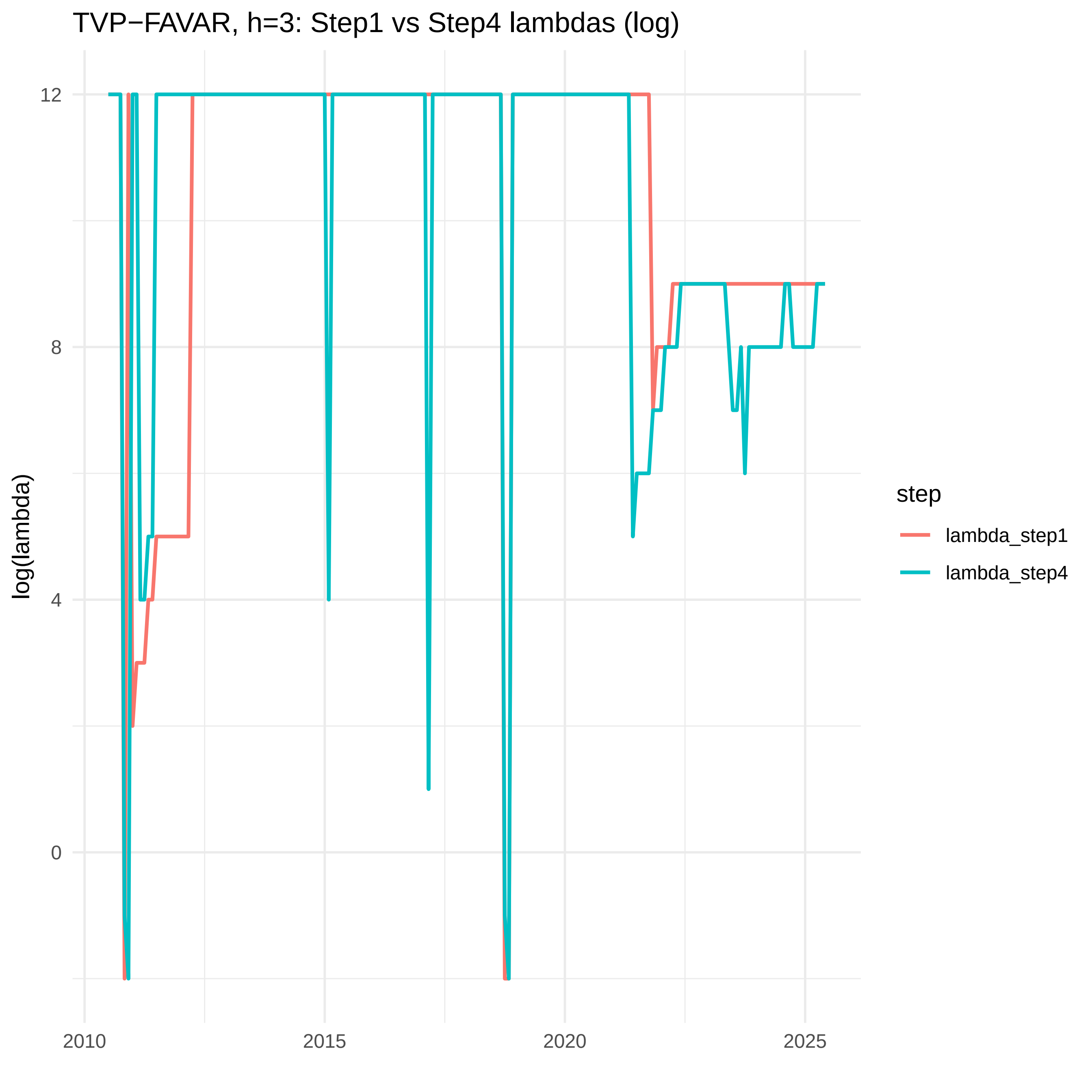


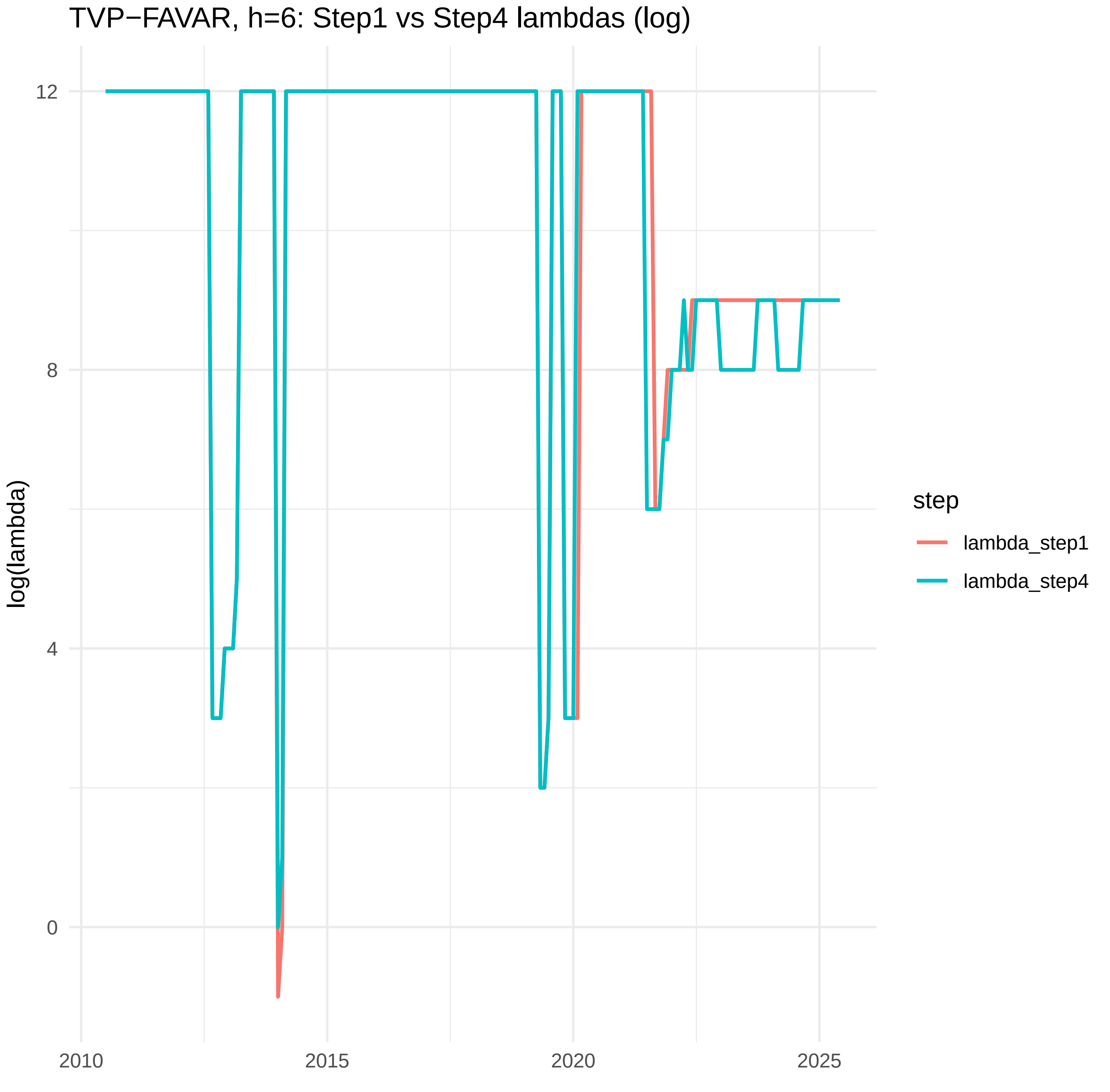
TVP−Factor, h=6: Step1 vs Step4 lambdas (log)
12
8
4
0
log(lambda)
2010
2015
2020
2025
step
lambda_step1
lambda_step4



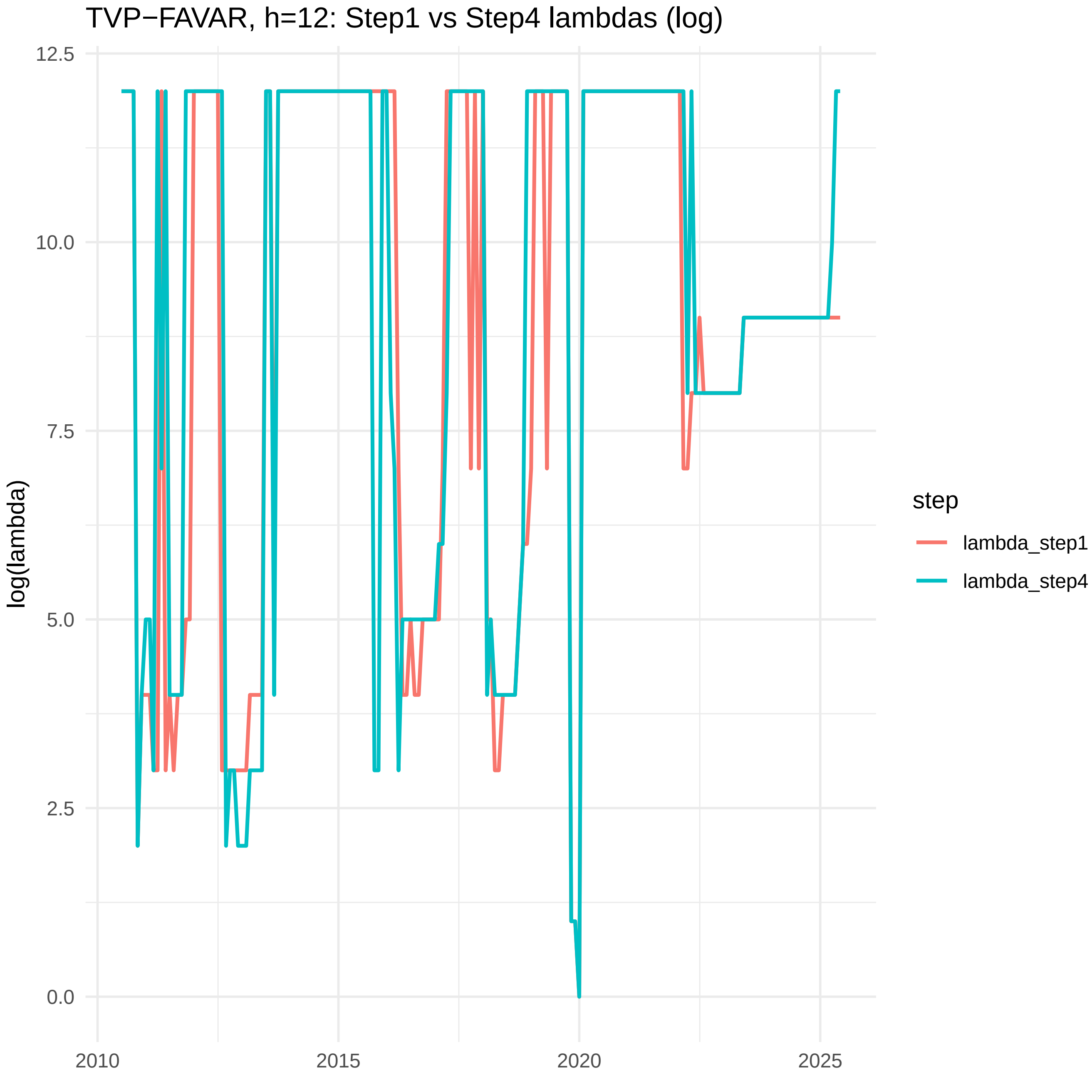
TVP−Factor, h=12: Step1 vs Step4 lambdas (log)
log(lambda)
12
8
4
0
2010
2015
2020
2025
step
lambda_step1
lambda_step4


TVP−FAVAR, h=1: Step1 vs Step4 lambdas (log)
log(lambda)
12
10
8
6
2010
2015
2020
2025
step
lambda_step1
lambda_step4

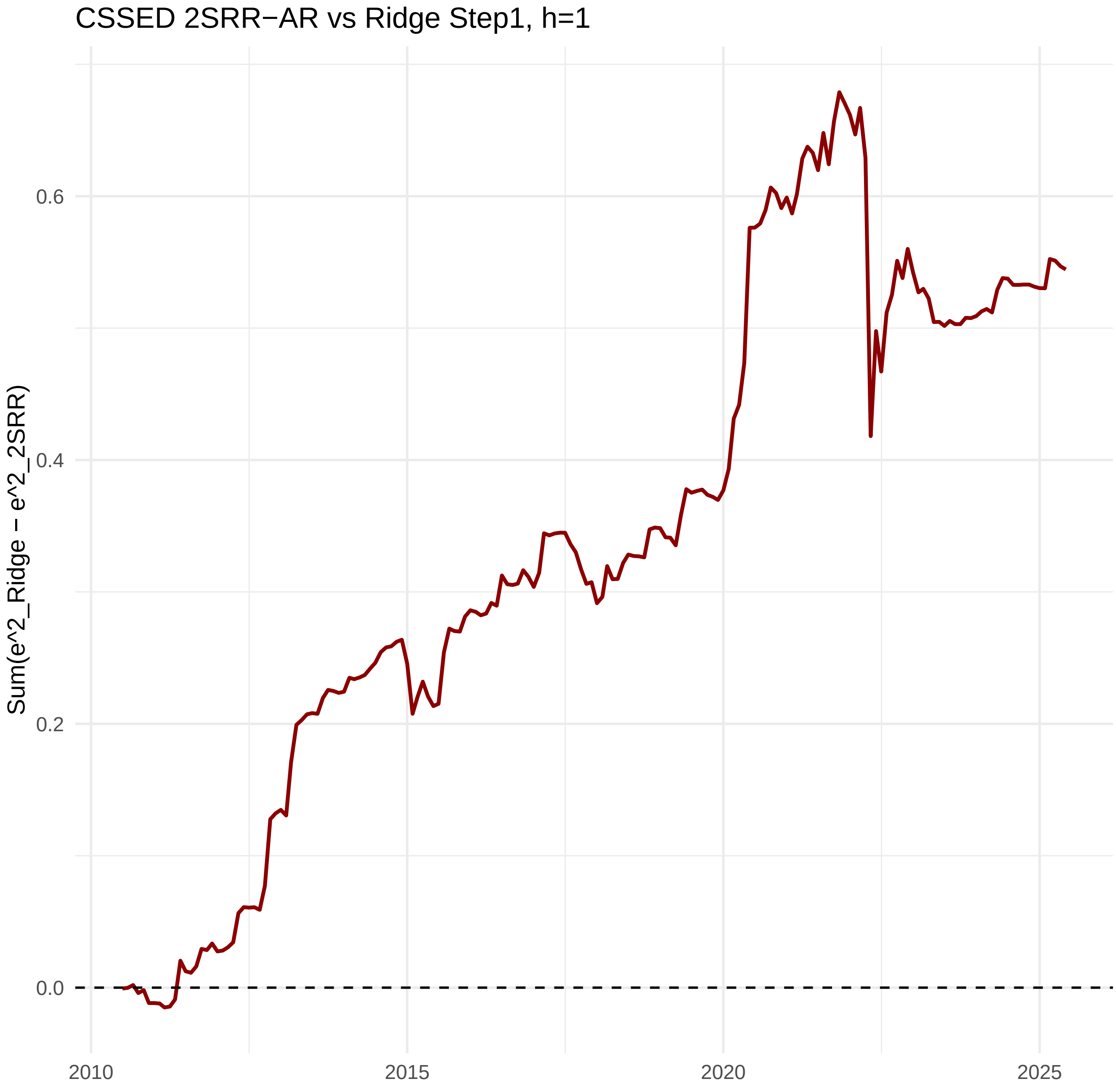

TVP−FAVAR, h=3: Step1 vs Step4 lambdas (log)
log(lambda)
12
8
4
0
2010
2015
2020
2025
step
lambda_step1
lambda_step4

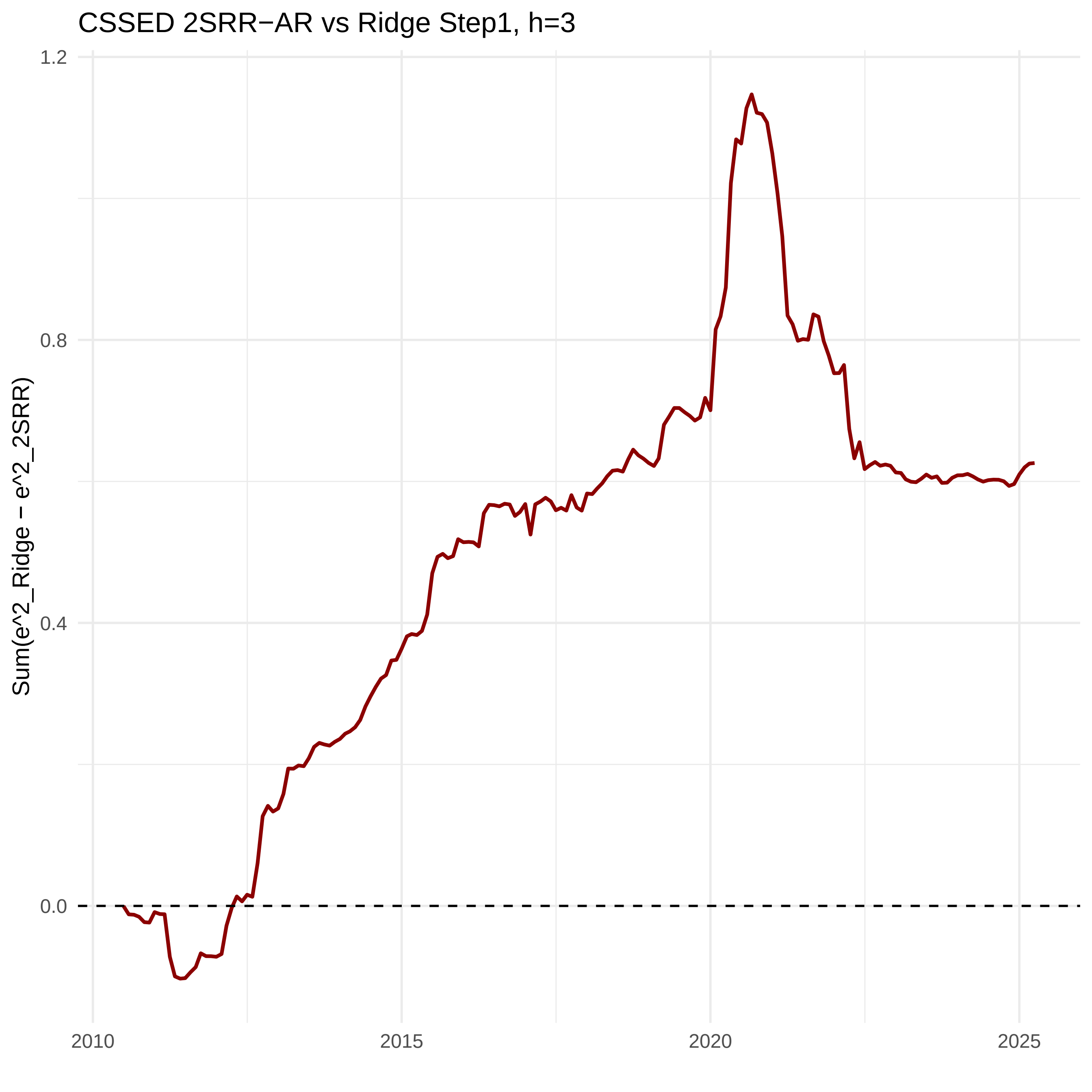


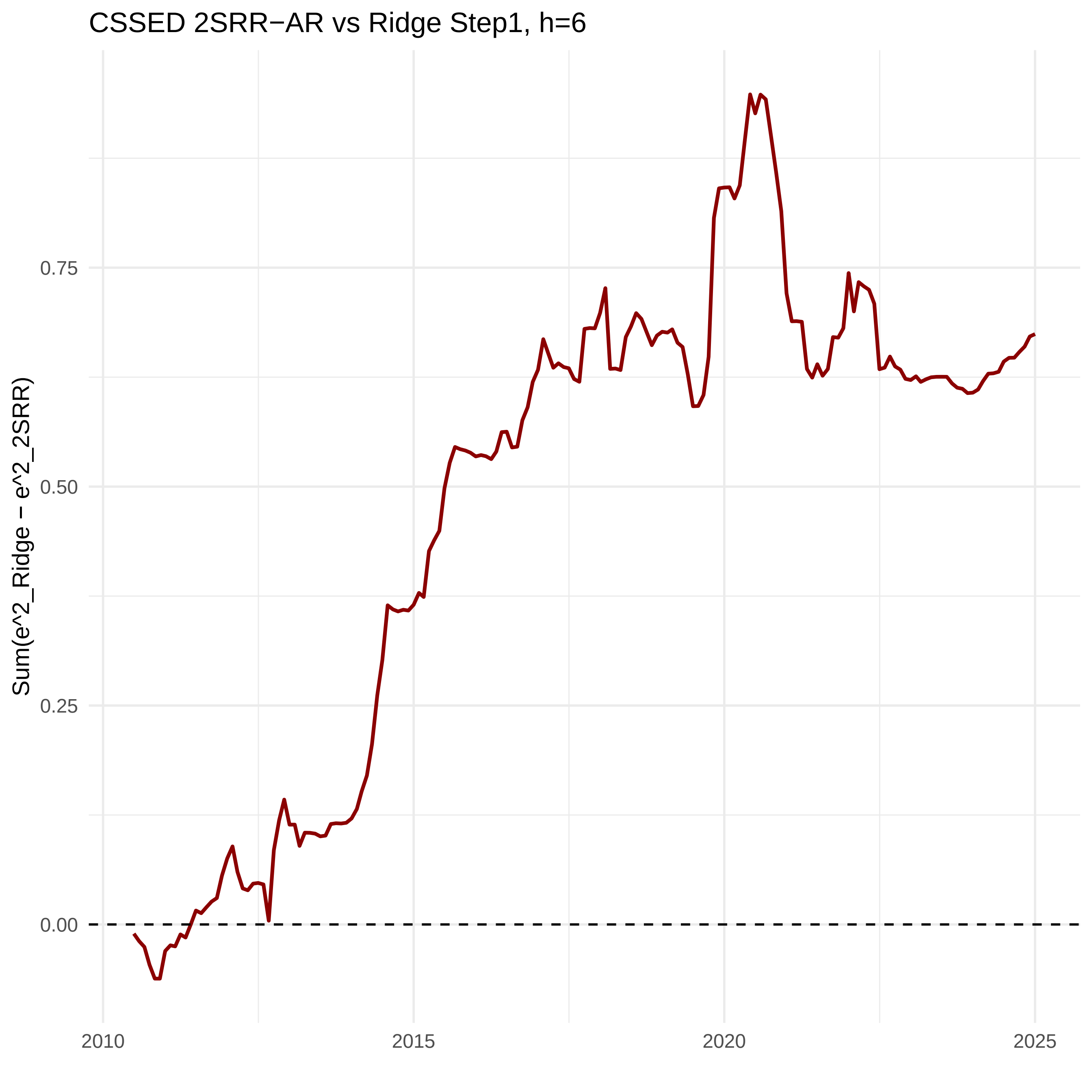
TVP−FAVAR, h=6: Step1 vs Step4 lambdas (log)
log(lambda)
12
8
4
0
2010
2015
2020
2025
step
lambda_step1
lambda_step4

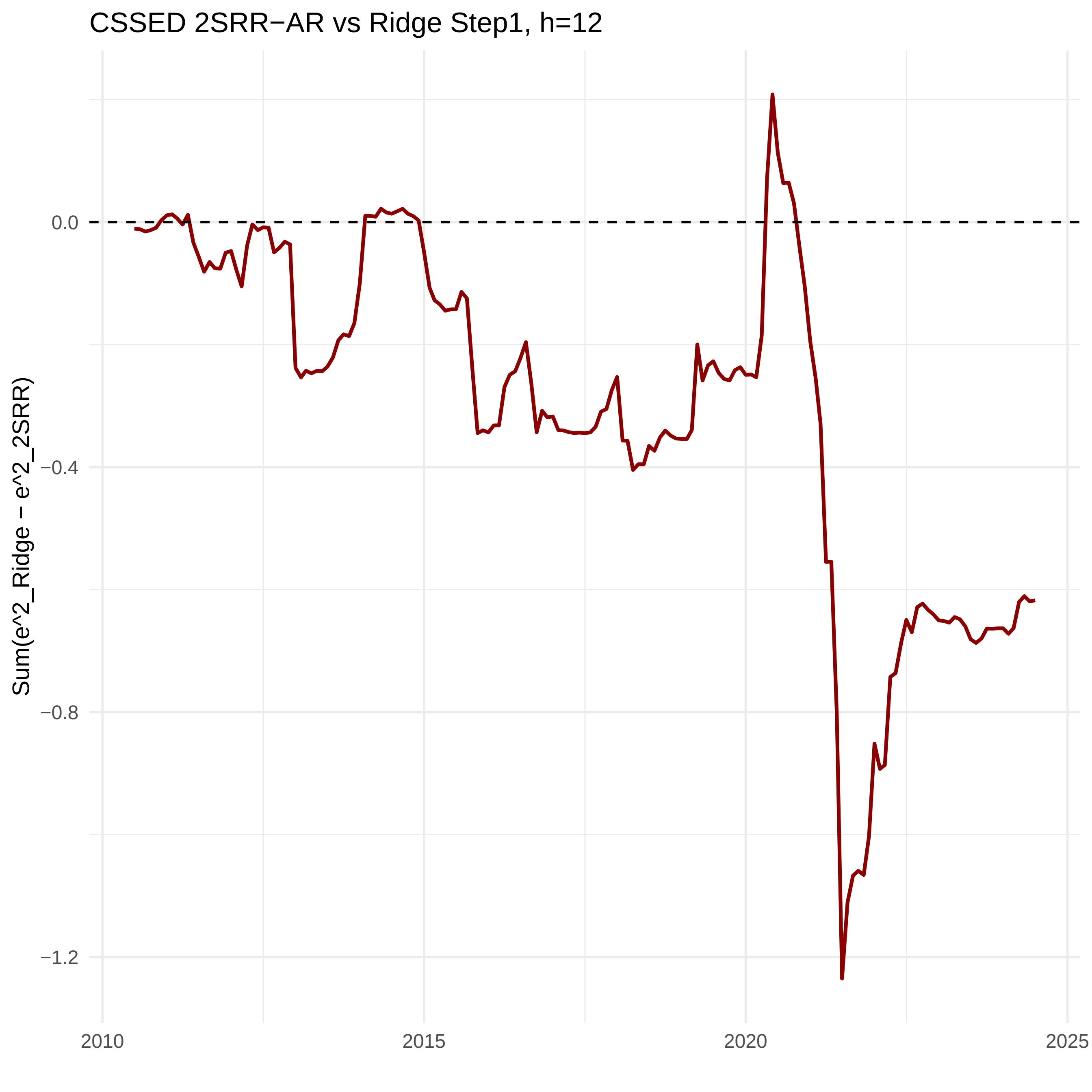


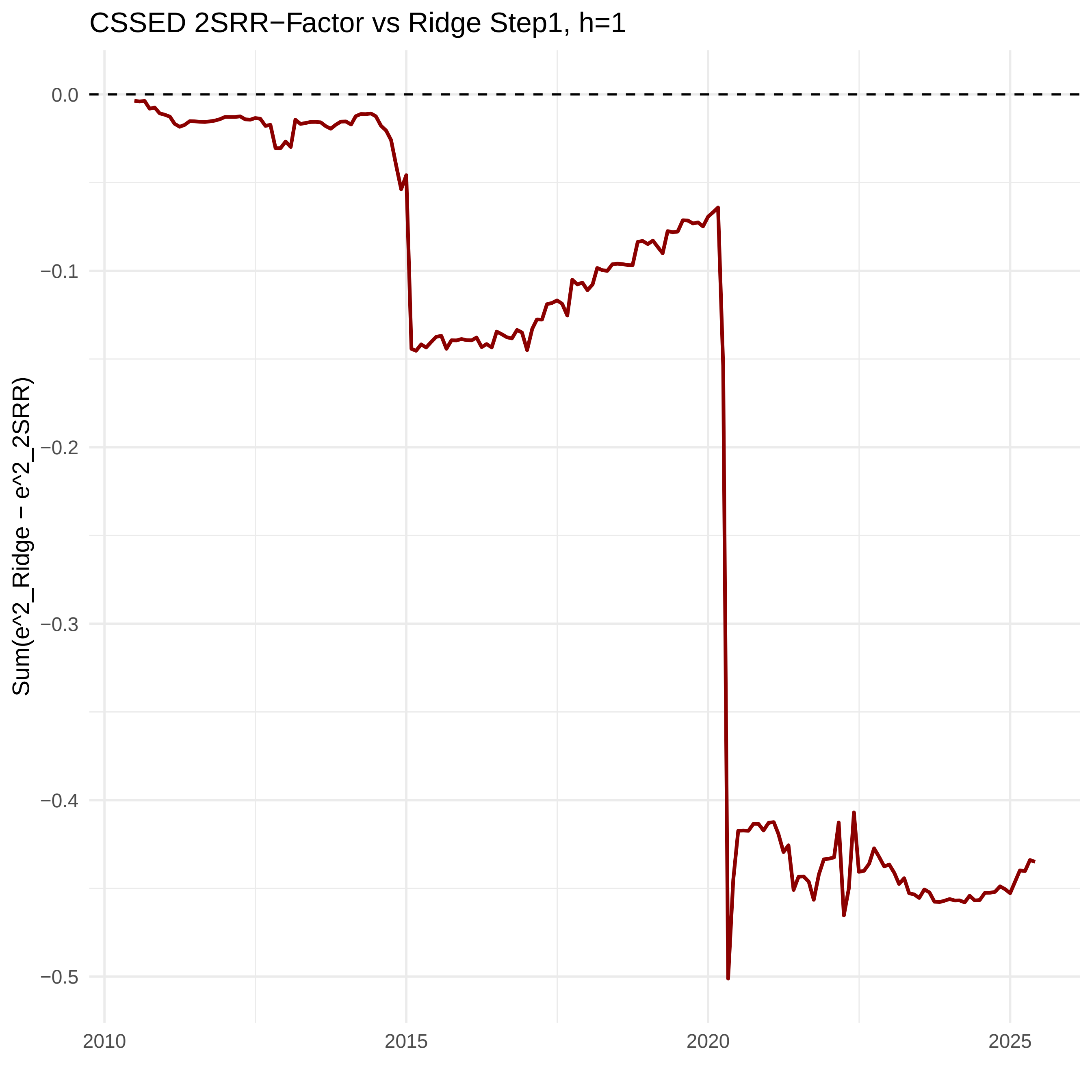
TVP−FAVAR, h=12: Step1 vs Step4 lambdas (log)
12.5
10.0
7.5
5.0
2.5
0.0
log(lambda)
2010
2015
2020
2025
step
lambda_step1
lambda_step4

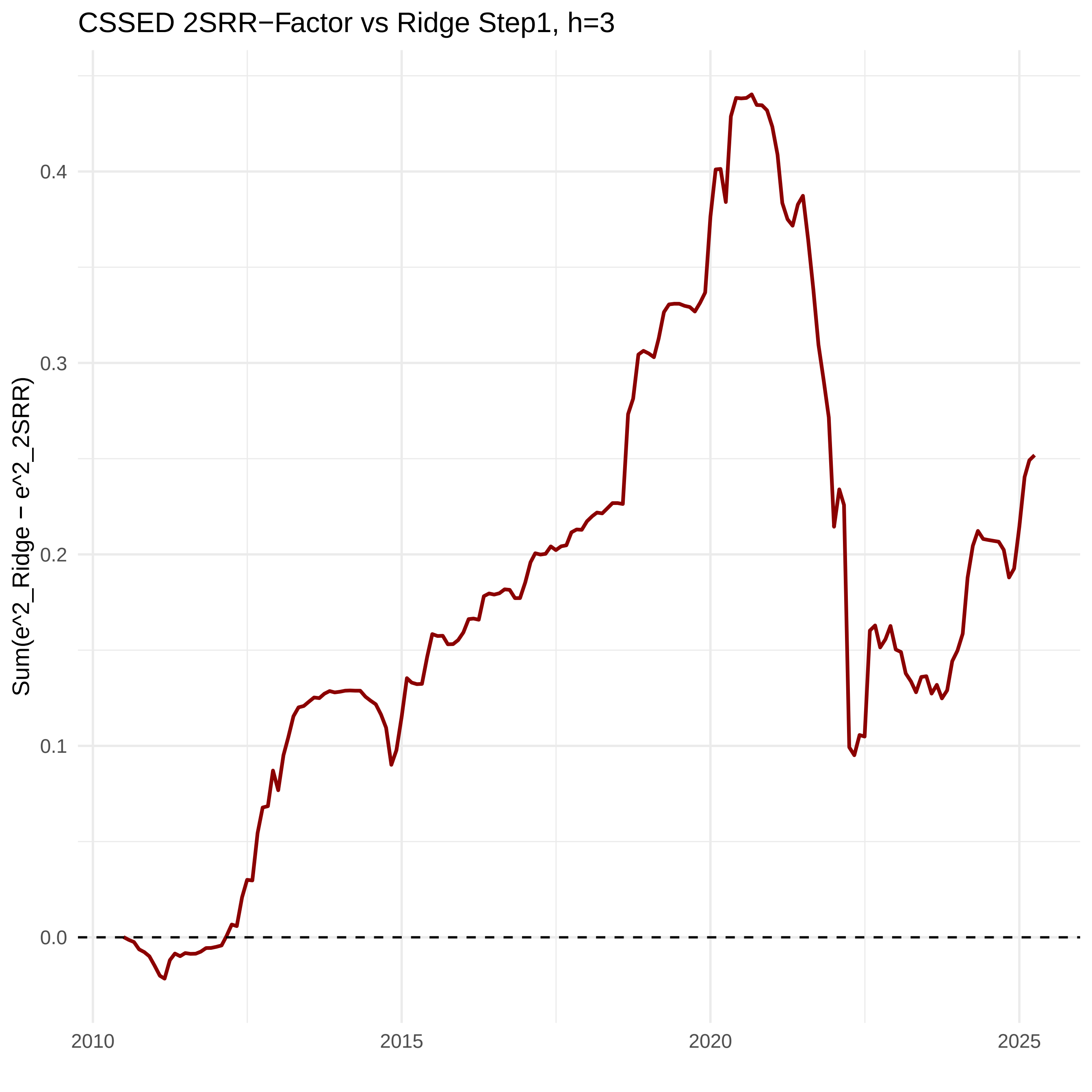

CSSED 2SRR−AR vs Ridge Step1, h=1
Sum(e^2_Ridge − e^2_2SRR)
0.6
0.4
0.2
0.0
2010
2015
2020
2025


CSSED 2SRR−AR vs Ridge Step1, h=3
Sum(e^2_Ridge − e^2_2SRR)
1.2
0.8
0.4
0.0
2010
2015
2020
2025

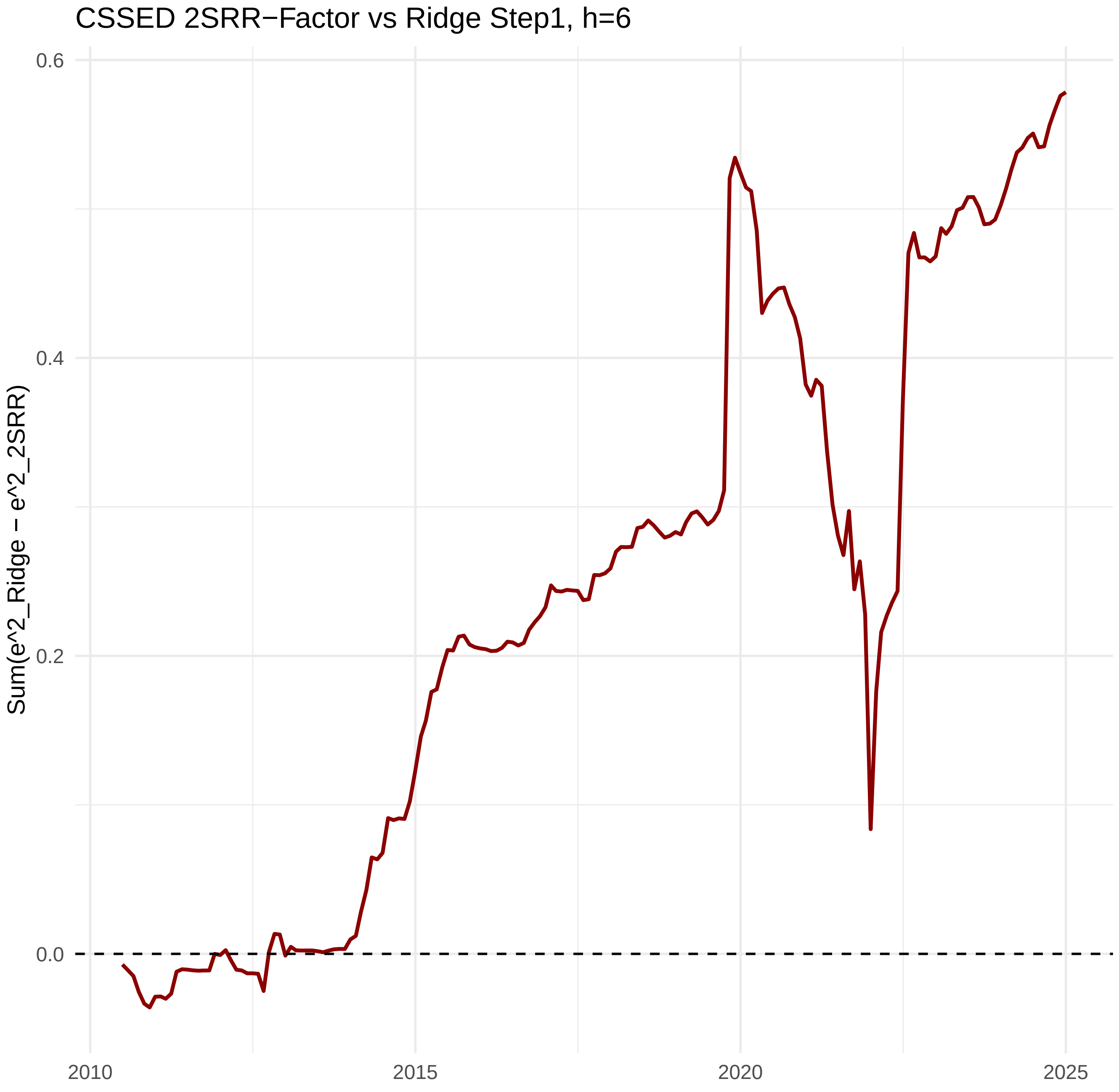

CSSED 2SRR−AR vs Ridge Step1, h=6
Sum(e^2_Ridge − e^2_2SRR)
0.75
0.50
0.25
0.00
2010
2015
2020
2025

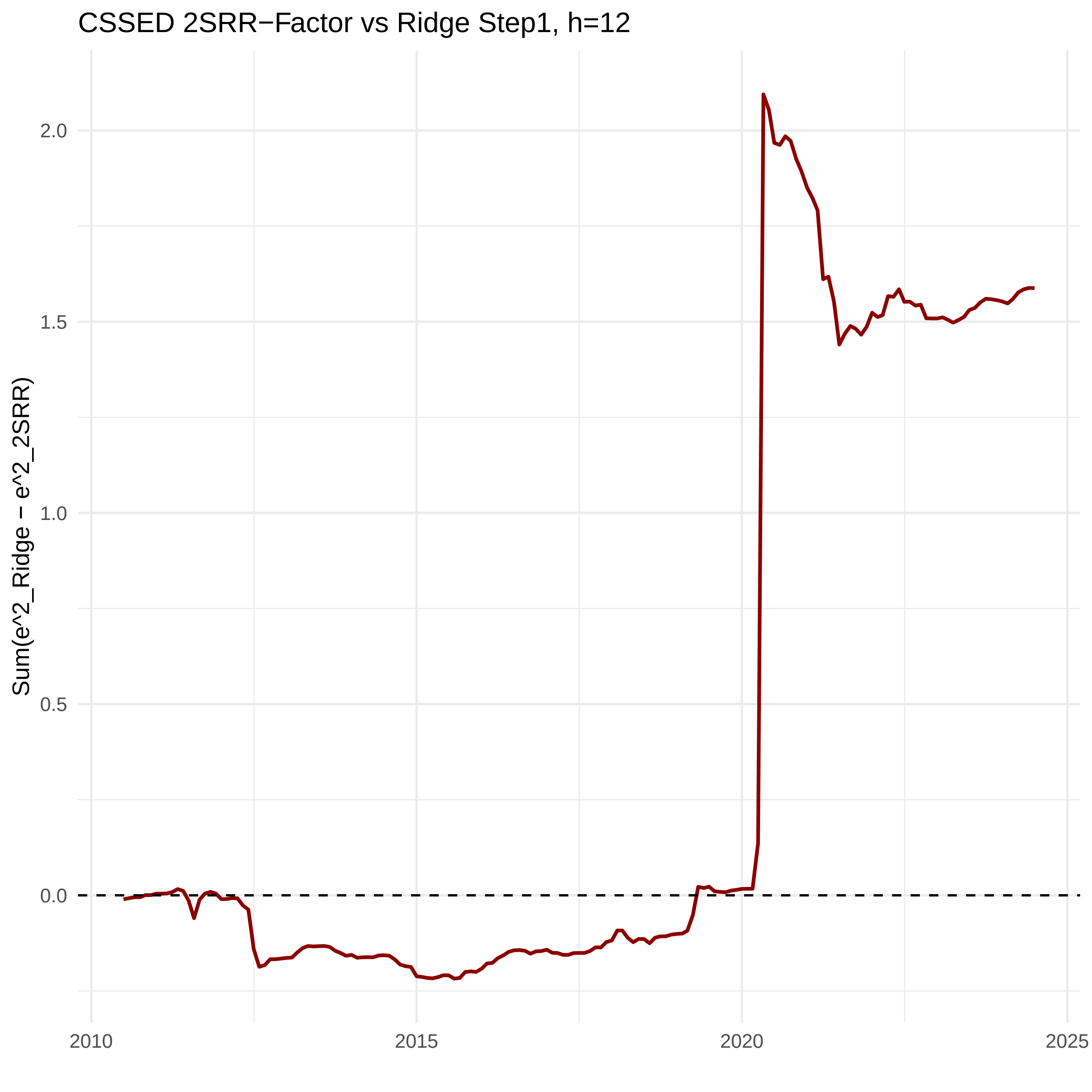

CSSED 2SRR−AR vs Ridge Step1, h=12
Sum(e^2_Ridge − e^2_2SRR)
0.0
−0.4
−0.8
−1.2
2010
2015
2020
2025

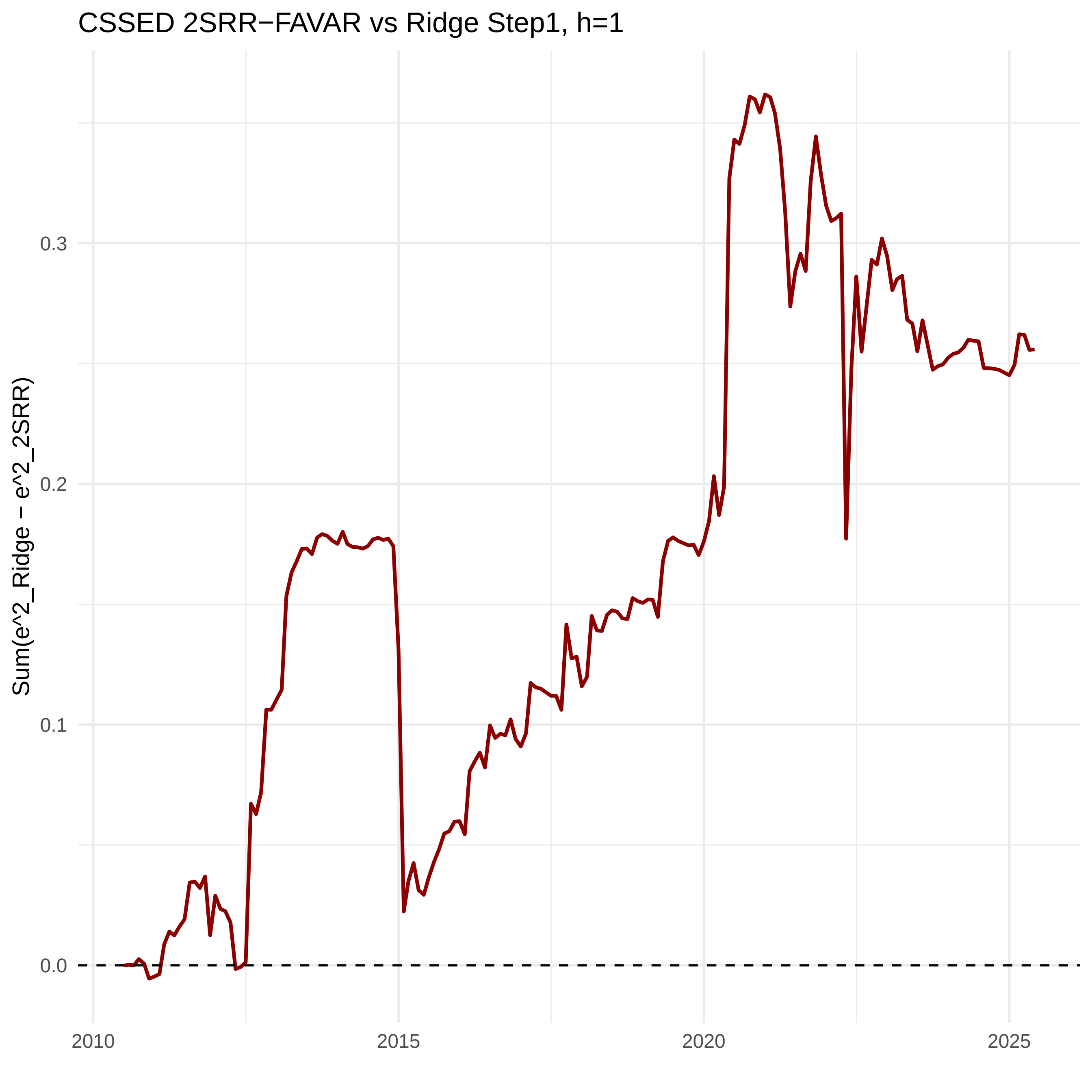

CSSED 2SRR−Factor vs Ridge Step1, h=1
Sum(e^2_Ridge − e^2_2SRR)
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
2010
2015
2020
2025



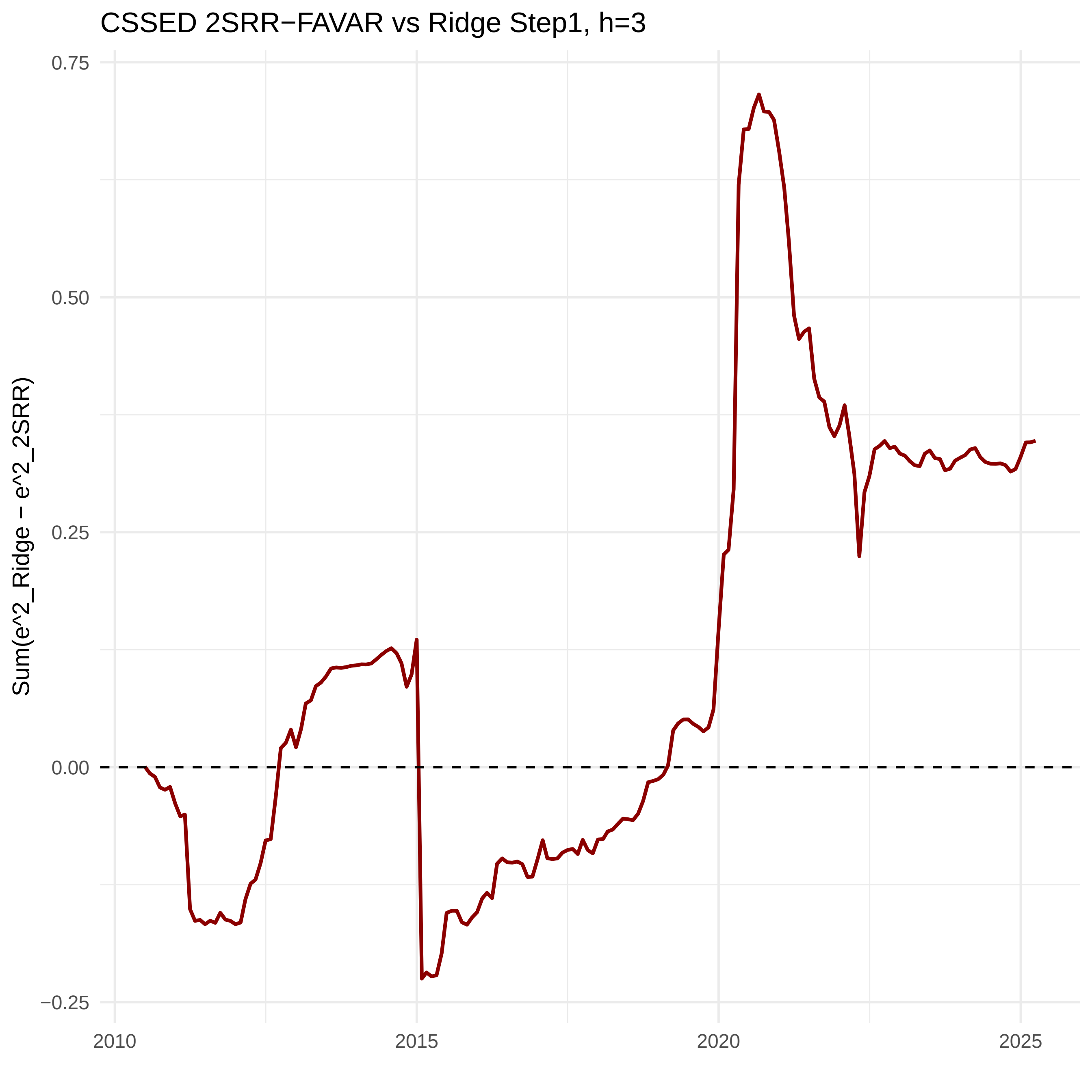
CSSED 2SRR−Factor vs Ridge Step1, h=3
Sum(e^2_Ridge − e^2_2SRR)
0.4
0.3
0.2
0.1
0.0
2010
2015
2020
2025


CSSED 2SRR−Factor vs Ridge Step1, h=6
Sum(e^2_Ridge − e^2_2SRR)
0.6
0.4
0.2
0.0
2010
2015
2020
2025

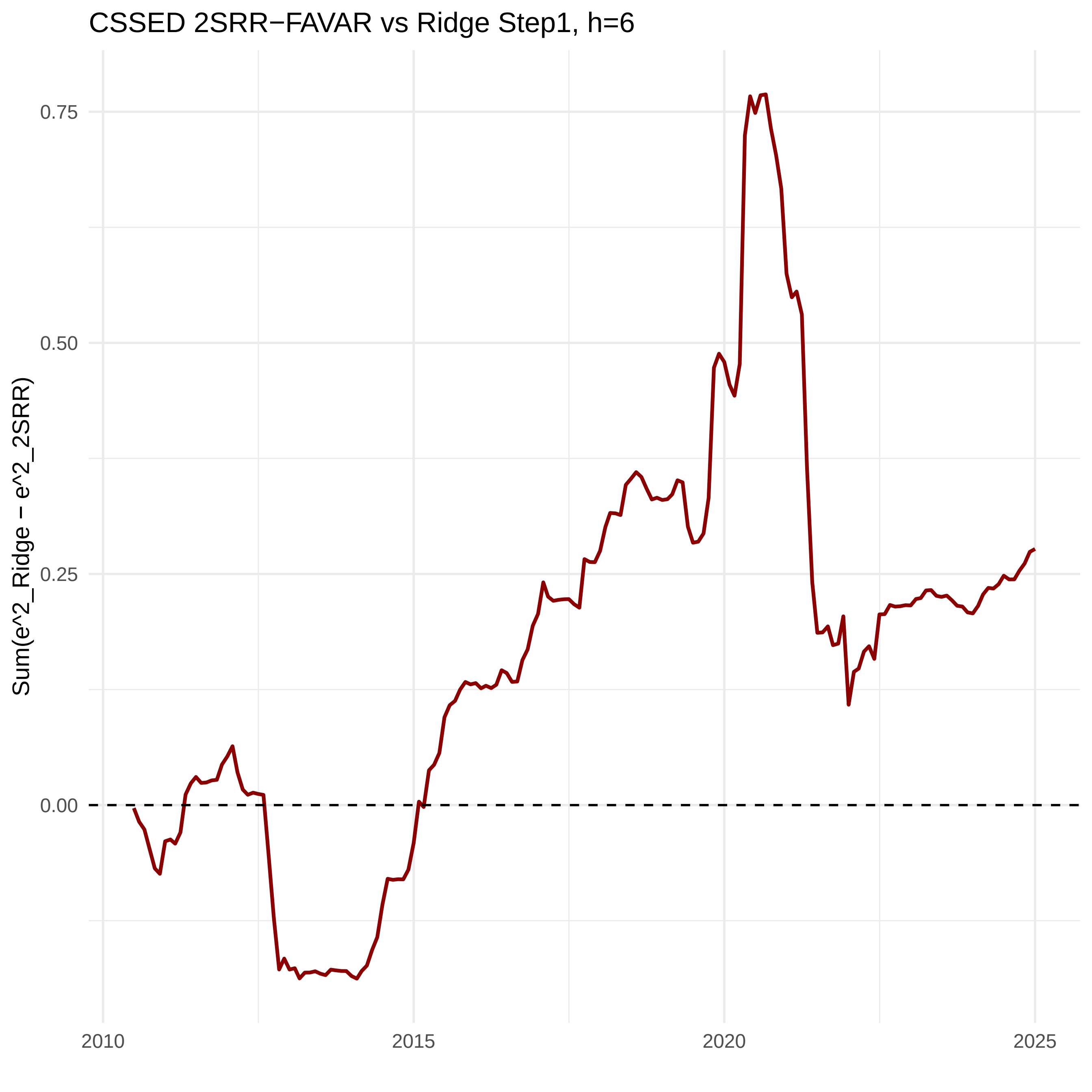

CSSED 2SRR−Factor vs Ridge Step1, h=12
Sum(e^2_Ridge − e^2_2SRR)
2.0
1.5
1.0
0.5
0.0
2010
2015
2020
2025

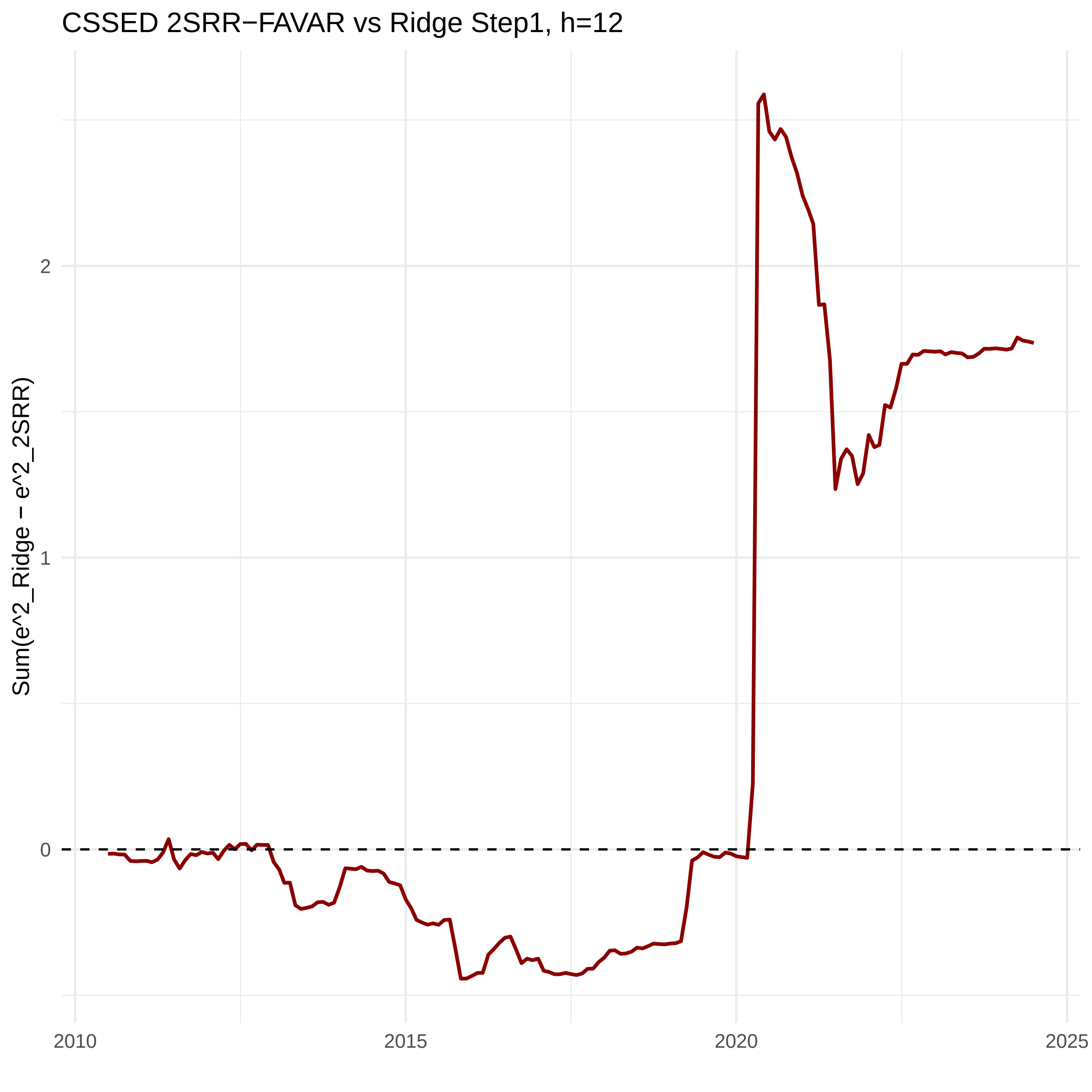


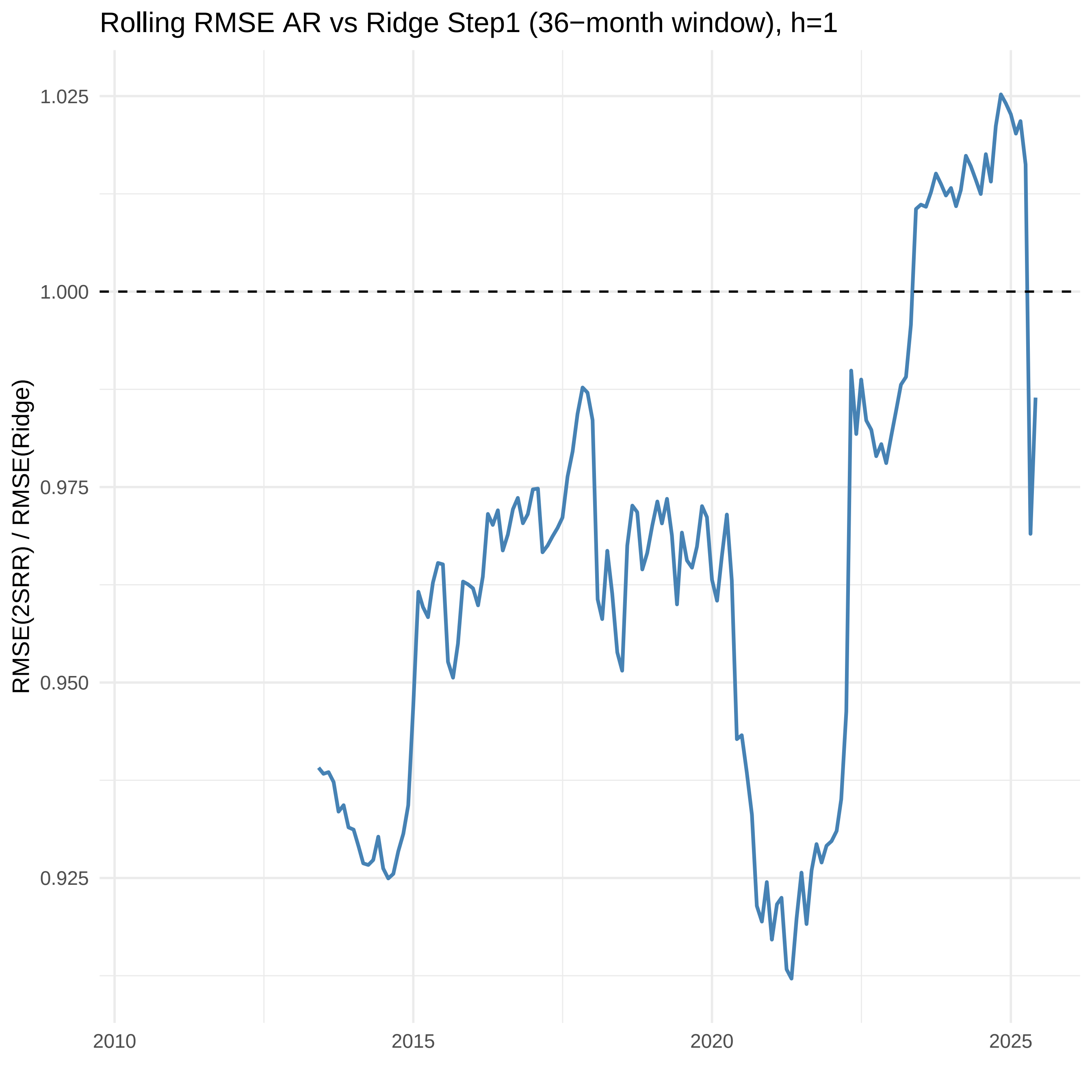
CSSED 2SRR−FAVAR vs Ridge Step1, h=1
Sum(e^2_Ridge − e^2_2SRR)
0.3
0.2
0.1
0.0
2010
2015
2020
2025

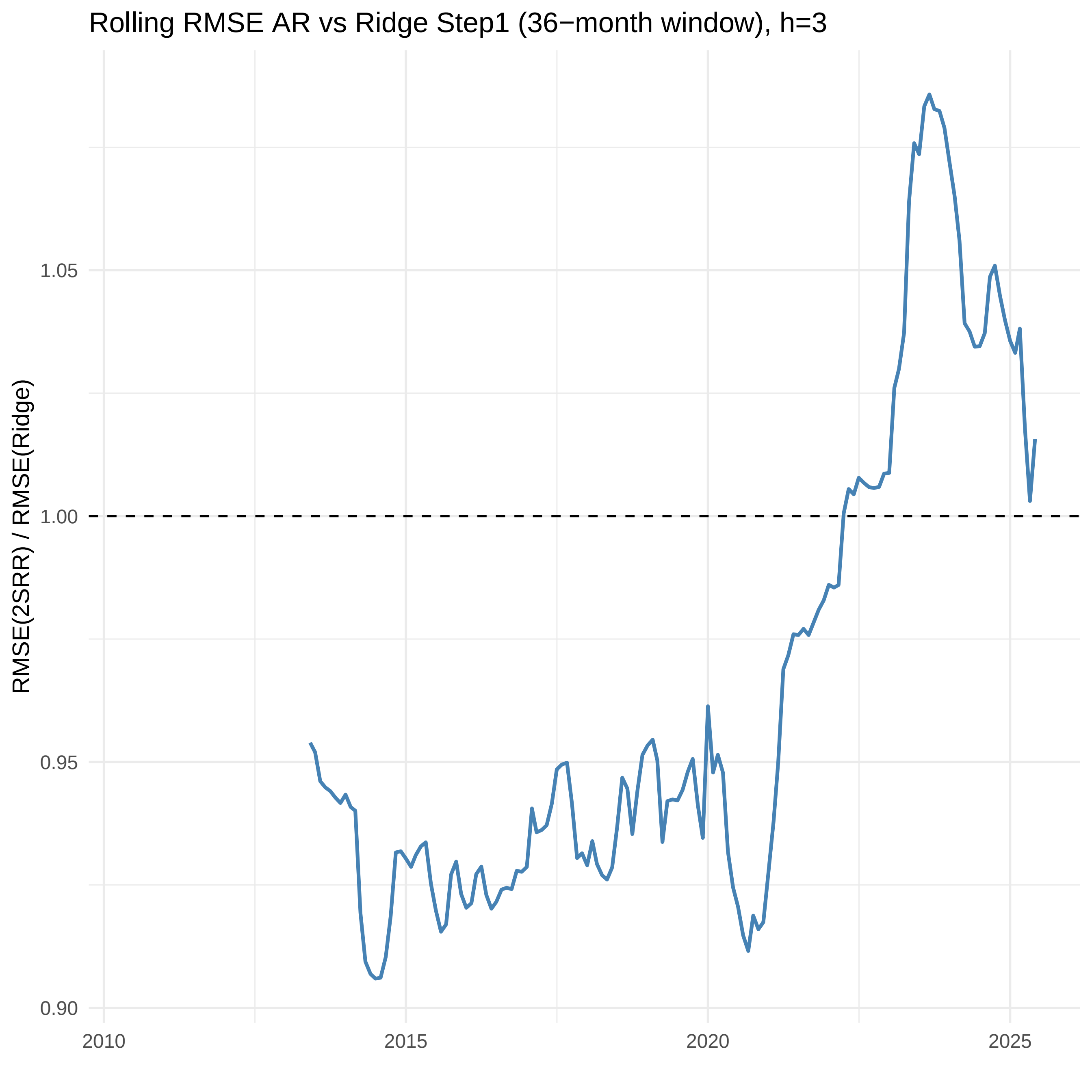


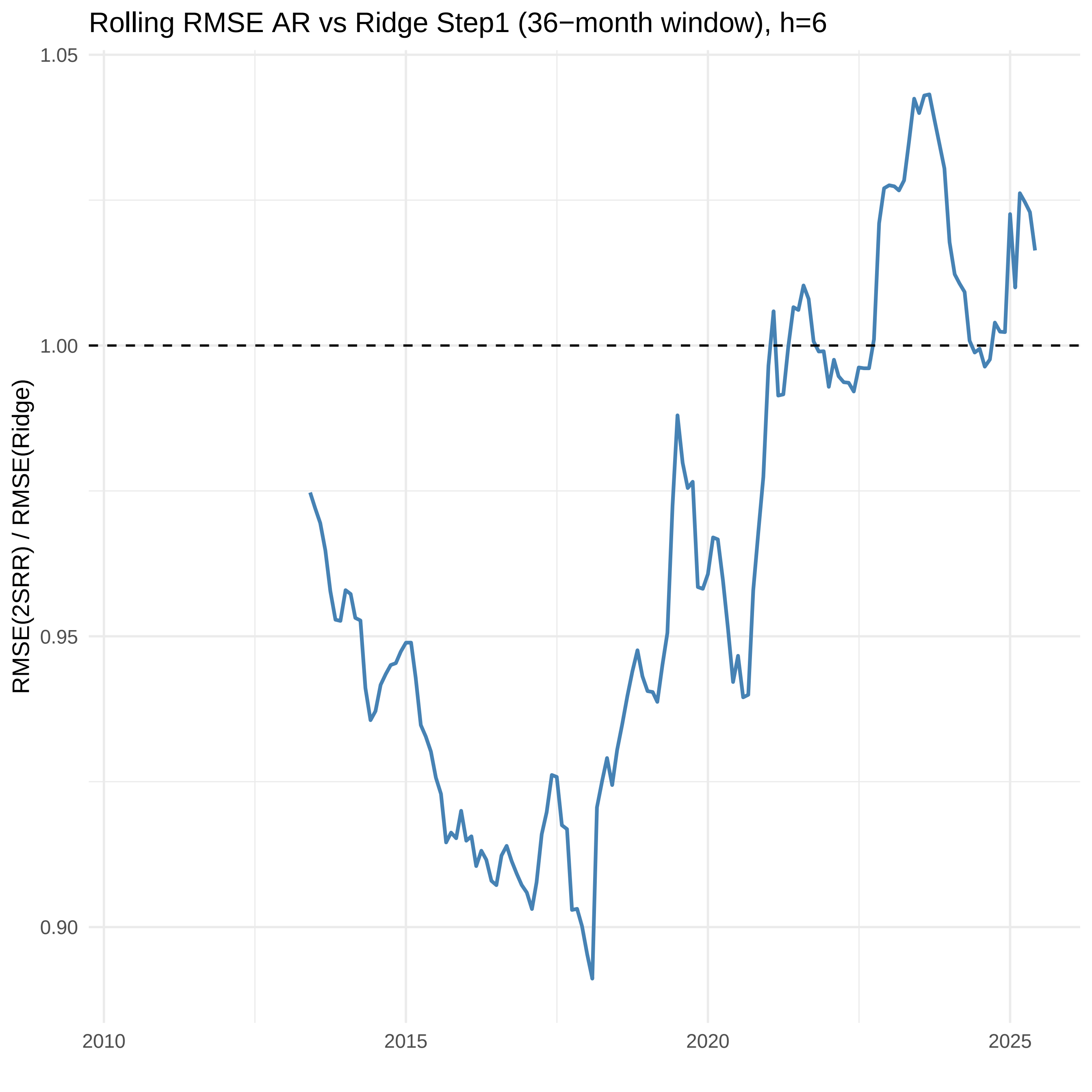
CSSED 2SRR−FAVAR vs Ridge Step1, h=3
Sum(e^2_Ridge − e^2_2SRR)
0.75
0.50
0.25
0.00
−0.25
2010
2015
2020
2025

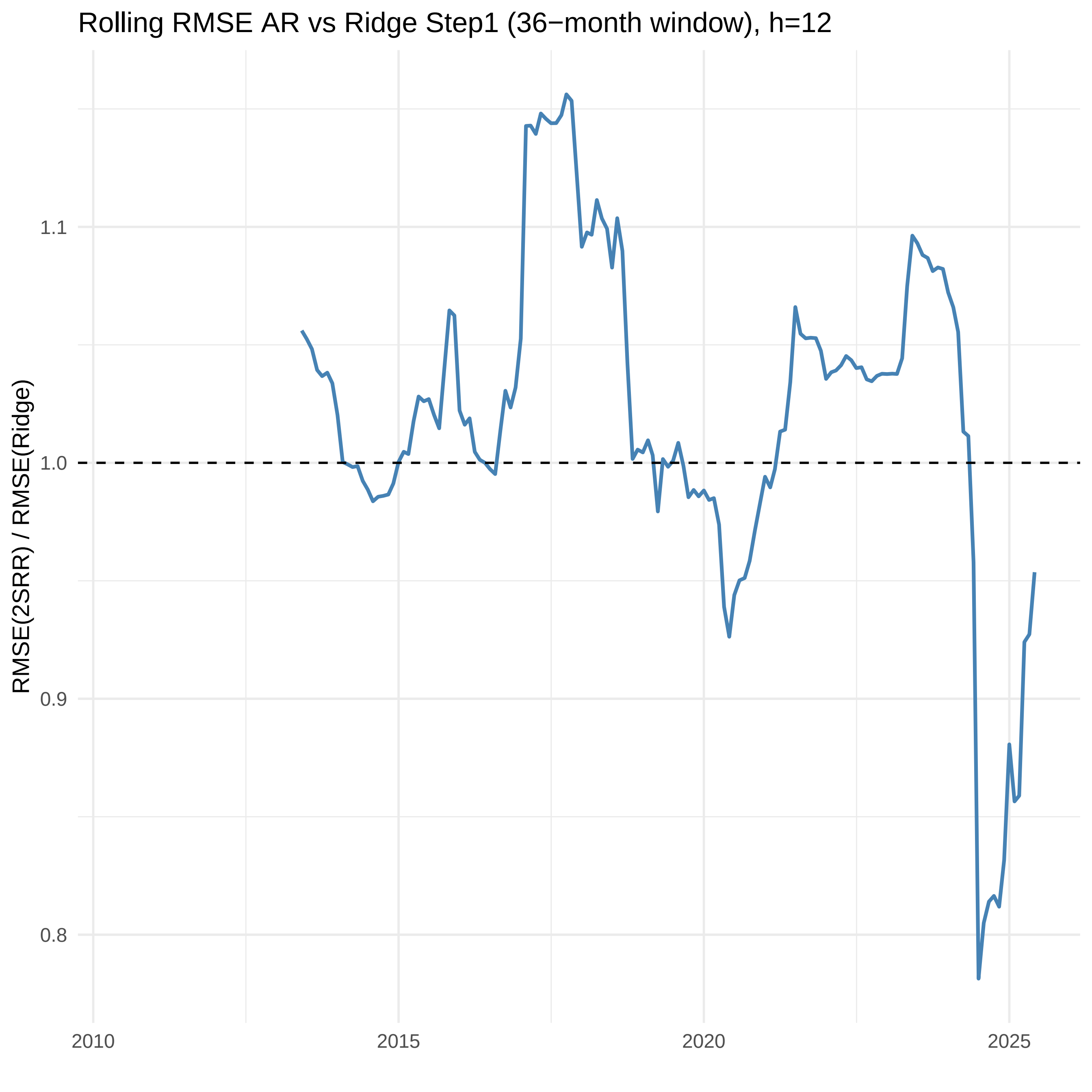

CSSED 2SRR−FAVAR vs Ridge Step1, h=6
Sum(e^2_Ridge − e^2_2SRR)
0.75
0.50
0.25
0.00
2010
2015
2020
2025


CSSED 2SRR−FAVAR vs Ridge Step1, h=12
Sum(e^2_Ridge − e^2_2SRR)
2
1
0
2010
2015
2020
2025

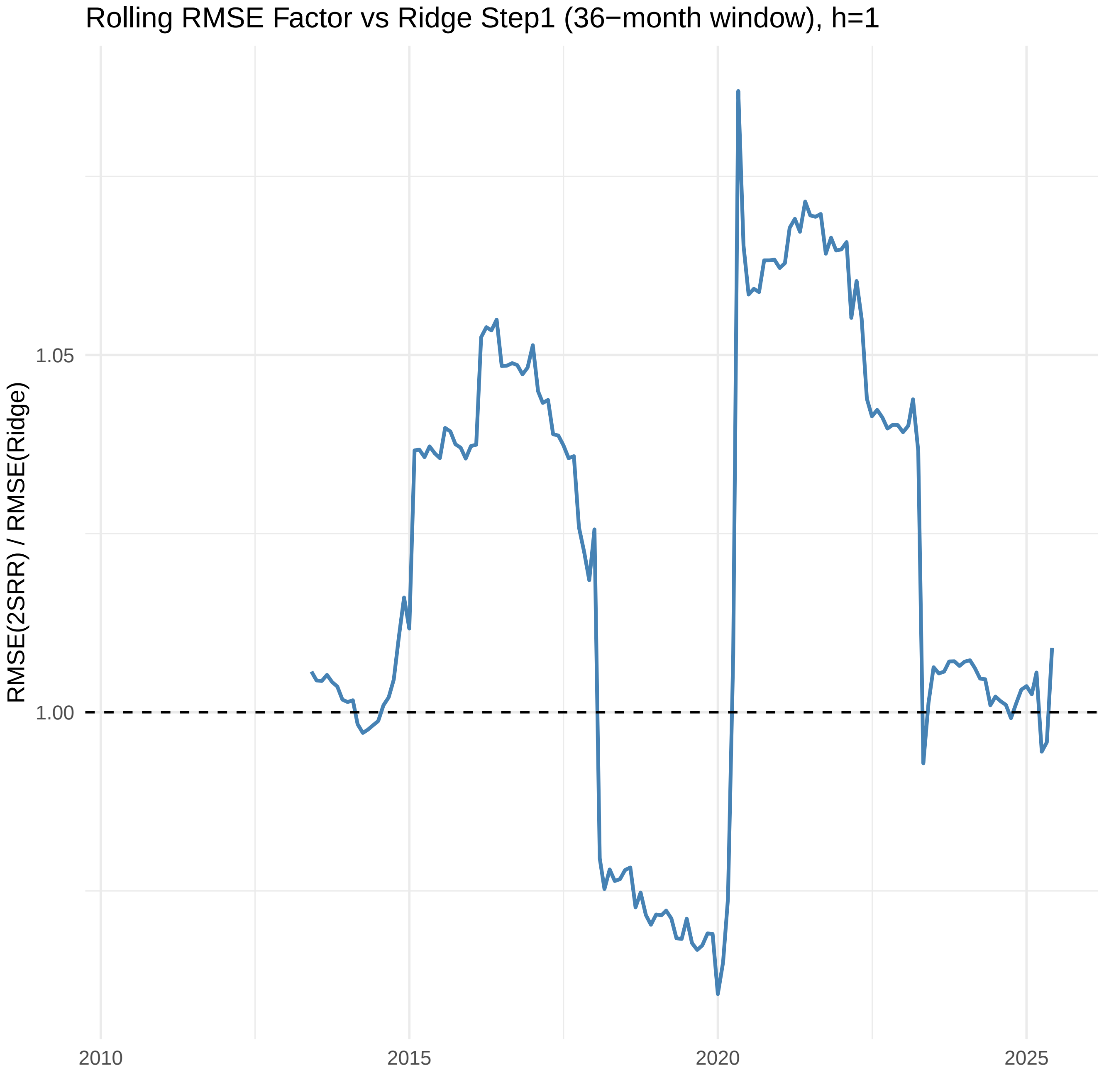

Rolling RMSE AR vs Ridge Step1 (36−month window), h=1
RMSE(2SRR) / RMSE(Ridge)
1.025
1.000
0.975
0.950
0.925
2010
2015
2020
2025

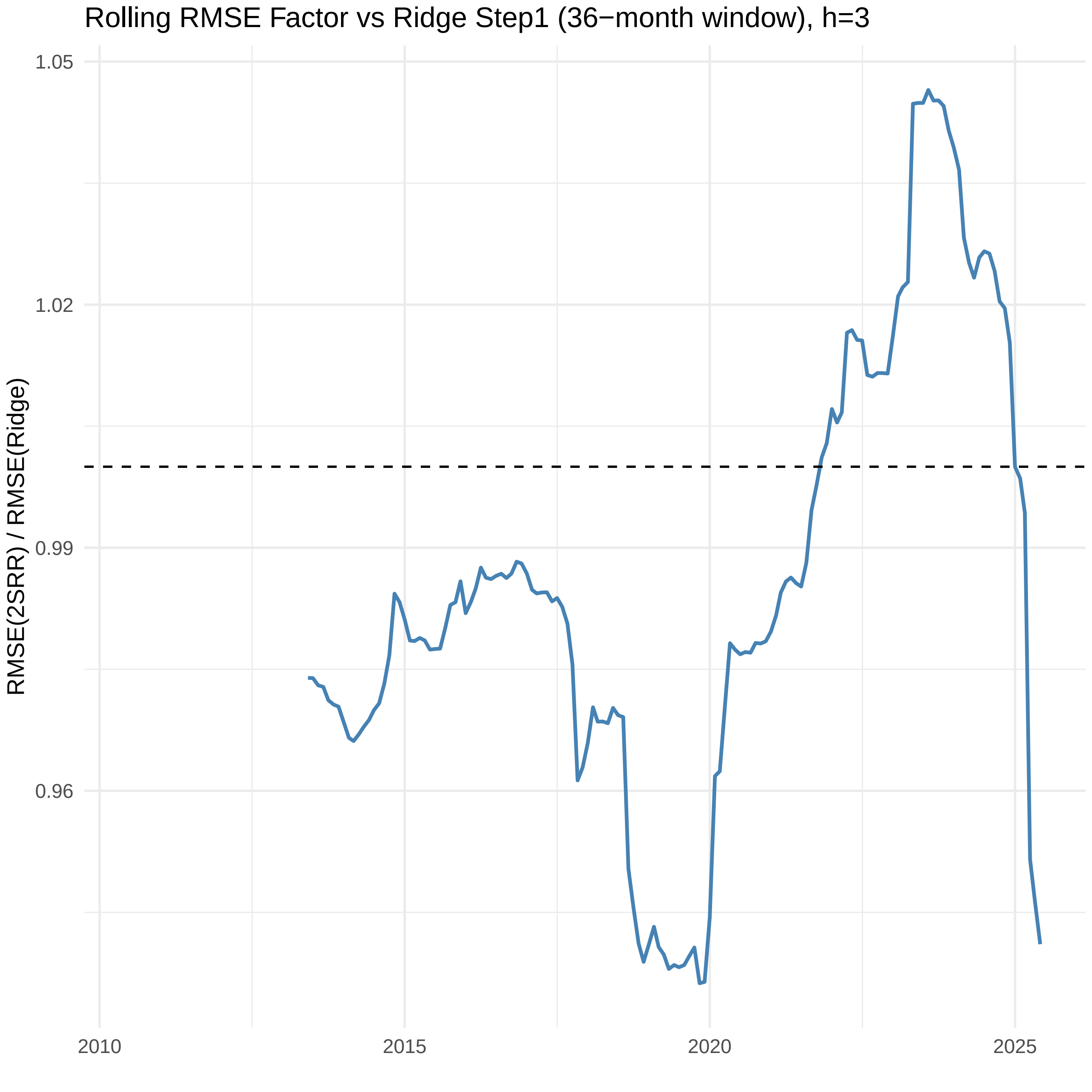

Rolling RMSE AR vs Ridge Step1 (36−month window), h=3
RMSE(2SRR) / RMSE(Ridge)
1.05
1.00
0.95
0.90
2010
2015
2020
2025

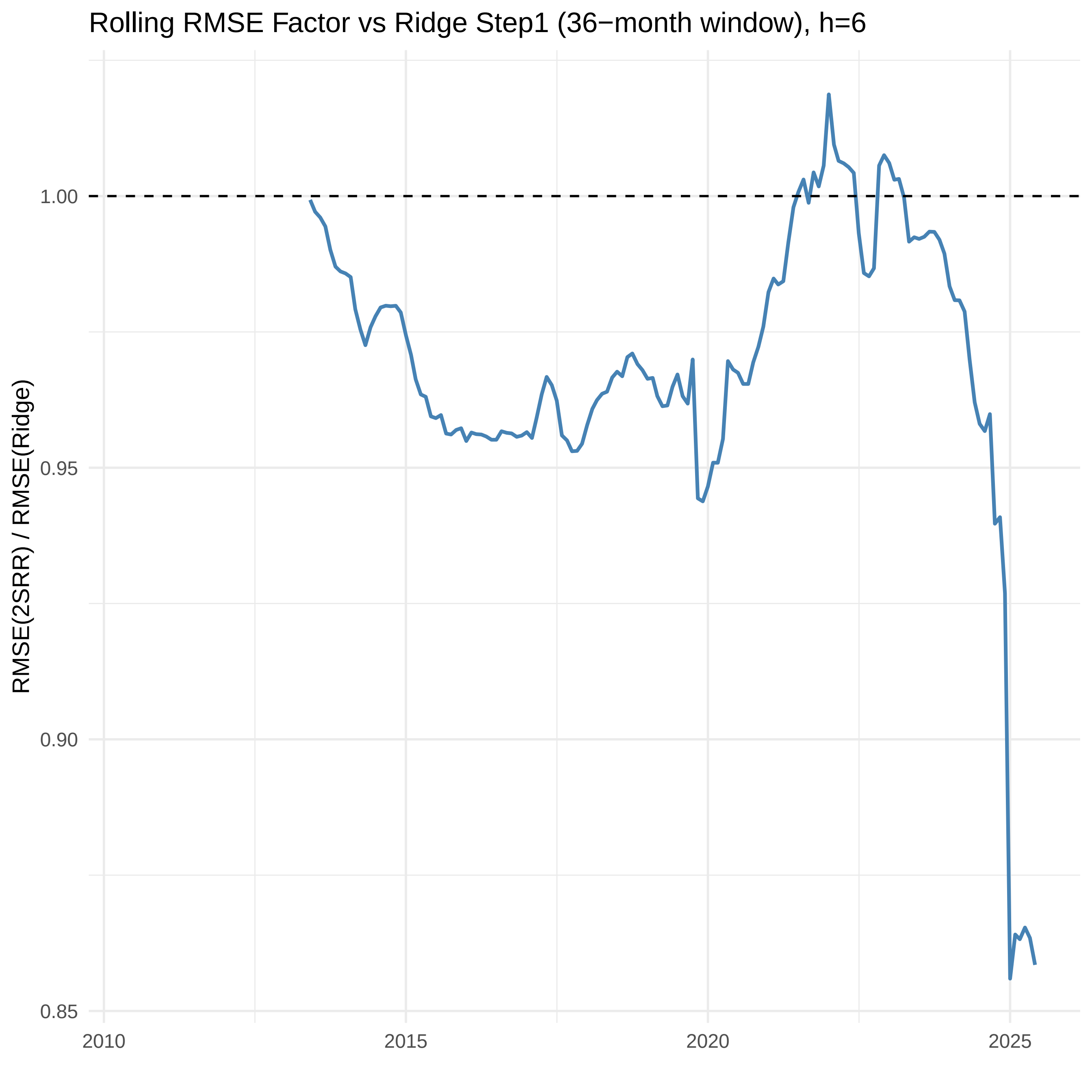

Rolling RMSE AR vs Ridge Step1 (36−month window), h=6
RMSE(2SRR) / RMSE(Ridge)
1.05
1.00
0.95
0.90
2010
2015
2020
2025

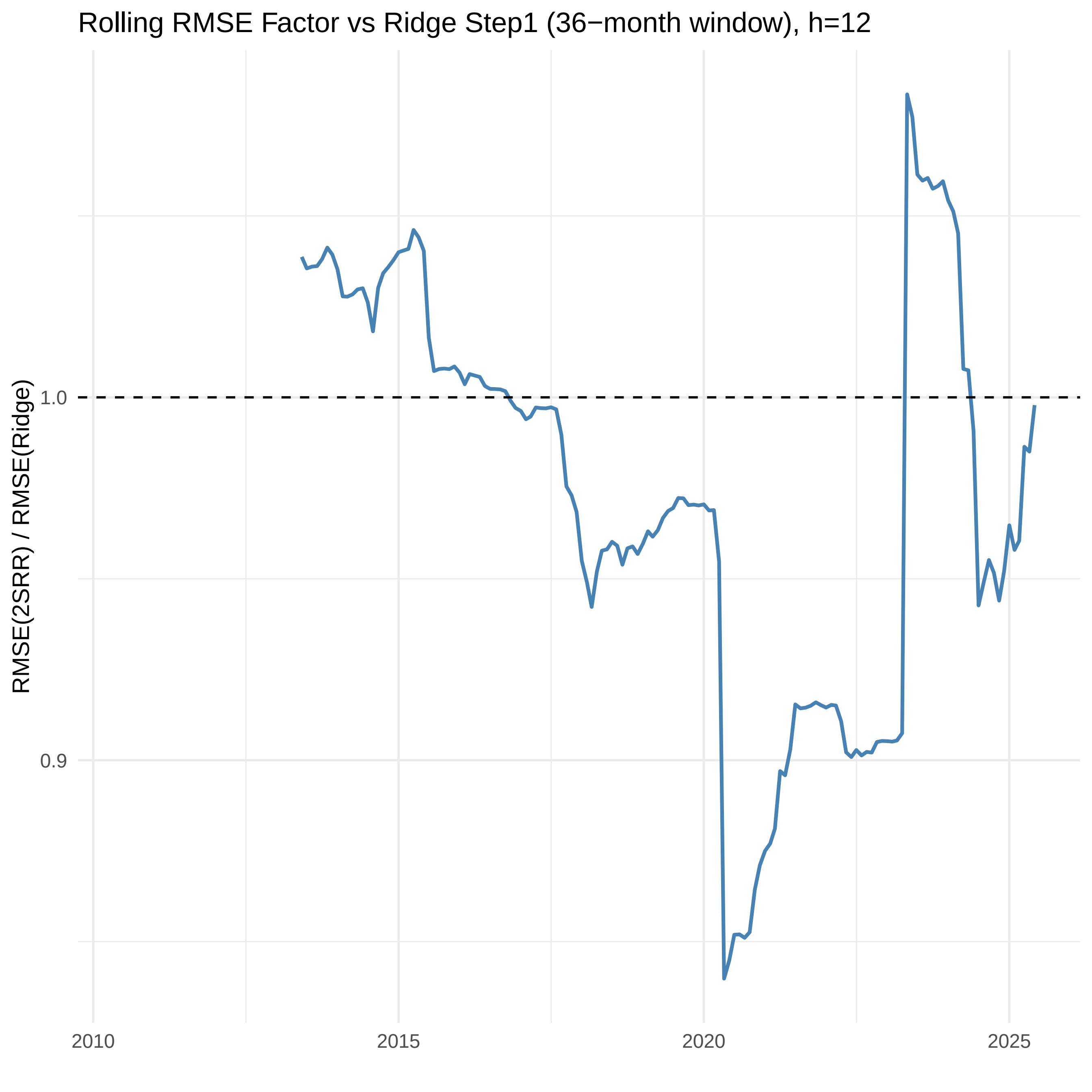


Rolling RMSE AR vs Ridge Step1 (36−month window), h=12
RMSE(2SRR) / RMSE(Ridge)
1.1
1.0
0.9
0.8
2010
2015
2020
2025

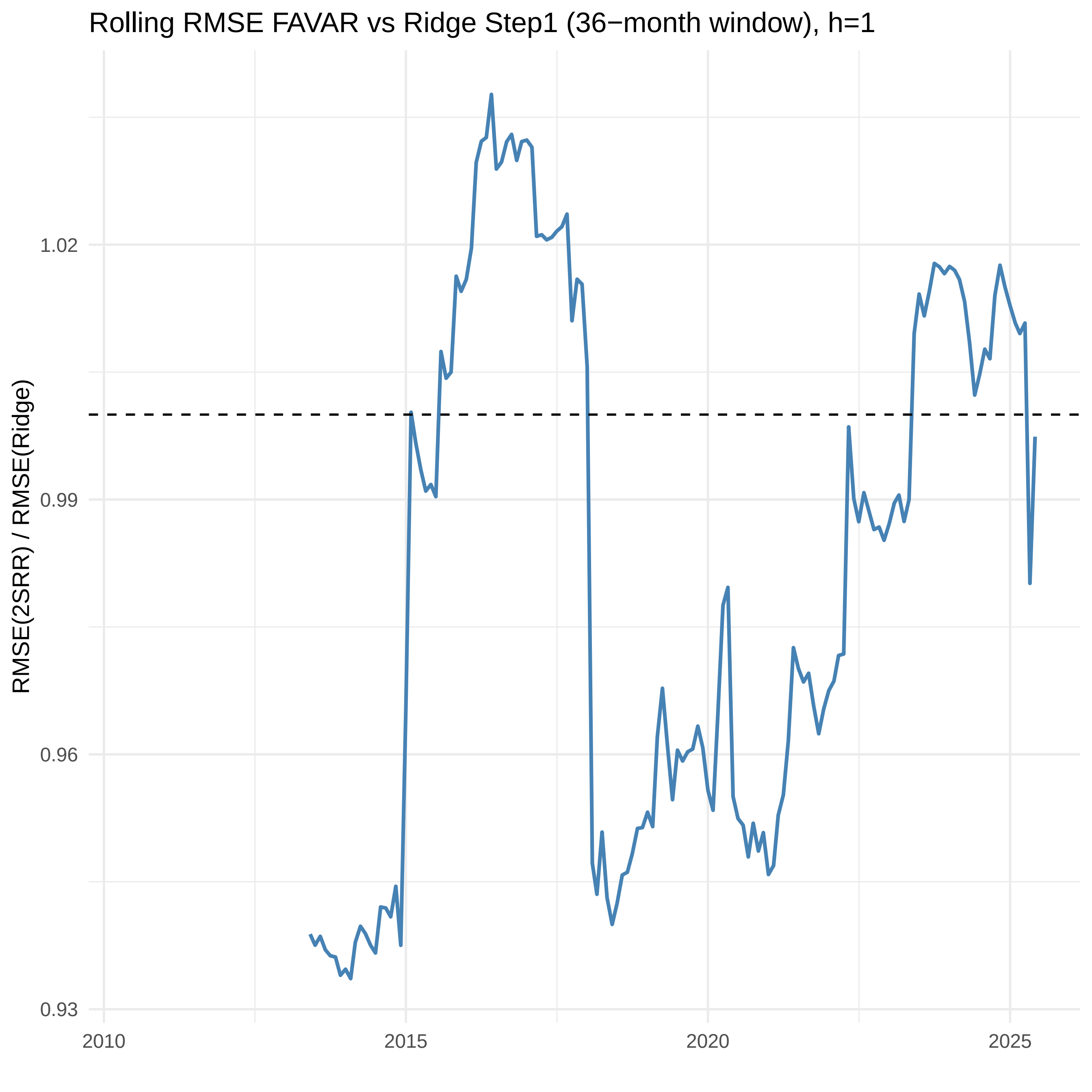

Rolling RMSE Factor vs Ridge Step1 (36−month window), h=1
RMSE(2SRR) / RMSE(Ridge)
1.05
1.00
2010
2015
2020
2025


Rolling RMSE Factor vs Ridge Step1 (36−month window), h=3
RMSE(2SRR) / RMSE(Ridge)
1.05
1.02
0.99
0.96
2010
2015
2020
2025

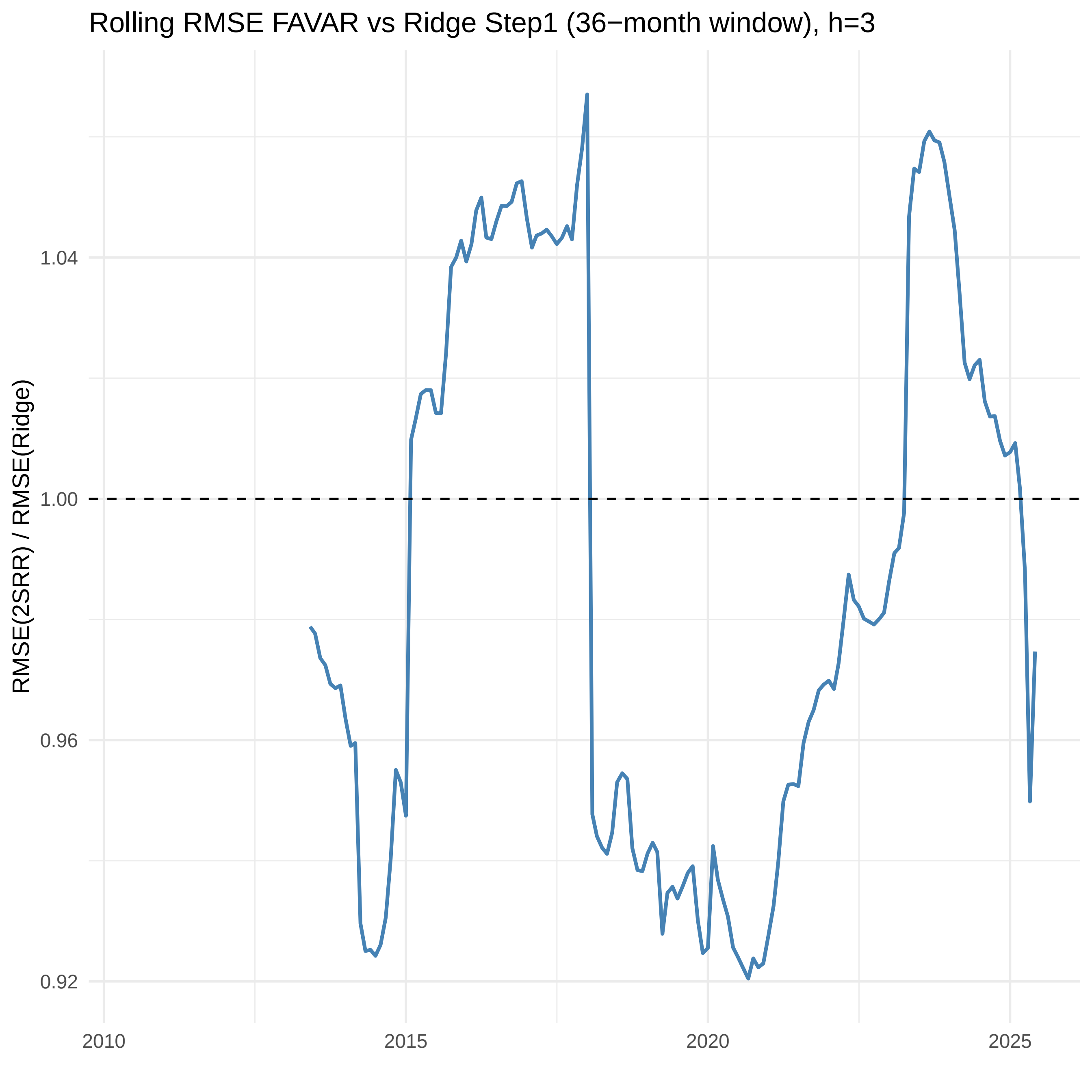

Rolling RMSE Factor vs Ridge Step1 (36−month window), h=6
RMSE(2SRR) / RMSE(Ridge)
1.00
0.95
0.90
0.85
2010
2015
2020
2025

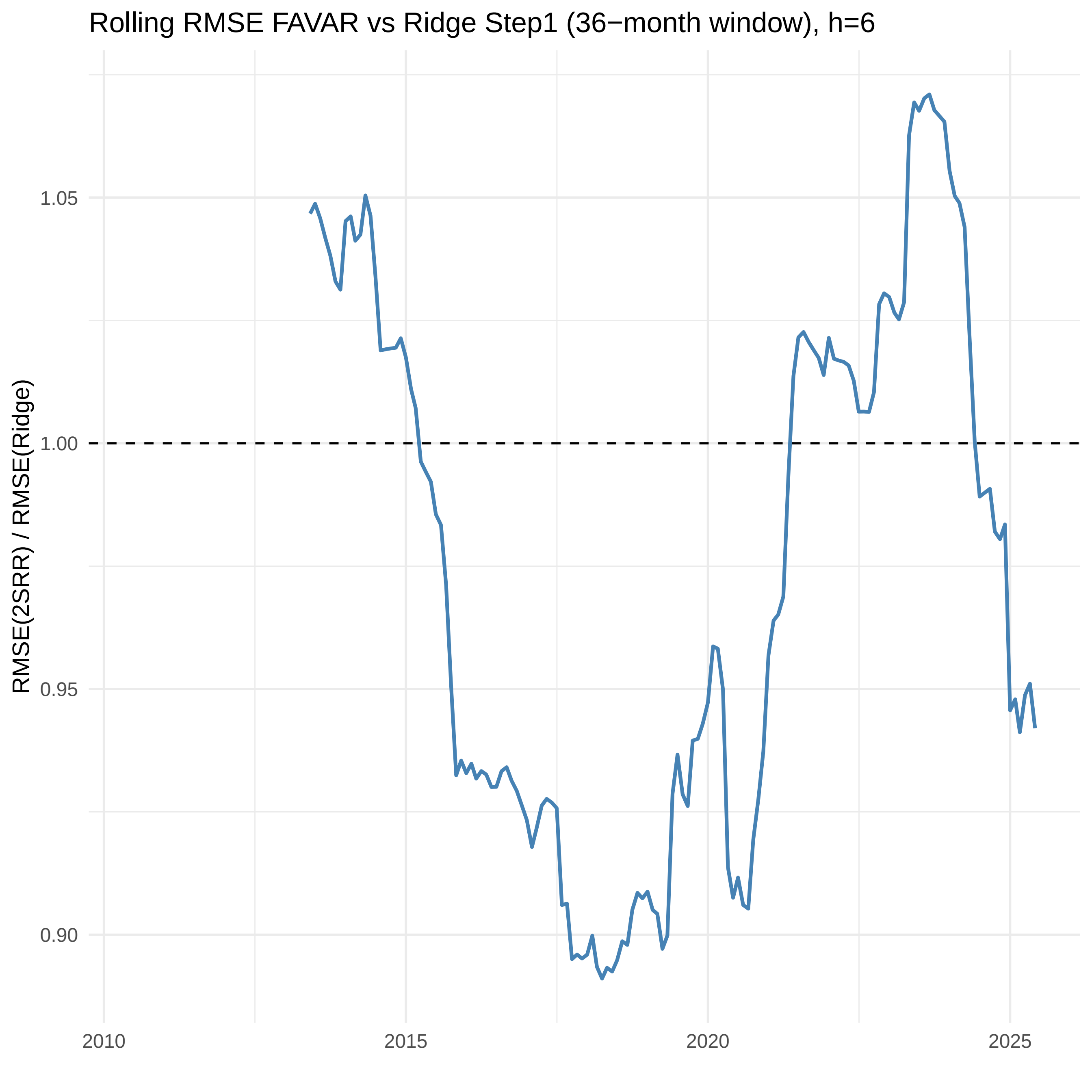


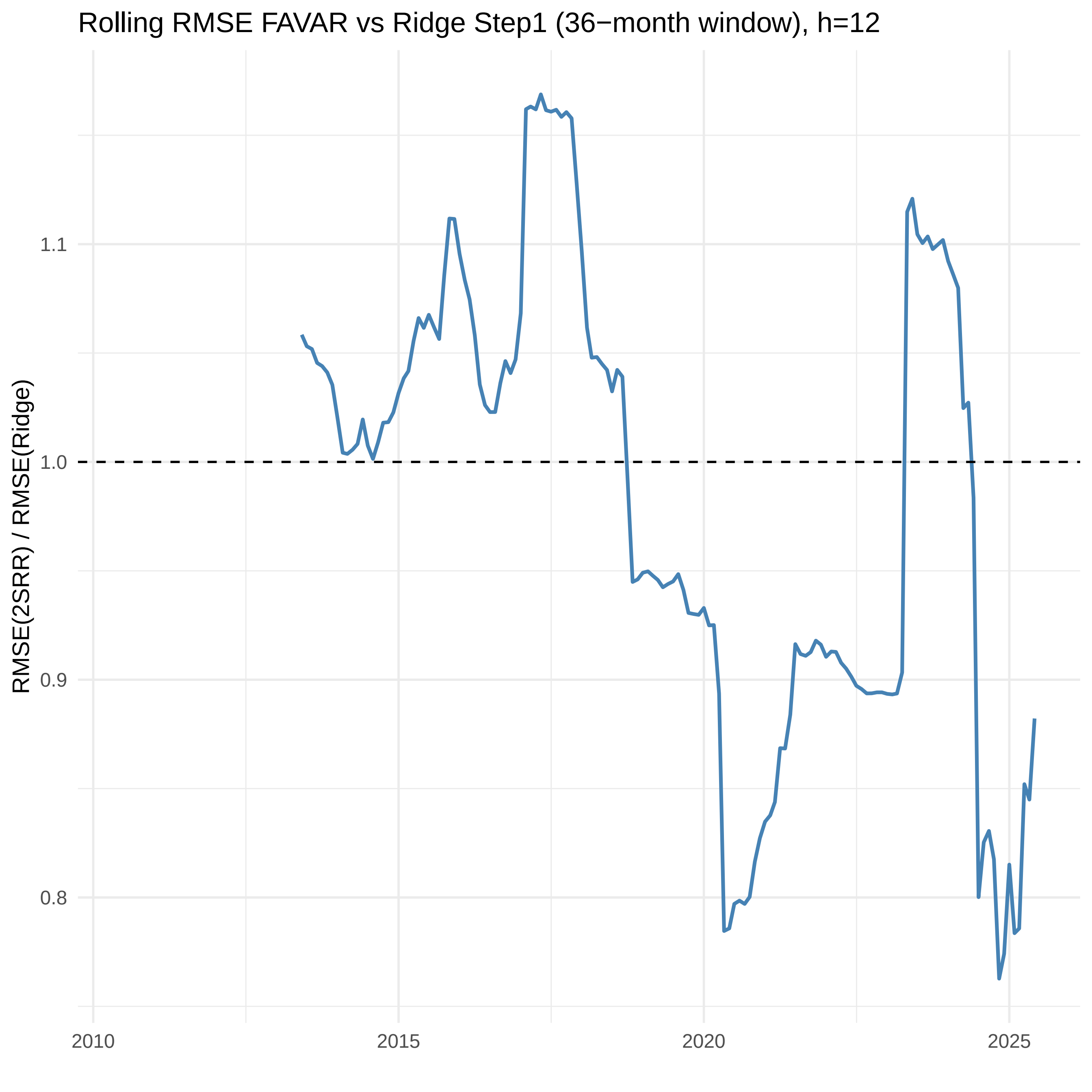
Rolling RMSE Factor vs Ridge Step1 (36−month window), h=12
RMSE(2SRR) / RMSE(Ridge)
1.0
0.9
2010
2015
2020
2025

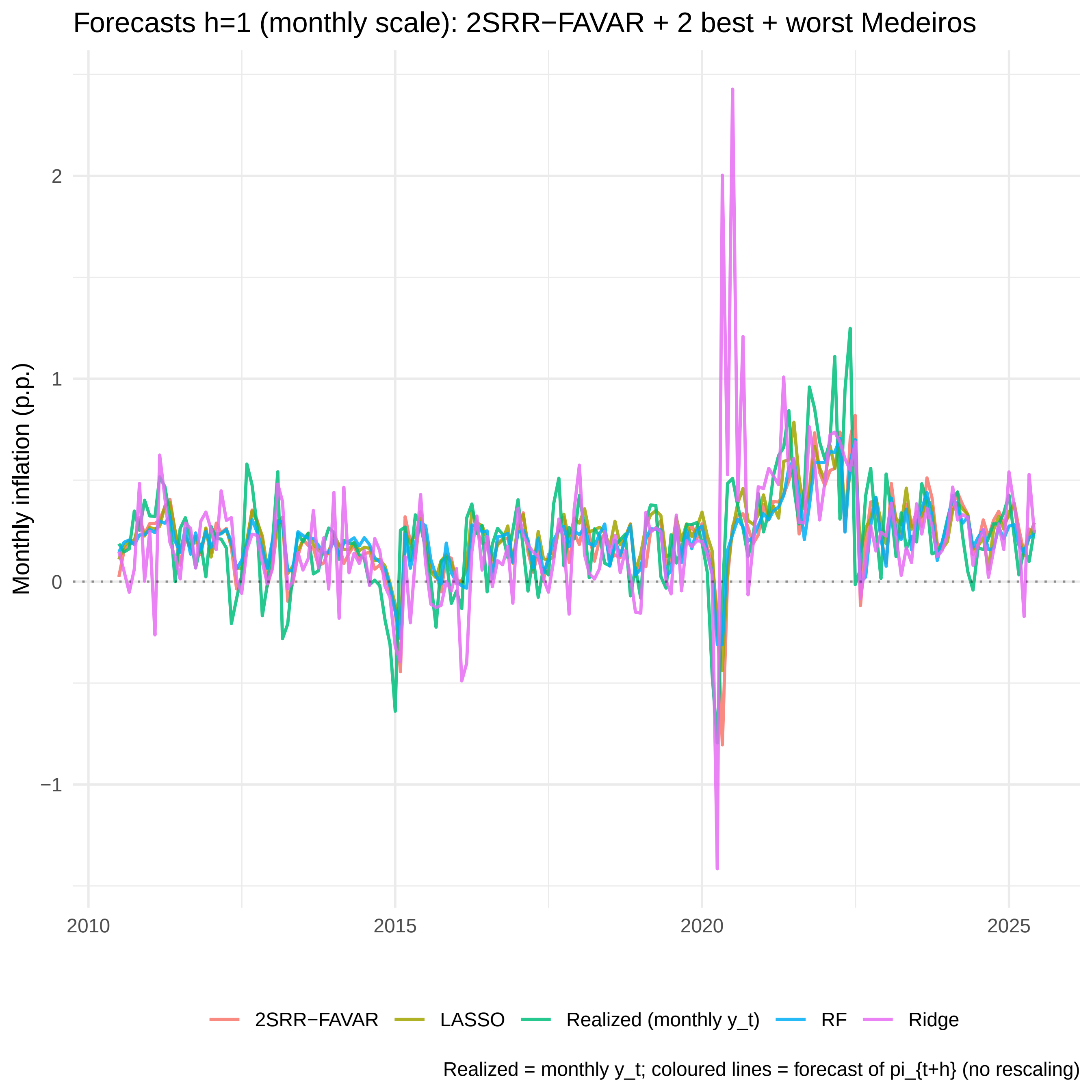


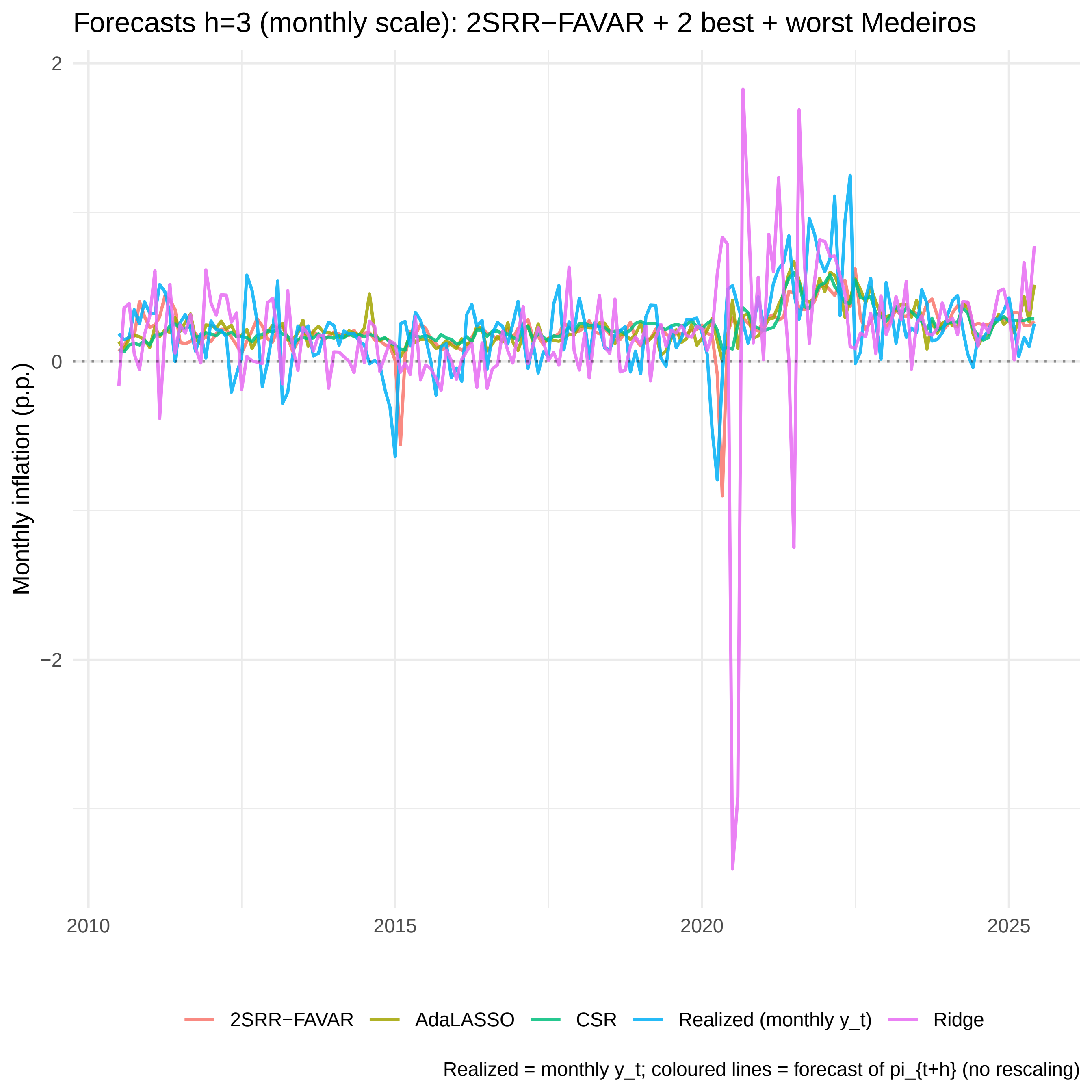
Rolling RMSE FAVAR vs Ridge Step1 (36−month window), h=1
RMSE(2SRR) / RMSE(Ridge)
1.02
0.99
0.96
0.93
2010
2015
2020
2025

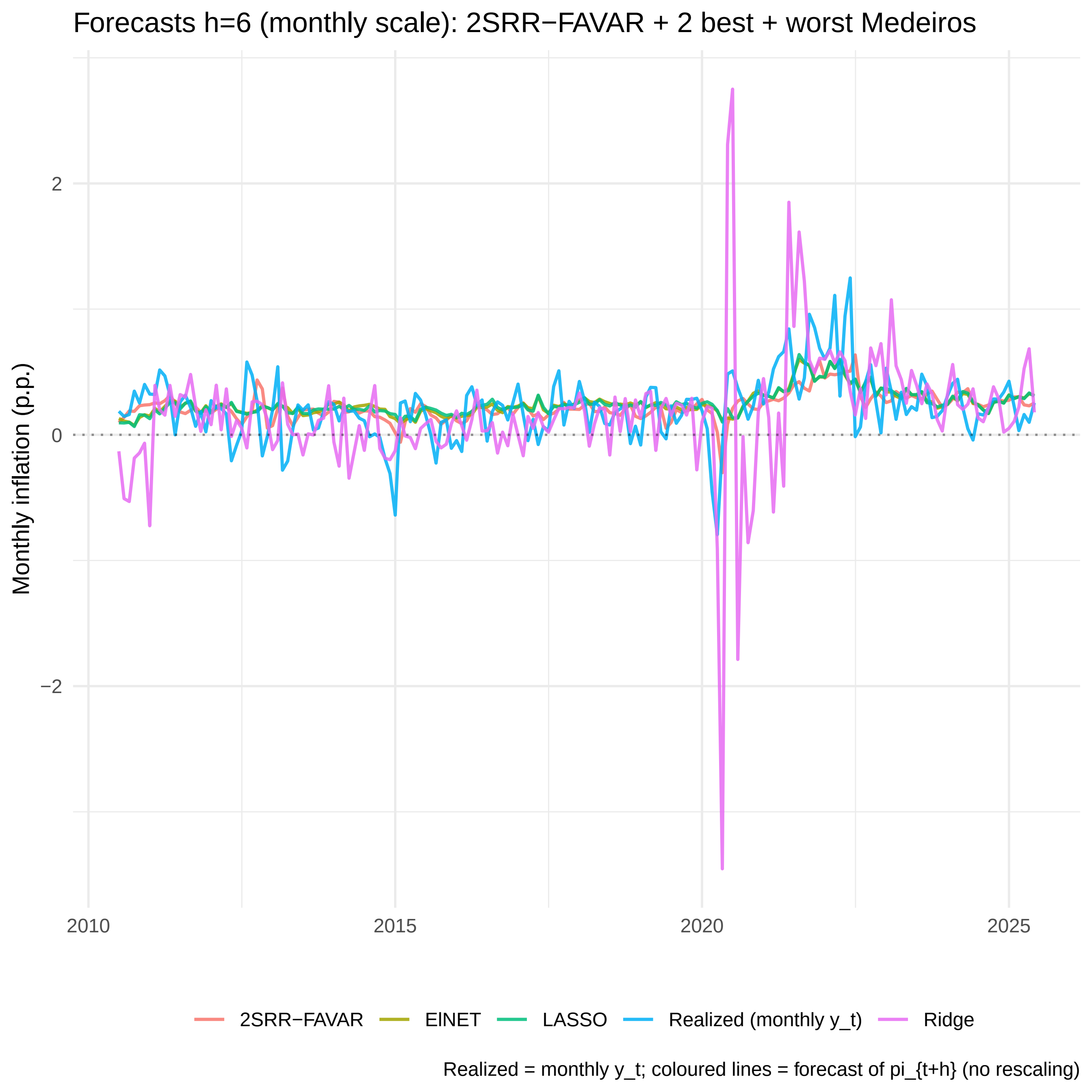

Rolling RMSE FAVAR vs Ridge Step1 (36−month window), h=3
RMSE(2SRR) / RMSE(Ridge)
1.04
1.00
0.96
0.92
2010
2015
2020
2025

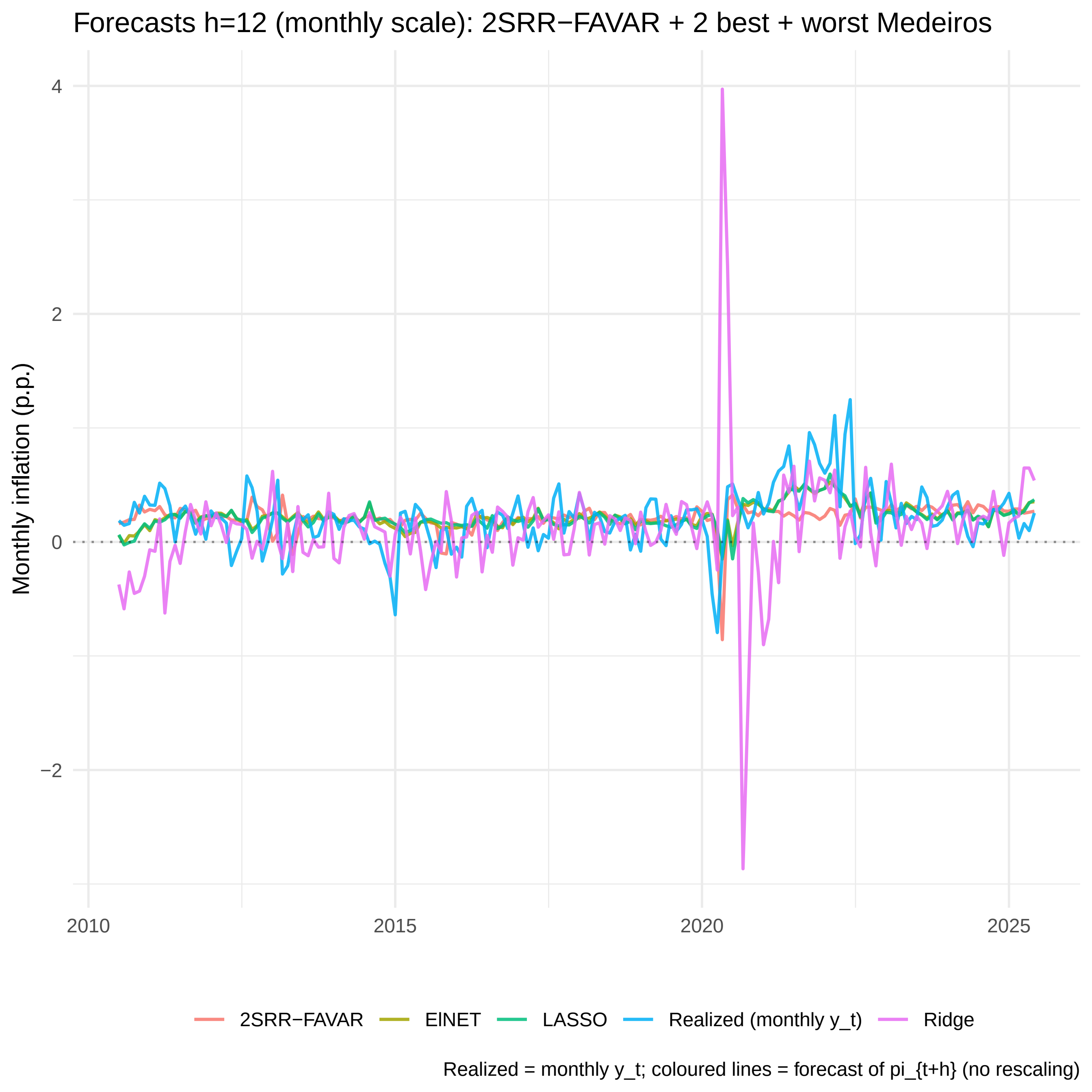

Rolling RMSE FAVAR vs Ridge Step1 (36−month window), h=6
RMSE(2SRR) / RMSE(Ridge)
1.05
1.00
0.95
0.90
2010
2015
2020
2025

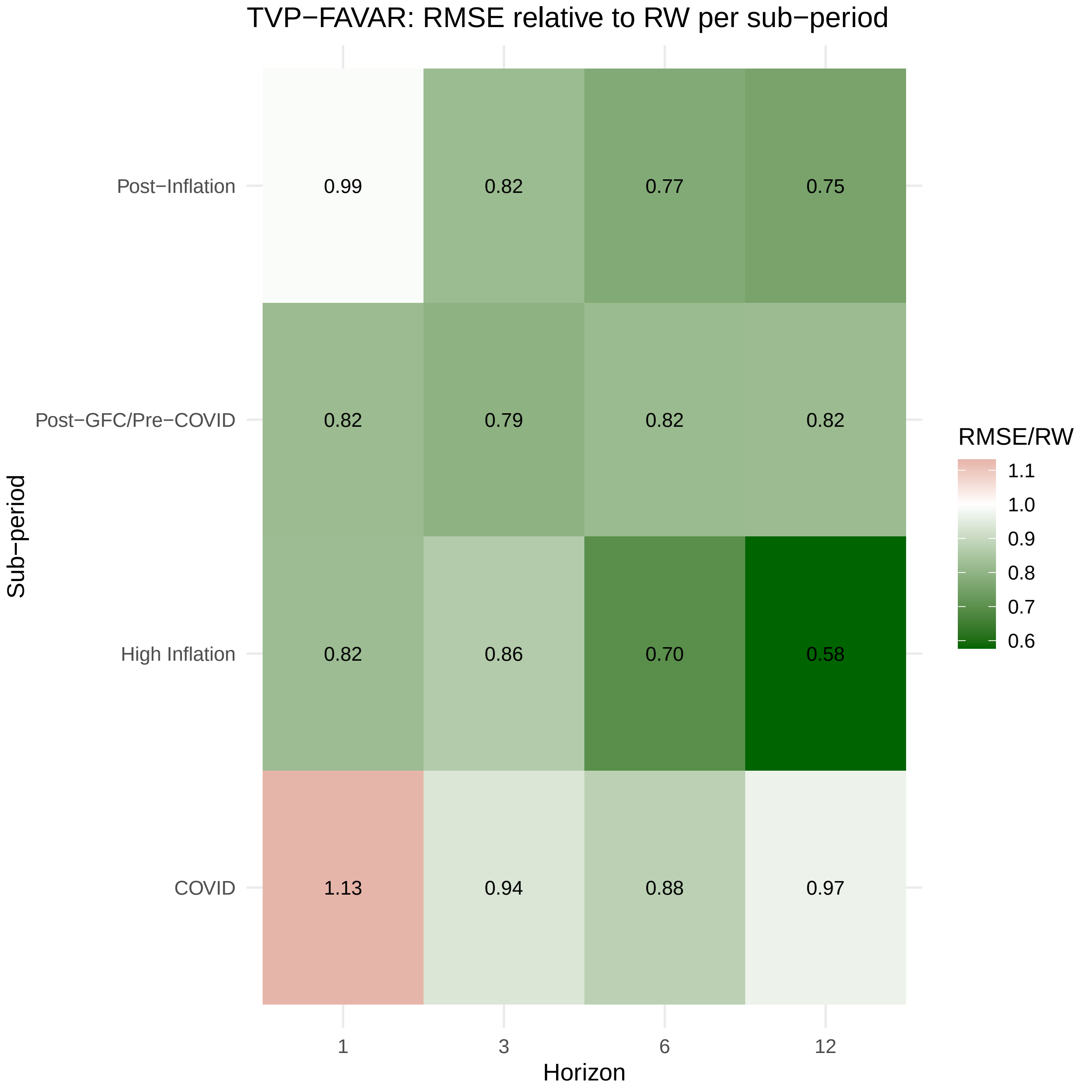

Rolling RMSE FAVAR vs Ridge Step1 (36−month window), h=12
RMSE(2SRR) / RMSE(Ridge)
1.1
1.0
0.9
0.8
2010
2015
2020
2025


Forecasts h=1 (monthly scale): 2SRR−FAVAR + 2 best + worst Medeiros
Monthly inflation (p.p.)
2
1
0
−1
2010
2015
2020
2025
2SRR−FAVAR
LASSO
Realized (monthly y_t)
RF
Ridge
Realized = monthly y_t; coloured lines = forecast of pi_{t+h} (no rescaling)

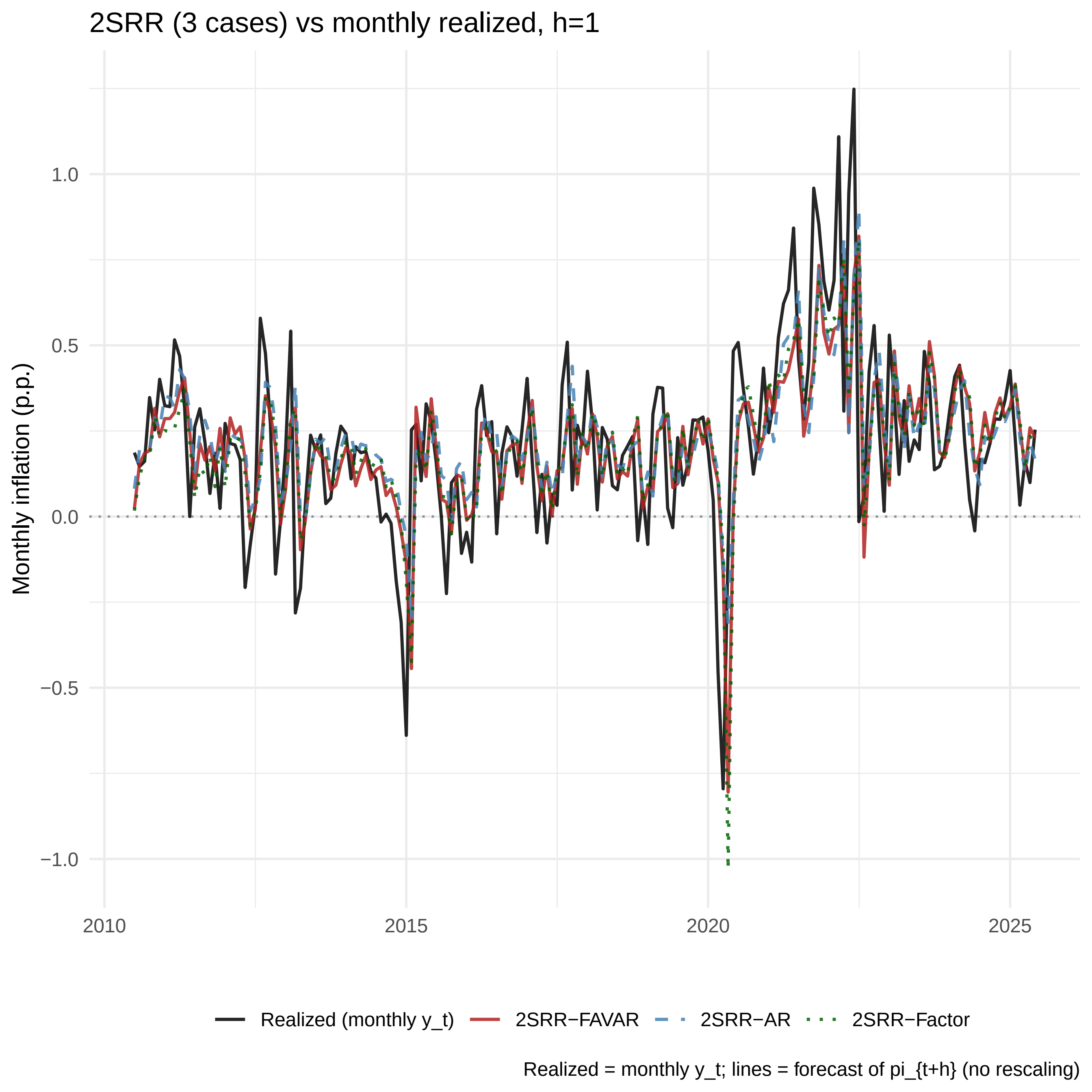

Forecasts h=3 (monthly scale): 2SRR−FAVAR + 2 best + worst Medeiros
Monthly inflation (p.p.)
2
0
−2
2010
2015
2020
2025
2SRR−FAVAR
AdaLASSO
CSR
Realized (monthly y_t)
Ridge
Realized = monthly y_t; coloured lines = forecast of pi_{t+h} (no rescaling)

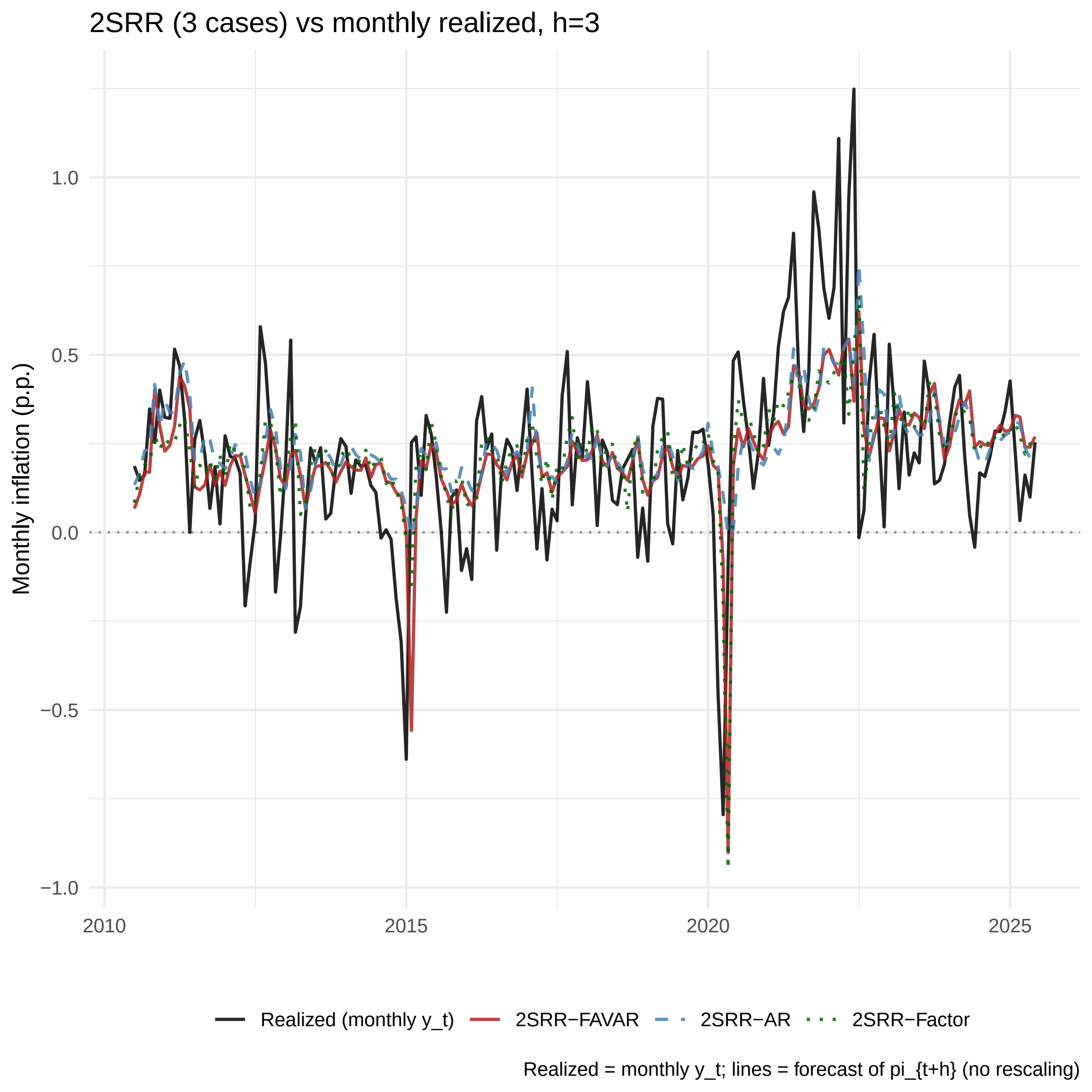

Forecasts h=6 (monthly scale): 2SRR−FAVAR + 2 best + worst Medeiros
Monthly inflation (p.p.)
2
0
−2
2010
2015
2020
2025
2SRR−FAVAR
ElNET
LASSO
Realized (monthly y_t)
Ridge
Realized = monthly y_t; coloured lines = forecast of pi_{t+h} (no rescaling)

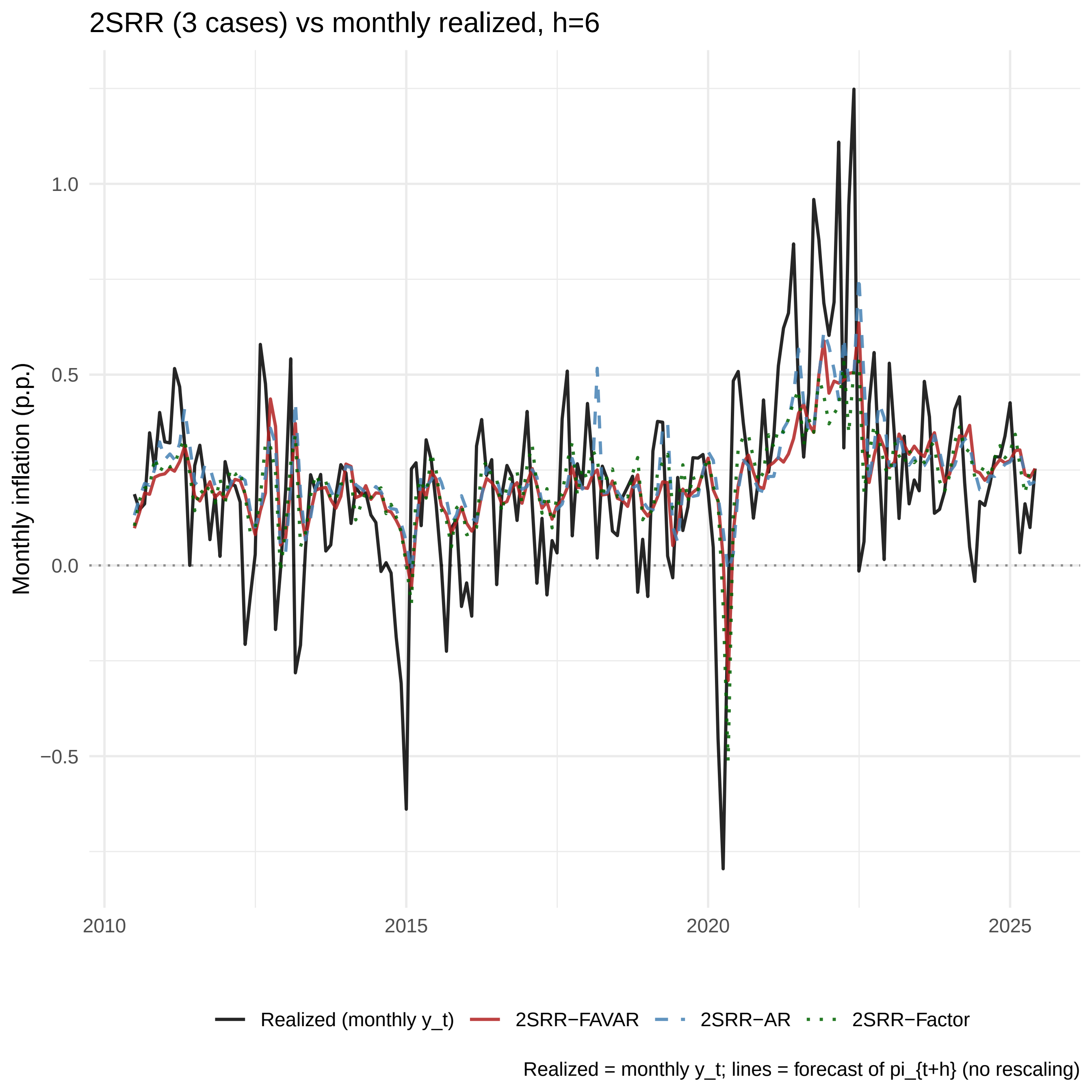

Forecasts h=12 (monthly scale): 2SRR−FAVAR + 2 best + worst Medeiros
Monthly inflation (p.p.)
4
2
0
−2
2010
2015
2020
2025
2SRR−FAVAR
ElNET
LASSO
Realized (monthly y_t)
Ridge
Realized = monthly y_t; coloured lines = forecast of pi_{t+h} (no rescaling)

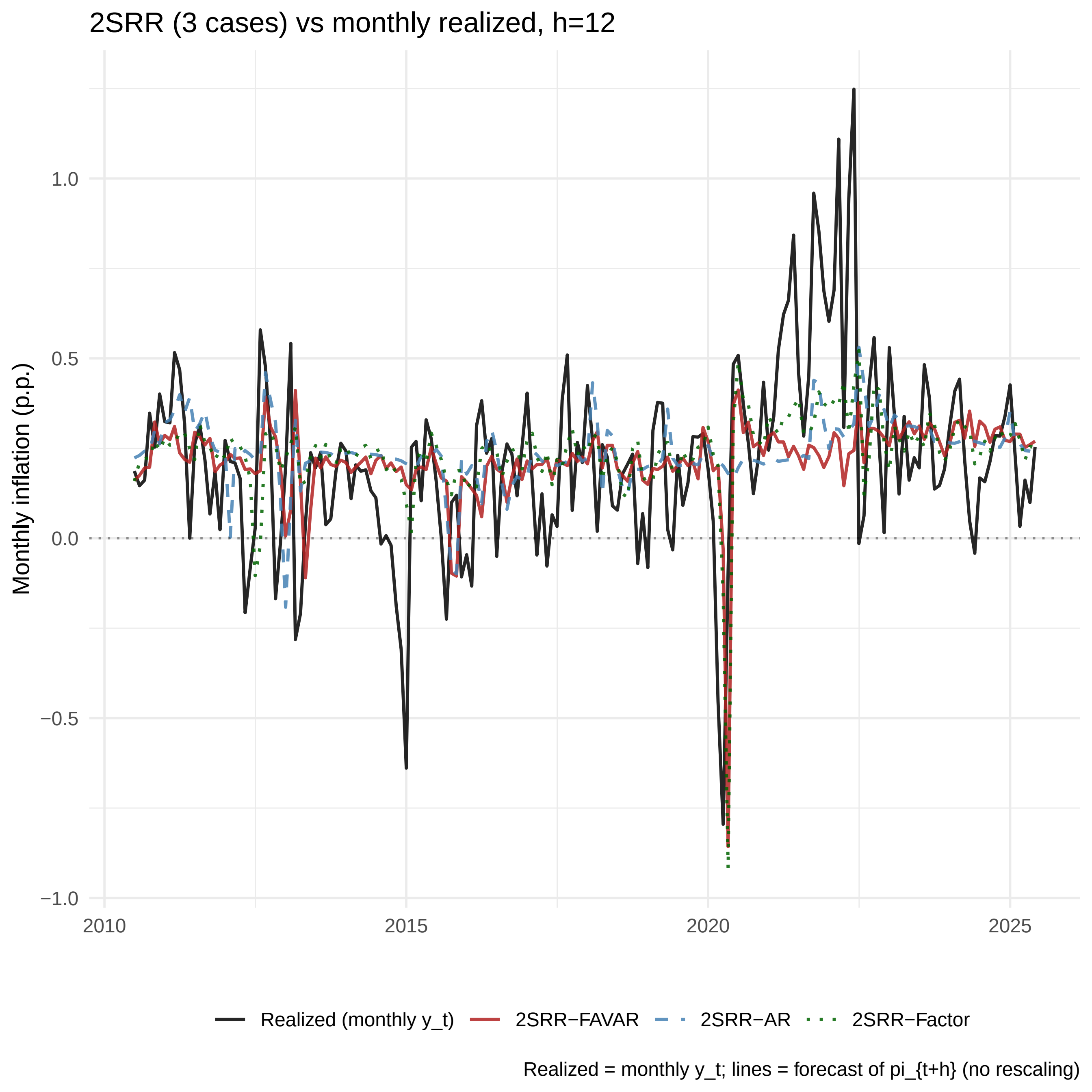


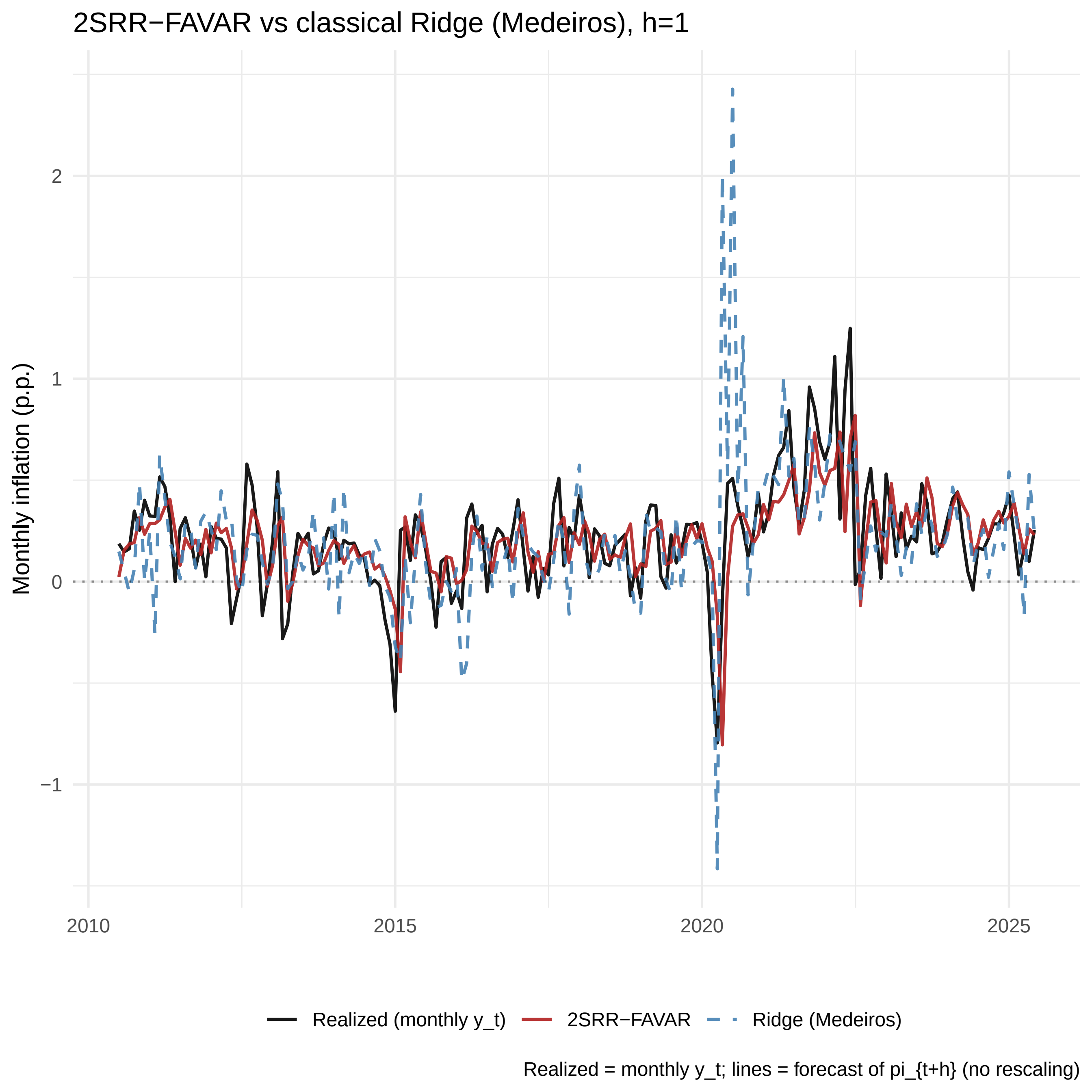
TVP−FAVAR: RMSE relative to RW per sub−period
Post−Inflation
Post−GFC/Pre−COVID
High Inflation
COVID
Sub−period
0.99
0.82
0.77
0.75
0.82
0.79
0.82
0.82
0.82
0.86
0.70
0.58
1.13
0.94
0.88
0.97
1
3
6
12
Horizon
RMSE/RW
1.1
1.0
0.9
0.8
0.7
0.6


2SRR (3 cases) vs monthly realized, h=1
Monthly inflation (p.p.)
1.0
0.5
0.0
−0.5
−1.0
2010
2015
2020
2025
Realized (monthly y_t)
2SRR−FAVAR
2SRR−AR
2SRR−Factor
Realized = monthly y_t; lines = forecast of pi_{t+h} (no rescaling)

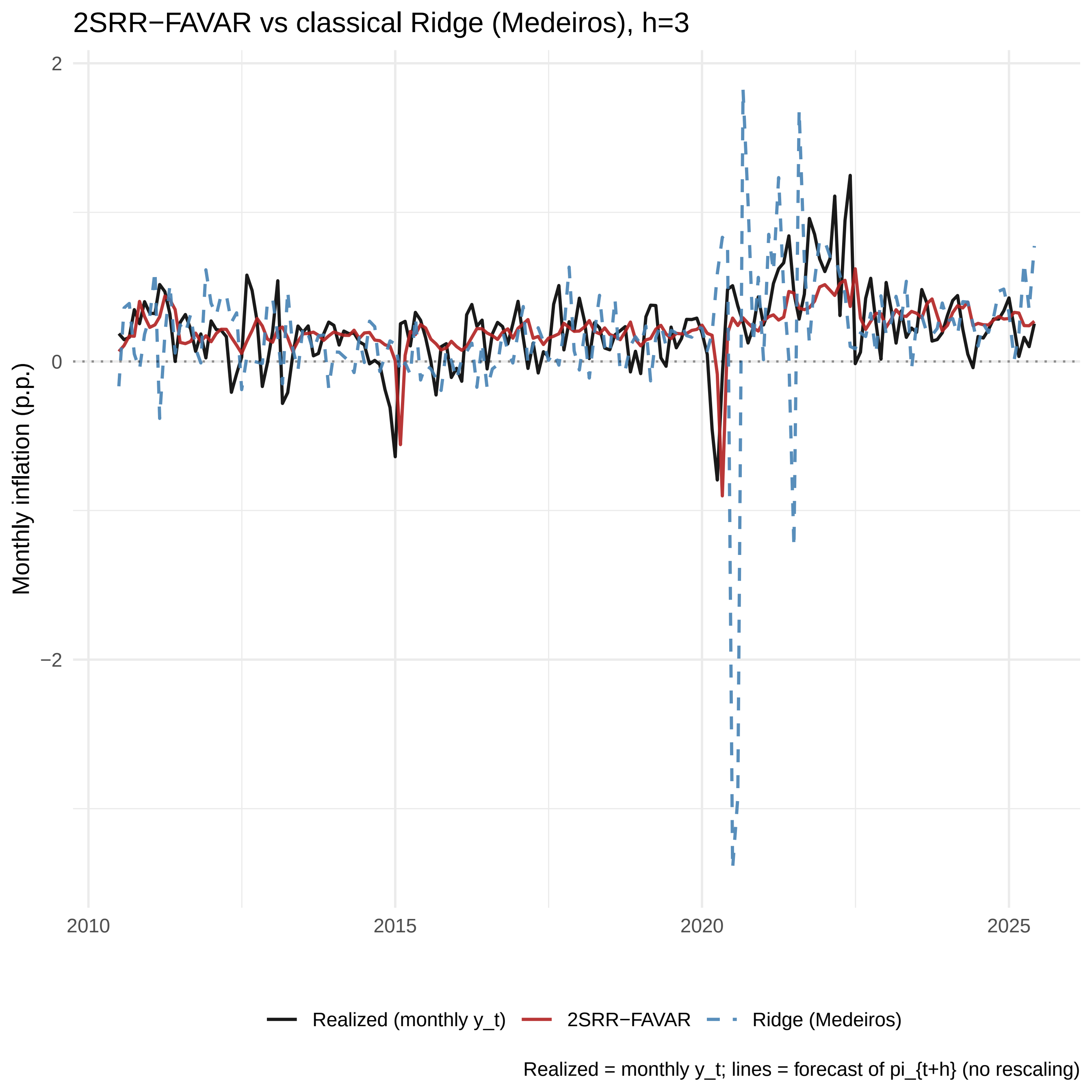

2SRR (3 cases) vs monthly realized, h=3
Monthly inflation (p.p.)
1.0
0.5
0.0
−0.5
−1.0
2010
2015
2020
2025
Realized (monthly y_t)
2SRR−FAVAR
2SRR−AR
2SRR−Factor
Realized = monthly y_t; lines = forecast of pi_{t+h} (no rescaling)

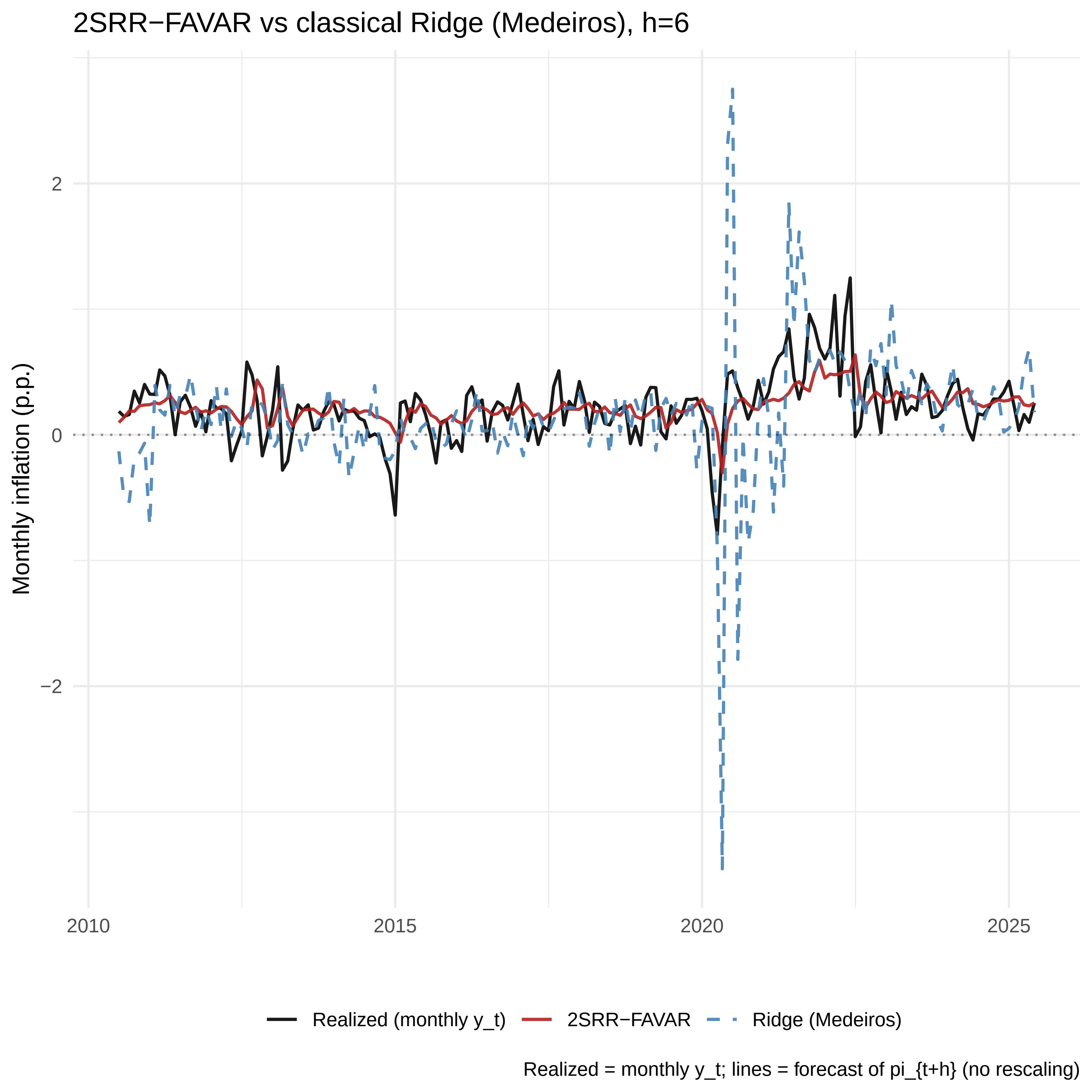

2SRR (3 cases) vs monthly realized, h=6
Monthly inflation (p.p.)
1.0
0.5
0.0
−0.5
2010
2015
2020
2025
Realized (monthly y_t)
2SRR−FAVAR
2SRR−AR
2SRR−Factor
Realized = monthly y_t; lines = forecast of pi_{t+h} (no rescaling)

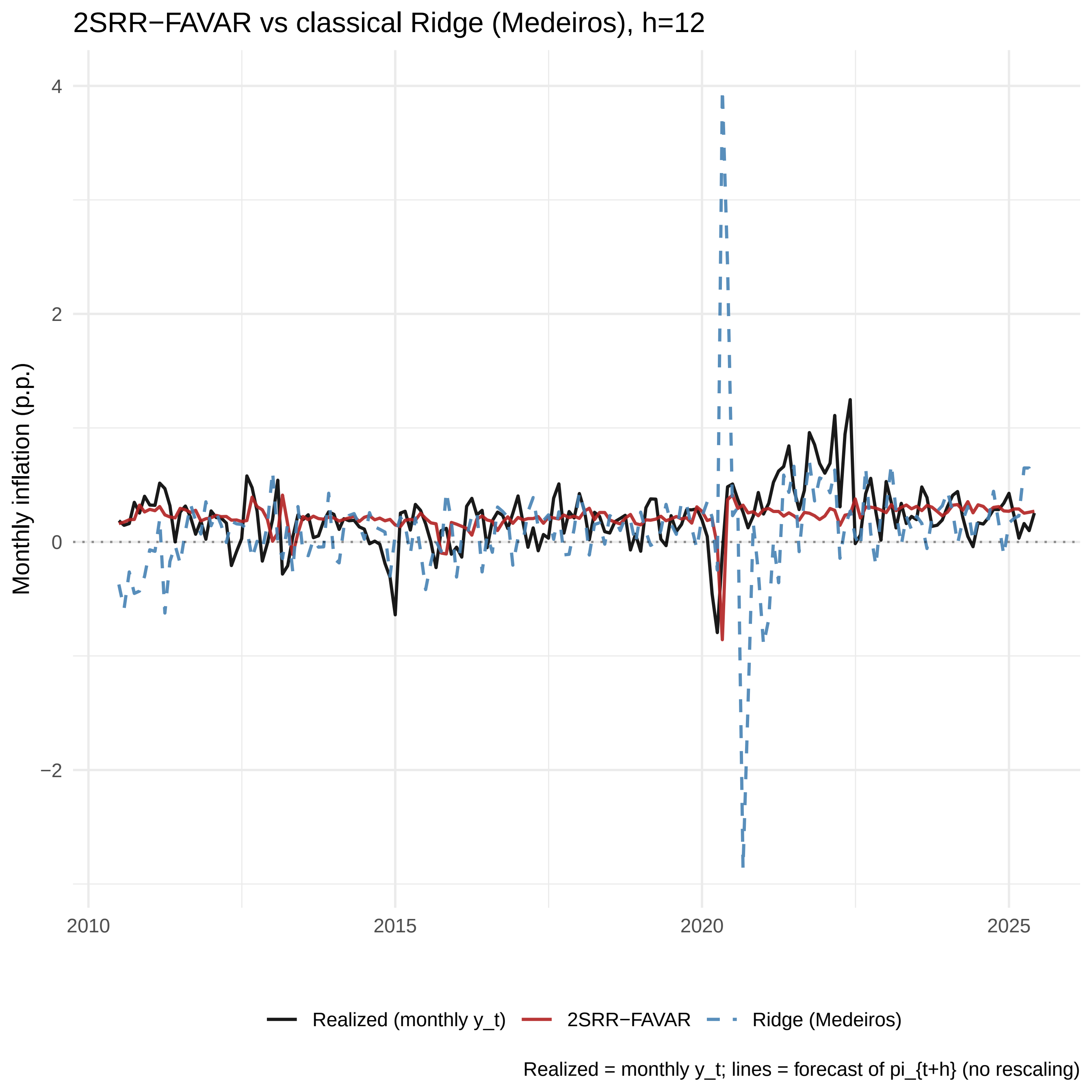

2SRR (3 cases) vs monthly realized, h=12
Monthly inflation (p.p.)
1.0
0.5
0.0
−0.5
−1.0
2010
2015
2020
2025
Realized (monthly y_t)
2SRR−FAVAR
2SRR−AR
2SRR−Factor
Realized = monthly y_t; lines = forecast of pi_{t+h} (no rescaling)

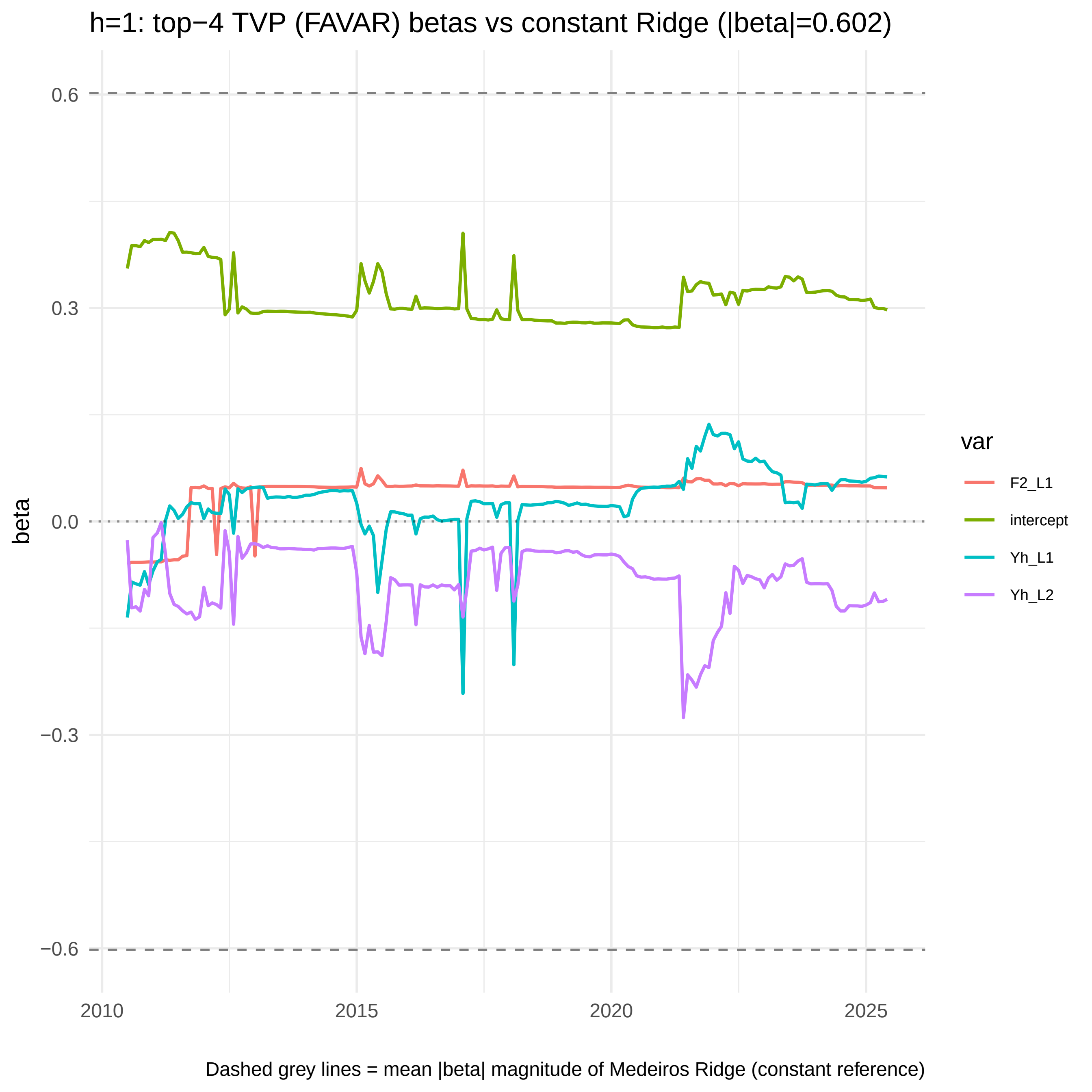


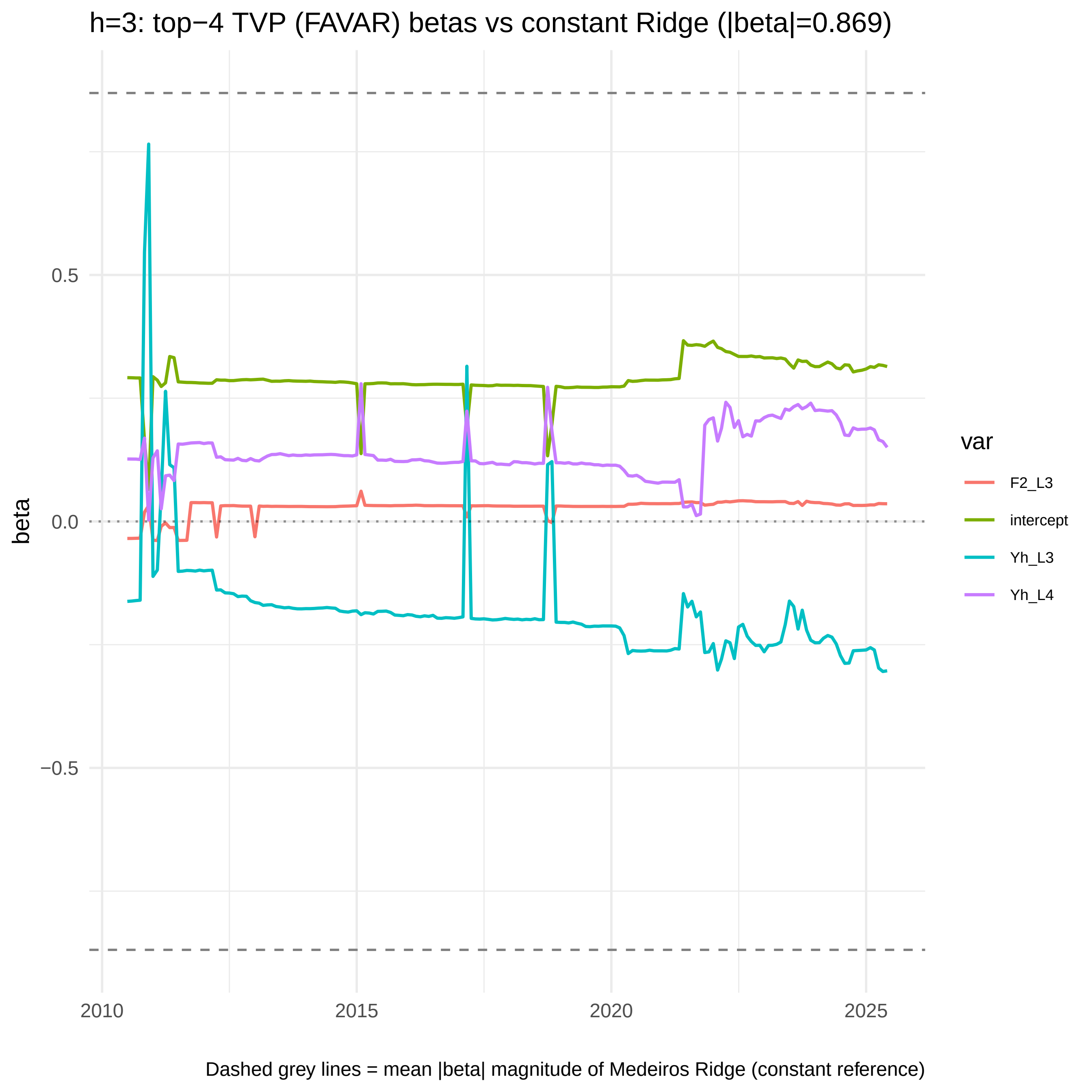
2SRR−FAVAR vs classical Ridge (Medeiros), h=1
Monthly inflation (p.p.)
2
1
0
−1
2010
2015
2020
2025
Realized (monthly y_t)
2SRR−FAVAR
Ridge (Medeiros)
Realized = monthly y_t; lines = forecast of pi_{t+h} (no rescaling)

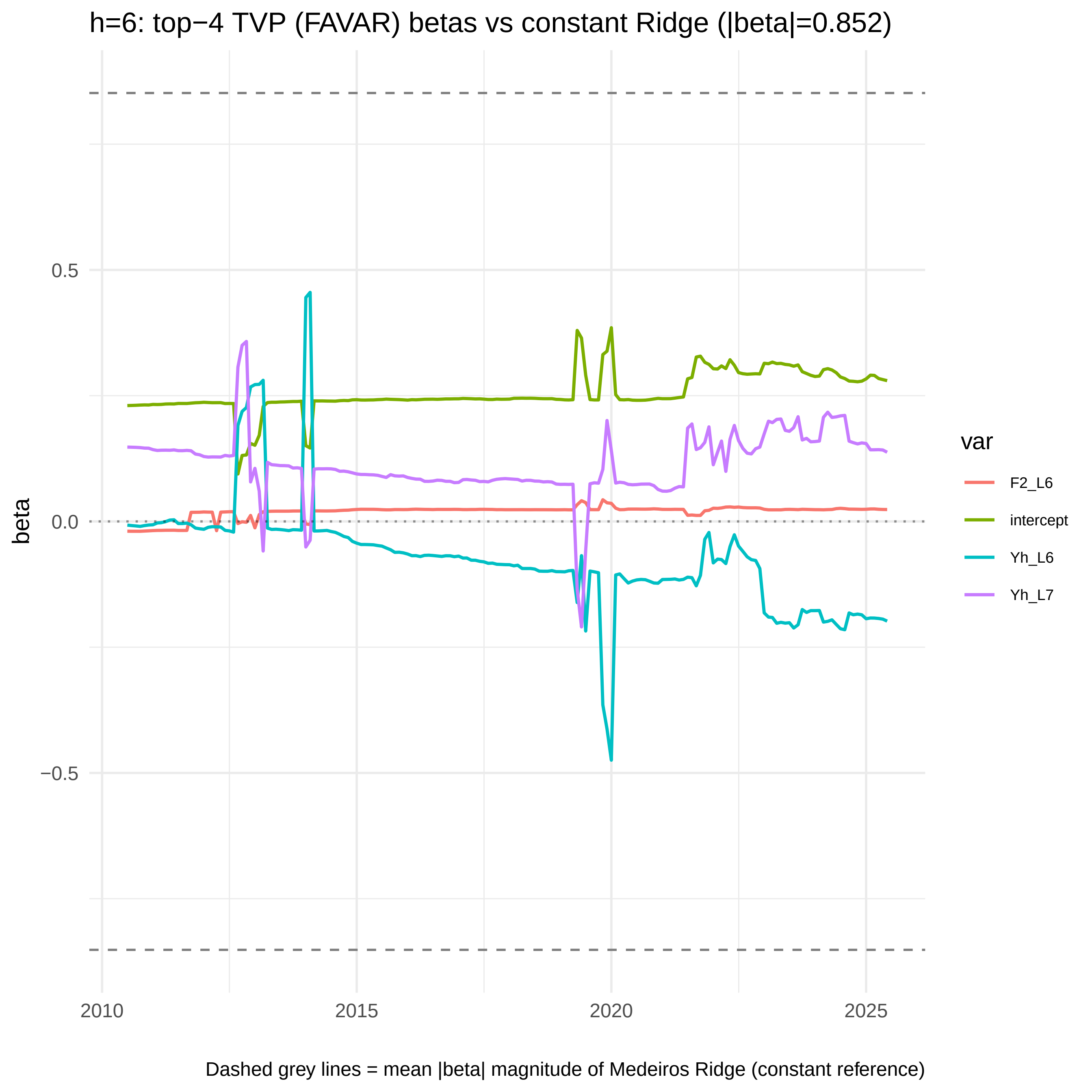

2SRR−FAVAR vs classical Ridge (Medeiros), h=3
Monthly inflation (p.p.)
2
0
−2
2010
2015
2020
2025
Realized (monthly y_t)
2SRR−FAVAR
Ridge (Medeiros)
Realized = monthly y_t; lines = forecast of pi_{t+h} (no rescaling)

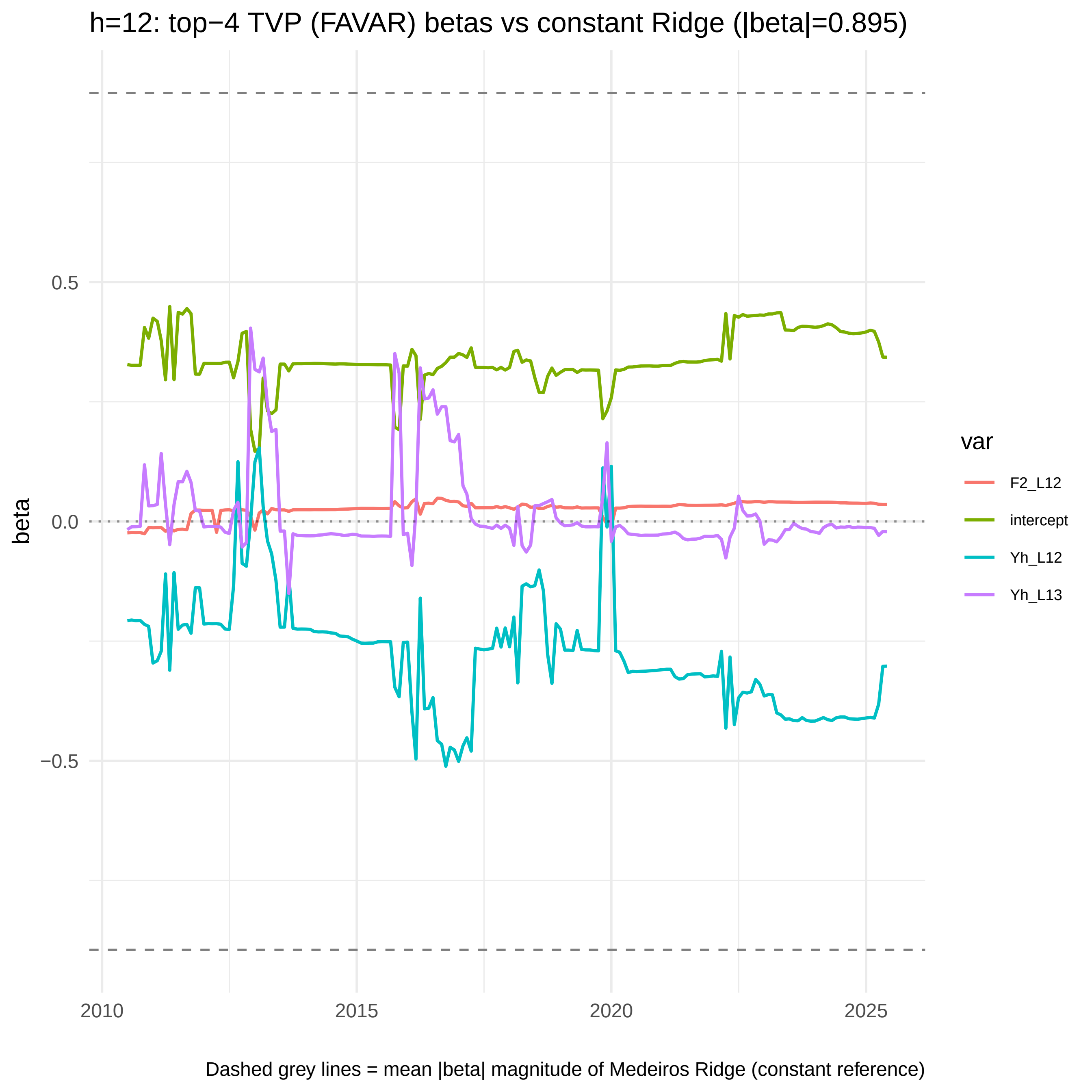

2SRR−FAVAR vs classical Ridge (Medeiros), h=6
Monthly inflation (p.p.)
2
0
−2
2010
2015
2020
2025
Realized (monthly y_t)
2SRR−FAVAR
Ridge (Medeiros)
Realized = monthly y_t; lines = forecast of pi_{t+h} (no rescaling)


2SRR−FAVAR vs classical Ridge (Medeiros), h=12
Monthly inflation (p.p.)
4
2
0
−2
2010
2015
2020
2025
Realized (monthly y_t)
2SRR−FAVAR
Ridge (Medeiros)
Realized = monthly y_t; lines = forecast of pi_{t+h} (no rescaling)


h=1: top−4 TVP (FAVAR) betas vs constant Ridge (|beta|=0.602)
beta
0.6
0.3
0.0
−0.3
−0.6
2010
2015
2020
2025
var
F2_L1
intercept
Yh_L1
Yh_L2
Dashed grey lines = mean |beta| magnitude of Medeiros Ridge (constant reference)

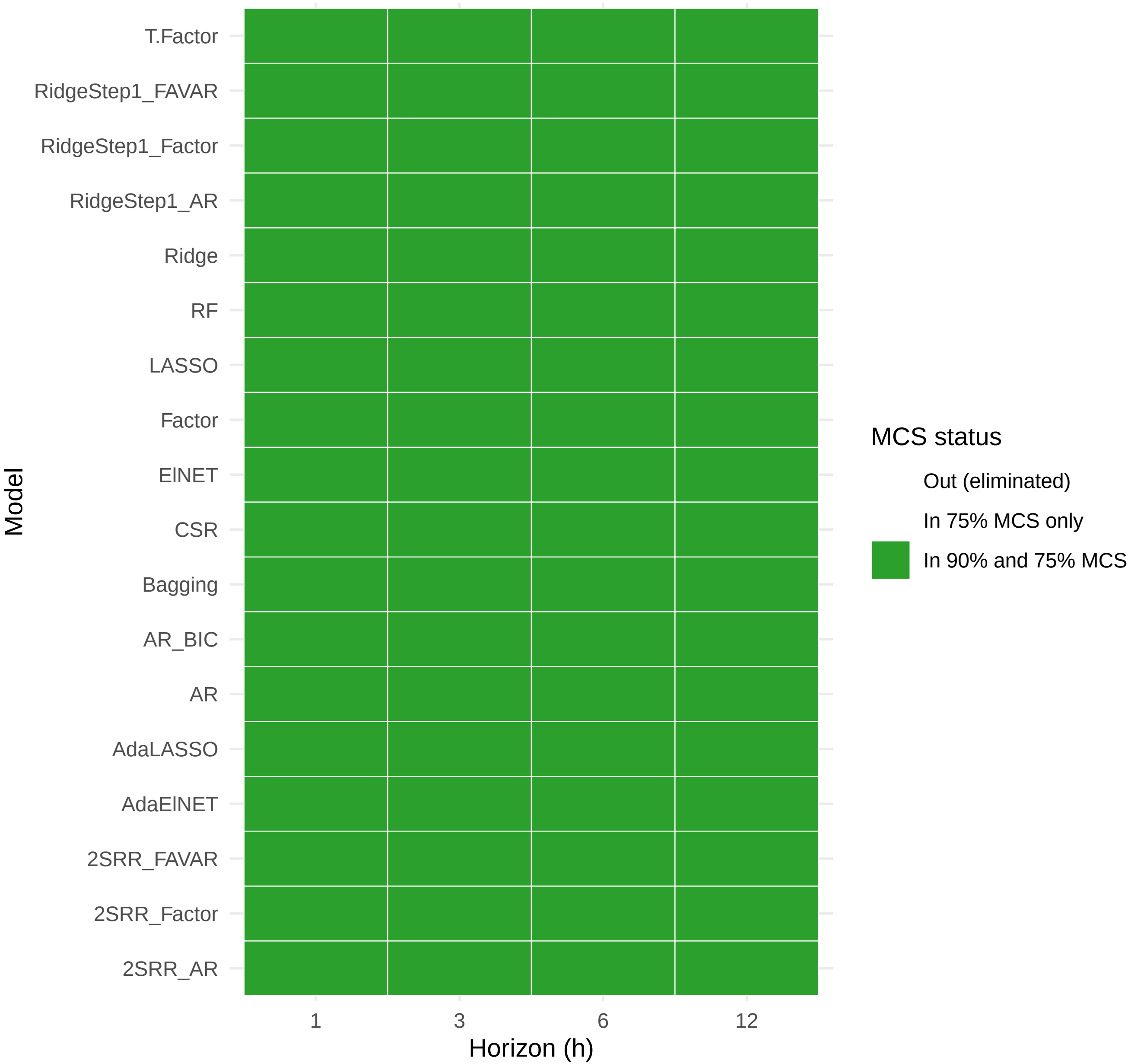

h=3: top−4 TVP (FAVAR) betas vs constant Ridge (|beta|=0.869)
beta
0.5
0.0
−0.5
2010
2015
2020
2025
var
F2_L3
intercept
Yh_L3
Yh_L4
Dashed grey lines = mean |beta| magnitude of Medeiros Ridge (constant reference)

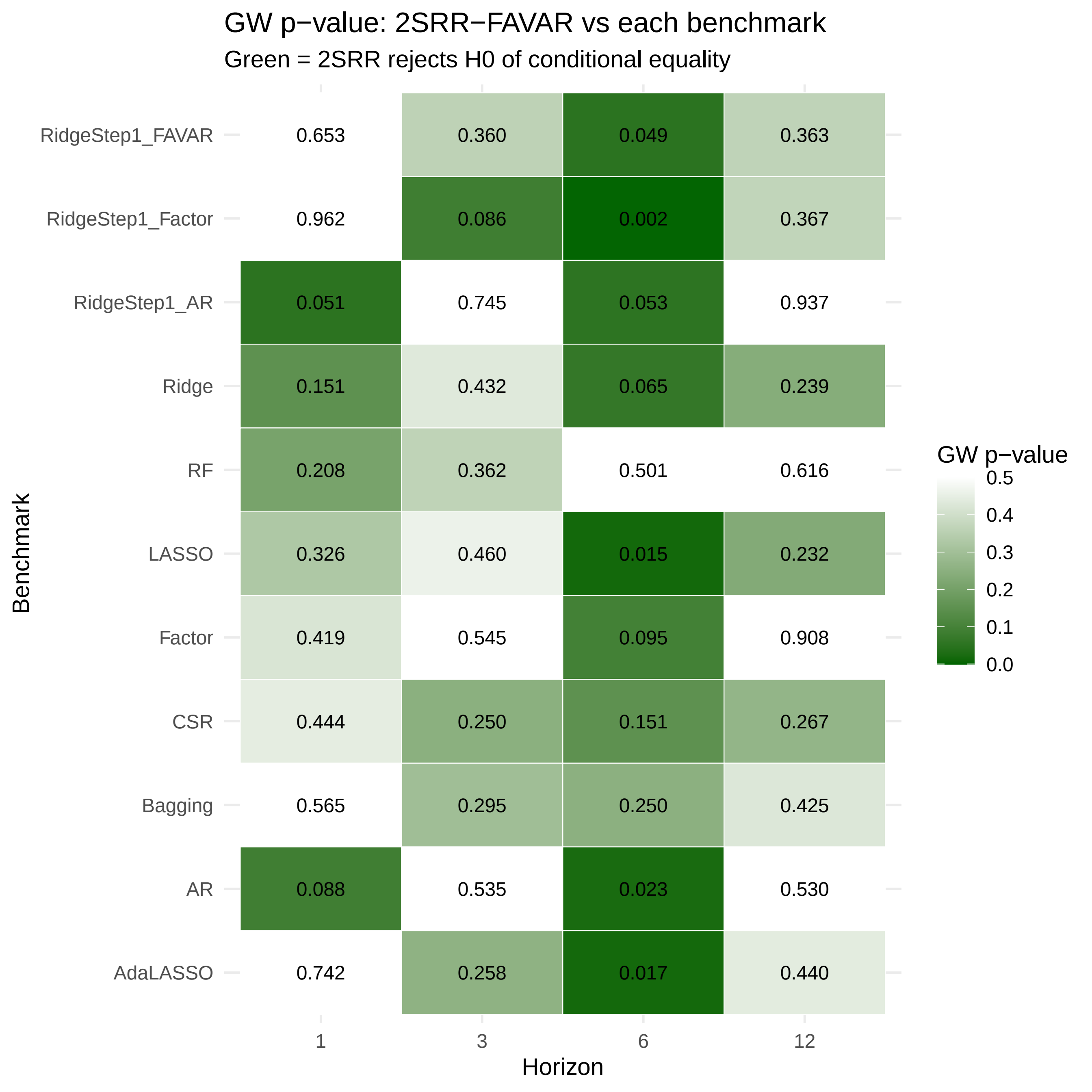

h=6: top−4 TVP (FAVAR) betas vs constant Ridge (|beta|=0.852)
beta
0.5
0.0
−0.5
2010
2015
2020
2025
var
F2_L6
intercept
Yh_L6
Yh_L7
Dashed grey lines = mean |beta| magnitude of Medeiros Ridge (constant reference)


h=12: top−4 TVP (FAVAR) betas vs constant Ridge (|beta|=0.895)
beta
0.5
0.0
−0.5
2010
2015
2020
2025
var
F2_L12
intercept
Yh_L12
Yh_L13
Dashed grey lines = mean |beta| magnitude of Medeiros Ridge (constant reference)

Model Confidence Set per horizon

Green = both MCS sets; orange = only 75% set; grey = eliminated




GW p−value: 2SRR−FAVAR vs each benchmark
Green = 2SRR rejects H0 of conditional equality
Benchmark
RidgeStep1_FAVAR
RidgeStep1_Factor
RidgeStep1_AR
Ridge
RF
LASSO
Factor
CSR
Bagging
AR
AdaLASSO
Horizon
1
3
6
12
0.653 0.360 0.049 0.363
0.962 0.086 0.002 0.367
0.051 0.745 0.053 0.937
0.151 0.432 0.065 0.239
0.208 0.362 0.501 0.616
0.326 0.460 0.015 0.232
0.419 0.545 0.095 0.908
0.444 0.250 0.151 0.267
0.565 0.295 0.250 0.425
0.088 0.535 0.023 0.530
0.742 0.258 0.017 0.440
GW p−value
0.5
0.4
0.3
0.2
0.1
0.0